Interpretable Multivariate Conformal Prediction with Fast Transductive Standardization

Yunjie Fan BRIANFAN@USC.EDU
Department of Mathematics
University of Southern California
Los Angeles, CA, USA

Matteo Sesia SESIA@MARSHALL.USC.EDU
Department of Data Sciences and Operations
Thomas Lord Department of Computer Science
University of Southern California
Los Angeles, CA, USA

Abstract

We propose a conformal prediction method for constructing tight simultaneous prediction intervals for multiple, potentially related, numerical outputs given a single input. This method can be combined with any multi-target regression model and guarantees finite-sample coverage. It is computationally efficient and yields informative prediction intervals even with limited data. The core idea is a novel *coordinate-wise* standardization procedure that makes residuals across output dimensions directly comparable, estimating suitable scaling parameters using the calibration data themselves. This requires neither modeling cross-output dependence nor auxiliary sample splitting. Implementing this idea requires overcoming technical challenges associated with transductive or full conformal prediction. Experiments on simulated and real data demonstrate that this method can produce tighter prediction intervals than existing baselines while maintaining valid simultaneous coverage.

Keywords: interpretability, multivariate analysis, statistical inference, supervised learning, uncertainty quantification.

1 Introduction

1.1 Motivation and Preview of Contributions

We are interested in jointly quantifying uncertainty when predicting several, possibly dependent, continuous outcomes from a shared set of input features. This problem is motivated by many applications where data-driven decisions depend on several unknown variables rather than on any one of them in isolation. For example, a financial institution may use available customer data to jointly forecast that customer's future credit-line utilization, monthly spending volume, and credit score. In this and many other contexts, decision quality hinges on the joint reliability of all predicted values, and an output that is incorrect in even one dimension could lead to adverse consequences.

Conformal prediction is a popular and broadly applicable framework for uncertainty quantification in predictive settings (Vovk et al., 2005; Angelopoulos and Bates, 2023). It derives its main strengths from the ability to augment the output of any machine-learning model with *prediction sets* equipped with finite-sample guarantees, without relying on strong assumptions on the data distribution. Its sole assumption is the availability of labeled data

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that are *exchangeable* with the test point of interest, a weaker requirement than the standard assumption of i.i.d. sampling. For a *univariate* setting, where the goal is to predict a single continuous-valued outcome, there now exist several well-established conformal prediction methods that produce valid, computationally fast, and easy-to-interpret prediction intervals (Lei et al., 2018; Romano et al., 2019; Sesia and Romano, 2021).

Extending conformal prediction to the *multivariate* setting is a non-trivial problem that has attracted considerable recent interest, spurring a variety of solutions (Dheur et al., 2025). The fundamental challenge is to aggregate a vector of generalized residuals—whose components may differ substantially in scale and variability—into a scalar-valued *nonconformity score* leading to *prediction sets* that should satisfy four key desiderata: (i) finite-sample coverage, (ii) informativeness or tightness, (iii) computational tractability, and (iv) interpretability. Interpretability can take different meanings depending on the application, and in some settings it is consistent with prediction sets having complex shapes. Nevertheless, in many practical scenarios, hyper-rectangular prediction sets—or, equivalently, jointly valid prediction intervals—provide the most natural and transparent way to convey uncertainty to practitioners. This paper focuses on such settings.

Existing methods achieve only a subset of these desiderata. The simplest approach to obtain hyper-rectangular prediction sets is to apply univariate conformal prediction to each output and then take the Cartesian product of the resulting intervals (Neeven and Smirnov, 2018); however, a conservative Bonferroni adjustment to the marginal miscoverage levels is needed to guarantee joint coverage; see Algorithm S1 in Appendix B. This strategy is very easy to implement, but the Bonferroni correction is often too conservative to be practically satisfactory, as it adopts a worst-case view of the dependence structure across output dimensions. By contrast, we aim to achieve valid coverage with joint prediction intervals that are as tight as possible; see Figure 1 for a sketch of this goal.

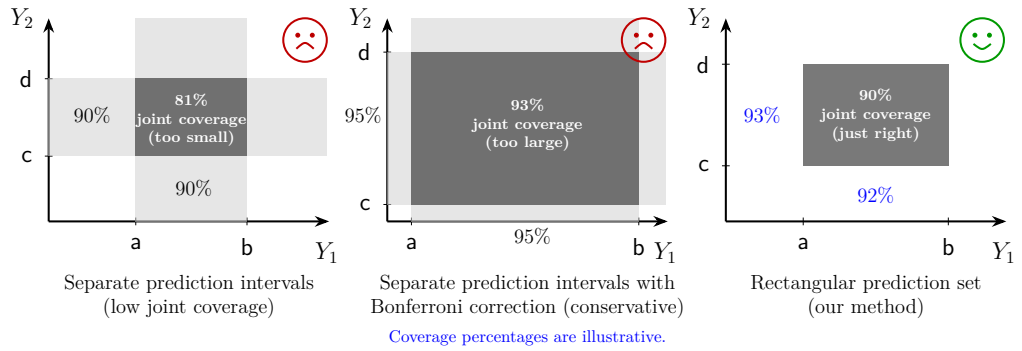


Figure 1: Two-dimensional toy example depicting graphically the main objective of this work: constructing *informative* and *interpretable* hyper-rectangular conformal prediction sets for multivariate outcomes. Left: univariate approaches can produce marginal prediction intervals for each output separately, which do not guarantee joint coverage. Center: a Bonferroni correction can easily restore valid joint coverage, at the cost of being overly conservative. Right: our goal is to efficiently construct hyper-rectangular prediction sets that are easy to interpret and retain valid coverage; doing so involves technical challenges.

A second class of methods reduces multivariate residuals to scalar-valued non-conformity scores, for example by applying an ℓ_2 or ℓ_∞ norm, applies univariate conformal calibration to these scores, and maps the resulting score threshold back into the original output space; see Algorithm S2 in Appendix B. While computationally efficient and able to produce hyper-rectangular prediction sets when using the ℓ_∞ norm, these approaches can be dominated by the noisiest dimension, leading to prediction sets that are unnecessarily wide along low-uncertainty directions.

More sophisticated approaches have been developed to directly minimize the volume of the resulting prediction sets, for example using optimal transport, learned embeddings, copulas, or generative models to describe the dependence structure across residual dimensions. However, these methods aim to solve a problem that is different from that considered in this paper, since they are explicitly designed to produce non-rectangular prediction sets. Moreover, they often introduce substantial computational overhead and require additional independent training data to provide finite-sample guarantees.

In this paper we focus on targeting what we believe is the core difficulty in many applications: the mismatch in the *marginal* scale and variability of residuals across output dimensions. To address this challenge, we develop a method based on transductive coordinate-wise residual standardization. This restores comparability across coordinates and enables the construction of tight hyper-rectangular prediction sets that guarantee finite-sample coverage, without necessitating additional training data or prohibitive computational overhead. Remarkably, our method does *not* need to explicitly model residual dependencies across different dimensions, unlike existing approaches focusing on non-rectangular prediction sets.

Making this idea operational requires overcoming several hurdles, which we resolve through a sequence of increasingly practical steps. We begin by formulating an ideal oracle procedure that assumes access to the test point's true outcome vector and uses this information to standardize the multi-dimensional residuals in the calibration set without violating the exchangeability conditions that are necessary to guarantee finite-sample coverage. Although clearly impractical, this oracle is valuable to guide the design of our practical method. Inspired by the oracle procedure, we introduce two increasingly sophisticated data-driven approaches. The first provides a more conservative approximation of the oracle, whereas the second achieves greater statistical efficiency by building on top of the first. Through extensive numerical experiments, we demonstrate that this progression of ideas culminates in a practical method producing prediction sets satisfying all our desiderata.

1.2 Related Work

Extending conformal prediction to multivariate outcomes has attracted substantial attention in recent years, leading to the development of a wide range of alternative methods.

A prominent line of work seeks minimum-volume prediction sets with flexible geometries by modeling dependence across output dimensions. This includes copula-based approaches (Messoudi et al., 2021; Park and Cho, 2025; Sun and Yu, 2023), non-parametric density estimation methods (Sadinle et al., 2019; Izbicki et al., 2022; Wang et al., 2023; Dheur et al., 2025), parametric density estimation methods (Johnstone and Cox, 2021; Braun et al., 2025a,b; Tumu et al., 2024), optimal-transport techniques (Thurin et al., 2025; Klein et al., 2025), and flow-based generative models (Colombo, 2024; Lee et al., 2025; Fang et al.,

2025). While these approaches can produce smaller prediction sets than our method, the resulting sets often have complex shapes that are more difficult to interpret. Moreover, these approaches require additional model training on independent data to provide finite-sample guarantees, and many also involve substantial computational overhead.

To the best of our knowledge, the idea of directly addressing heterogeneity across output dimensions via marginal residual rescaling remains relatively under-explored. Baheri and Amiri Shahbazi (2025) pursue a related idea, but rely on a conservative Bonferroni-type correction. The approach of Sampson and Chan (2024) is more closely related to ours, but their method requires a base region of predictions, which one would need to estimate through an additional independent dataset if not provided directly through model estimation; avoiding this requirement is the main technical challenge addressed in our paper. As we show empirically, the methods of Sampson and Chan (2024) are often considerably less efficient than ours—particularly in small-sample regimes—resulting in less stable and less informative prediction sets.

Our solution can be viewed as implementing a special instance of transductive, or full, conformal prediction (Vovk et al., 2005; Vovk, 2013). In contrast to the more commonly used split conformal methods, full conformal prediction repeatedly incorporates the test point, augmented with different candidate labels, directly into a learning algorithm. Although this strategy is computationally prohibitive in general, especially for continuous-valued numerical outcomes and complex algorithms, it can be implemented efficiently in some cases by exploiting structural properties of the underlying predictive model. Accordingly, our paper relates to a broader literature on making full conformal inference computationally practical in special cases, including approaches tailored to sparse generalized linear models (Lei, 2019; Guha et al., 2023) and methods that leverage stability or other forms of structure in the predictive model (Ndiaye and Takeuchi, 2019; Abad et al., 2022; Martinez et al., 2023; Li, 2024). Unlike those works, which focus primarily on accelerating model refitting, we assume a fixed, pre-trained model and instead aim to improve efficiency by avoiding calibration data splitting while explicitly standardizing residuals across output dimensions. Both the problem setting and the proposed methodology are therefore novel.

More broadly, our approach is also related to recent conformal methods that reuse calibration data for different purposes, such as enabling early stopping to control overfitting when training deep learning models (Liang et al., 2023) and leveraging data-adaptive grouping to improve conditional coverage (Zhou and Sesia, 2024; Zhang and Candès, 2024). In contrast to these works, our focus is on addressing heterogeneity across targets in multivariate regression while obtaining interpretable, rectangular prediction sets.

1.3 Outline

This paper is organized as follows. Section 2 introduces notation, formal problem statement, and the required assumptions, along with a description of an ideal population oracle approach that has access to population-level information. Section 3 presents our methodology, starting with a transductive oracle that knows the unobserved test outcome, followed by the general construction of our method along with its computable implementations. Section 4 validates our method empirically and compares its performance to that of existing approaches using both simulated and real data. Section 5 discusses some limitations and possible directions

for future research. The appendices contain the remaining mathematical proofs, additional algorithmic details, and further numerical results.

2 Problem Setup and Motivating Constructions

In this section, we present the formal mathematical formulation of our problem of interest along with the motivation of our solution and discuss all the necessary background to understand our method to be introduced in Section 3.

Notation For any $m \in \mathbb{N}$, we use $[m] := \{1, \ldots, m\}$ to denote the set of m indices. For any $m \in \mathbb{N}$, $\alpha \in (0,1)$, and $(v_1, \ldots, v_m) \in \mathbb{R}^m$, we use $\hat{Q}_{1-\alpha}(v_1, \ldots, v_m)$ to denote the $\lceil (1-\alpha)(m+1) \rceil$-th smallest value of the augmented list $(v_1, \ldots, v_m, +\infty)$, counting repeated values separately. We write $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$.

2.1 Problem Setup

We consider a multi-output (multivariate) regression setting where the goal is to predict a vector $\mathbf{Y} = (Y_1, \ldots, Y_d)$ of d real-valued outputs given input features $\mathbf{X} \in \mathcal{X}$.

Assume that $(\mathbf{X}^1, \mathbf{Y}^1), \ldots, (\mathbf{X}^{n+1}, \mathbf{Y}^{n+1})$, with $n \geq 2$, are jointly exchangeable, with common marginal distribution $P_{X,Y}$. This includes the case where the observations are drawn i.i.d. from an arbitrary distribution $P_{X,Y}$. We observe the calibration data $\mathcal{D}_{\text{cal}} = \{(\mathbf{X}^i, \mathbf{Y}^i)\}_{i=1}^n$ and the test input (covariate) $\mathbf{X}^{n+1}$, while the corresponding outcome $\mathbf{Y}^{n+1}$ remains unobserved and to be predicted. Our paper focus on the following two inferential targets of $\mathbf{Y}^{n+1}$.

Marginal Coverage Using all of the observed data, the task is to construct a reasonably tight prediction set $\hat{C}(\mathbf{X}^{n+1}) \subseteq \mathbb{R}^d$ that contains the unknown output $\mathbf{Y}^{n+1}$ with high probability, guaranteeing *marginal coverage*:

$$\mathbb{P}\Big[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})\Big] \geq 1 - \alpha, \tag{1}$$

where $\alpha \in (0,1)$ is the user-specified miscoverage level. The term "marginal coverage" means that the guarantee in (1) averages over the joint randomness of the calibration samples $\mathcal{D}_{\text{cal}}$ and the test point $(\mathbf{X}^{n+1}, \mathbf{Y}^{n+1})$.

Stronger notions of guarantees, such as coverage conditional on a certain realization of $\mathbf{X}^{n+1}$, are generally unattainable in finite samples with finite, informative prediction sets and without more restrictive assumptions on the distribution (Foygel Barber et al., 2021). Methods that approximate conditional coverage (Romano et al., 2020; Gibbs et al., 2025) may be combined with our approach. Investigating such combinations is an interesting direction for future work but is beyond the scope of this paper.

Interpretability In addition to coverage, we seek prediction sets with a simple, interpretable geometry. For the standard residual choices considered in the next section, our method produces *hyper-rectangular* prediction sets, or Cartesian products of univariate prediction intervals:

$$\hat{C}(\mathbf{X}^{n+1}) = \bigtimes_{j=1}^{d} \left(\left[L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1}) \right] \cap \mathbb{R} \right), \tag{2}$$

where $[L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})]$ denotes the coordinate-wise interval for the jth output. We intersect each interval with $\mathbb{R}$ for mathematical completeness since the coordinatewise bounds are allowed to be infinite.

Prediction sets of the form (2) provide a separate interval to quantify the uncertainty of each output, making them easy to interpret in the context of multi-output prediction: they tell us exactly which component misses the true response when a prediction set fails, as displayed in Table 2. For such sets, the marginal coverage requirement in (1) is equivalent to

$$\mathbb{P}\left[Y_j^{n+1} \in [L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})] \text{ for all } j \in [d]\right] \geq 1 - \alpha.$$

Thus, although the coverage guarantee is marginal over the calibration and test data, it is simultaneous across all d output coordinates.

In the following subsections, we will discuss how the coverage and interpretability requirements could be achieved in the context of split conformal prediction.

2.2 Coordinate-Wise Generalized Residuals

In the univariate case, split conformal prediction uses a common choice of nonconformity score to measure disagreement between a candidate output and a pre-trained model's prediction and build prediction intervals through exchangeable calibration of the scores. Our construction extends this idea to multivariate outputs by defining a generalized residual for each output coordinate. These residuals are then calibrated jointly to obtain a prediction set for the full output vector.

We start by assuming access to a *pre-trained* joint predictive model $\hat{\mathbf{f}} = (\hat{f}_1, \ldots, \hat{f}_d)$, fitted on a training dataset independent of both the calibration set $\mathcal{D}_{\text{cal}}$ and the test point. We treat this model as a black box. The only requirement is that it allows us to compute a measurable coordinate-wise *generalized residual* map $\mathbf{E} : \mathcal{X} \times \mathbb{R}^d \to \mathbb{R}^d$ of the form

$$\mathbf{E}(\mathbf{X}, \mathbf{Y}) = \big(E_1(\mathbf{X}, Y_1), \ldots, E_d(\mathbf{X}, Y_d)\big).$$

Each component of the residual map assesses agreement between an output and the corresponding fitted prediction, with larger values indicating greater disagreement. Most residual maps have a coordinatewise lower bound at each input $\mathbf{x} \in \mathcal{X}$ to represent the scenario of perfect prediction, and some may have a coordinatewise upper bound depending on the data. We denote these two bounds here coordinatewise by $[a_j(\mathbf{x}), b_j(\mathbf{x})]$, where

$$a_j(\mathbf{x}) = \inf_{y_j \in \mathbb{R}} E_j(\mathbf{x}, y_j), \qquad b_j(\mathbf{x}) = \sup_{y_j \in \mathbb{R}} E_j(\mathbf{x}, y_j), \qquad \mathbf{x} \in \mathcal{X}.$$

In most cases, $b_j(\mathbf{X}^{n+1}) \equiv +\infty$ because the disagreement can be arbitrarily large for bad predictions. Thus, we will use $+\infty$ as a coordinatewise upper bound throughout this paper.

We supply two standard choices of residuals in the context of multivariate predictions here along with their coordinatewise lower bounds and upper bounds when no additional information is given from the data:

1. *Absolute residuals*: for a model $\hat{f}_j$ providing a point prediction of Y_j, such as an estimate of the conditional mean $\mathbb{E}[Y_j \mid \mathbf{X}]$, a standard choice is

   $$E_j(\mathbf{X}, Y_j) = |Y_j - \hat{f}_j(\mathbf{X})|$$

   (Lei et al., 2018); its coordinatewise lower bounds and upper bounds are simply $a_j(\mathbf{X}^{n+1}) \equiv 0$ and $b_j(\mathbf{X}^{n+1}) \equiv +\infty$ for all $j \in [d]$.

2. *Quantile residuals*: for models that estimate conditional quantiles, as in conformalized quantile regression (Romano et al., 2019), a natural choice is

$$E_j(\mathbf{X}, Y_j) = \max\{\hat{f}_j^{\mathrm{lo}}(\mathbf{X}) - Y_j,\ Y_j - \hat{f}_j^{\mathrm{hi}}(\mathbf{X})\},$$

where $\hat{f}_j = (\hat{f}_j^{\mathrm{lo}}, \hat{f}_j^{\mathrm{hi}})$ with $\hat{f}_j^{\mathrm{lo}}(\mathbf{X}) \leq \hat{f}_j^{\mathrm{hi}}(\mathbf{X})$ denotes the pair of estimated lower and upper conditional quantiles of $Y_j \mid \mathbf{X}$, for example at levels $\alpha/2$ and $1-\alpha/2$.

This residual measures the signed distance of Y_j to the nearest boundary of the interval $[\hat{f}_j^{\mathrm{lo}}(\mathbf{X}), \hat{f}_j^{\mathrm{hi}}(\mathbf{X})]$: positive values indicate that Y_j falls outside the interval, negative values indicate that it lies strictly inside, and zero indicates that it lies on the boundary. Hence, the coordinatewise lower bounds of this residual are actually input dependent, i.e.,

$$a_j(\mathbf{X}^{n+1}) = -\frac{1}{2}\left(\hat{f}_j^{\mathrm{hi}}(\mathbf{X}^{n+1}) - \hat{f}_j^{\mathrm{lo}}(\mathbf{X}^{n+1})\right),$$

and its coordinatewise upper bounds are still $b_j(\mathbf{X}^{n+1}) \equiv +\infty$.

Applying the generalized residual map to the calibration data gives $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i) = (E_1^i, \ldots, E_d^i)$ for all $i \in [n]$. We reserve $\mathbf{E}^{n+1} = \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{Y}^{n+1})$ for the unobserved residual vector of the test point. Because the same independently fitted residual map is applied to every observation, the calibration and test residual vectors remain exchangeable.

This exchangeability preservation alone is sufficient for the basic conformal coverage guarantee below. Additional regularity conditions used in our method will be introduced in Section 2.4.

2.3 Conformal Calibration with Marginal Coverage

Given the calibration residuals $\mathbf{E}^1, \ldots, \mathbf{E}^n$, we construct a prediction set by accepting candidate outputs whose residuals fall below an upper bound in each coordinate:

$$\hat{C}(\mathbf{X}^{n+1}) = \left\{\mathbf{y} \in \mathbb{R}^d : a_j(\mathbf{X}^{n+1}) \leq E_j(\mathbf{X}^{n+1}, y_j) \leq W_j \text{ for all } j \in [d]\right\}. \tag{3}$$

Here, $W_1, \ldots, W_d$ are suitable data-dependent upper bounds, possibly infinite, chosen to satisfy

$$\mathbb{P}\left[E_j(\mathbf{X}^{n+1}, Y_j^{n+1}) \leq W_j \text{ for all } j \in [d]\right] \geq 1 - \alpha. \tag{4}$$

By construction, this condition is equivalent to the marginal coverage requirement in (1). If the residual map is fixed, increasing the thresholds coordinatewise can only enlarge the prediction set, giving us a route to replace any unknown thresholds by computable ones.

For the two common choices of residuals introduced above, each coordinate of the prediction set in (3) becomes a prediction interval: (i) $\left[\hat{f}_j(\mathbf{X}^{n+1}) - W_j, \hat{f}_j(\mathbf{X}^{n+1}) + W_j\right]$ for absolute residuals and (ii) $\left[\hat{f}_j^{\mathrm{lo}}(\mathbf{X}^{n+1}) - W_j,\ \hat{f}_j^{\mathrm{hi}}(\mathbf{X}^{n+1}) + W_j\right]$ for quantile residuals, hence indicating the final hyper-rectangular geometry.

It is worth mentioning that quantile residuals could lead to empty prediction sets if

$$W_j < -\frac{1}{2}\left(\hat{f}_j^{\mathrm{hi}}(\mathbf{X}^{n+1}) - \hat{f}_j^{\mathrm{lo}}(\mathbf{X}^{n+1})\right)$$

for some $j \in [d]$, although marginal coverage ensures that the probability of an empty prediction set is at most α. Sesia and Candès (2020) presents several modifications to quantile residuals to prevent the existence of empty sets, but we stick to the common choice presented above in this paper.

Also, for a general residual map, our intended construction in (3) need not be hyper-rectangular in the outcome space. Due to the variability of tasks in reality, it might be inefective to enforce a hyper-rectangular geometry in the outcome space for every possible residual map. Our construction in (3) instead describes the prediction set through coordinatewise bounds on the residuals, therefore ensuring the corresponding acceptance region is hyper-rectangular in the residual space and offering a sense of interpretability.

Finding suitable (informative) thresholds $W_1, \ldots, W_d$ satisfying (4) is also non-trivial. A naive way to achieve (4) is to set $W_j = \hat{Q}_{1-\alpha/d}(E_j^1, \ldots, E_j^n)$, which is equivalent to applying univariate conformal prediction separately across all d outcomes, with a Bonferroni correction (α/d replacing α). This typically leads to very conservative prediction sets because it takes an overly pessimistic worst-case view of possible dependencies across outcomes.

In this paper, we pursue a more efficient approach, beginning with induced *scalar* nonconformity scores $S^i = \Phi(\mathbf{E}^i) \in \mathbb{R}$ for all $i \in [n]$, computed using a measurable scalarization map $\Phi : \mathbb{R}^d \to \mathbb{R}$ whose form will be discussed below. We similarly write $S^{n+1} = \Phi(\mathbf{E}^{n+1})$ for the unobserved test score. Then, the prediction set is constructed as

$$\hat{C}(\mathbf{X}^{n+1}) = \left\{ \mathbf{y} \in \mathbb{R}^d : \Phi(\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y})) \leq \hat{Q}_{1-\alpha}(S^1, \ldots, S^n) \right\}. \tag{5}$$

If we make sure Φ comes from a function class where, for each scalarization map Φ, there exists a threshold map $g = (g_1, \ldots, g_d) : \overline{\mathbb{R}} \to \overline{\mathbb{R}}^d$ such that

$$\Phi(\mathbf{e}) \leq c \iff e_j \leq g_j(c) \text{ for all } j \in [d], \qquad \mathbf{e} \in \mathbb{R}^d, \quad c \in \overline{\mathbb{R}},$$

then we can recover the form of (3) using thresholds $W_j = g_j(\hat{Q}_{1-\alpha}(S^1, \ldots, S^n))$. We focus on transformations with this property, including the standardized maximum introduced below, but the coverage guarantee holds for other choices of Φ, as demonstrated by the following theorem.

Theorem 1 (Generalized Marginal Coverage) *Assume $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ are exchangeable, with $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i) \in \mathbb{R}^d$ for all $i \in [n+1]$. Let $\Phi : \mathbb{R}^d \to \mathbb{R}$ be any fixed measurable function, and write $S^i = \Phi(\mathbf{E}^i)$. Then the prediction set in (5) satisfies $\mathbb{P}\left[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})\right] \geq 1 - \alpha$. Moreover, if $S^1, \ldots, S^{n+1}$ are almost surely distinct, then $\mathbb{P}\left[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})\right] \leq 1 - \alpha + \frac{1}{n+1}$. The same result holds for a random Φ if $(S^1, \ldots, S^{n+1})$ are exchangeable conditional on Φ.*

Although Theorem 1 establishes marginal coverage overall, the guarantee alone does not yield similar coverage probabilities across different outputs. Depending on the residual distribution and the choice of Φ, some intervals may be much wider than necessary, making the resulting uncertainty estimates less informative (see Figure 7 for an example).

Several recent works consider norm-based scalarizations, such as $\Phi(\mathbf{e}) = \|\mathbf{e}\|_p$ for $1 \leq p \leq \infty$ (Johnstone and Ndiaye, 2022). Although conceptually and computationally simple,

these can be dominated by a residual coordinate with a much larger scale than the others, leading to unnecessarily large acceptance regions in the remaining coordinates.

To obtain tighter prediction sets, recent works have proposed learning a transformation of the residuals before applying a norm-based scalarization, for example through optimal transport (Thurin et al., 2025; Klein et al., 2025). However, estimating these transport maps requires additional data splitting of the calibration dataset to preserve exchangeability and substantial computation, and the resulting prediction sets can have complicated geometries (even in the residual space).

We pursue a route similar to these two approaches, but instead of learning a complicated transformation on the observed data, we propose a simple and interpretable transformation that standardizes the residuals coordinatewise and takes their maximum across coordinates without taking absolute values. Our standardization also uses estimates from the observed data, but in a way that retains finite-sample marginal coverage and avoids additional data splitting. This construction also yields rectangular prediction sets for the standard residual choices above and aims to avoid unnecessarily conservative intervals in some output dimensions.

2.4 A Population-Standardized Oracle

To motivate our method, we first consider an oracle construction that has access to the coordinate-wise population means and standard deviations of the residuals. Although unavailable in practice, this construction illustrates how coordinate-wise standardization can mitigate heterogeneity in residual scales while preserving coverage and, for standard residual choices, rectangular interpretability.

Consider prediction sets of the form in (5), obtained by transforming the vector-valued residuals $\mathbf{E}^i$ using the following family of scalarization maps:

$$\Phi(\mathbf{t}; \boldsymbol{\beta}, \boldsymbol{\theta}) := \max_{1 \leq j \leq d} \frac{t_j - \beta_j}{\theta_j}, \qquad \mathbf{t} \in \mathbb{R}^d, \tag{6}$$

where $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_d)^\top$ and $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_d)^\top$, with $\theta_j > 0$ for all $j \in [d]$, are vectors of location and scale parameters, respectively. This transformation certainly has the coordinate-wise representation, discussed earlier in Section 2.3:

$$\Phi(\mathbf{t}; \boldsymbol{\beta}, \boldsymbol{\theta}) \leq c \iff t_j \leq c\theta_j + \beta_j \text{ for all } j \in [d], \qquad c \in \overline{\mathbb{R}},$$

with the convention $\pm\infty \cdot \theta_j + \beta_j = \pm\infty$.

Given these two parameter vectors, we define the prediction set at nominal level $1 - \alpha$ and test covariate $\mathbf{X}^{n+1}$ by

$$\begin{gathered} C(\mathbf{X}^{n+1}; \boldsymbol{\beta}, \boldsymbol{\theta}) = \left\{ \mathbf{y} \in \mathbb{R}^d : a_j(\mathbf{X}^{n+1}) \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\boldsymbol{\beta},\boldsymbol{\theta}}_{1-\alpha} \theta_j + \beta_j \text{ for all } j \in [d] \right\}, \\ \text{where} \quad \hat{Q}^{\boldsymbol{\beta},\boldsymbol{\theta}}_{1-\alpha} := \hat{Q}_{1-\alpha}\big(S^1_{\boldsymbol{\beta},\boldsymbol{\theta}}, \ldots, S^n_{\boldsymbol{\beta},\boldsymbol{\theta}}\big), \quad S^i_{\boldsymbol{\beta},\boldsymbol{\theta}} = \Phi(\mathbf{E}^i; \boldsymbol{\beta}, \boldsymbol{\theta}). \end{gathered} \tag{7}$$

This set achieves the desired coordinatewise structure defined in (3), with $W_j = \hat{Q}^{\boldsymbol{\beta},\boldsymbol{\theta}}_{1-\alpha} \theta_j + \beta_j$, and is hyper-rectangular for the standard residual choices described above. Therefore,

Theorem 1 guarantees that $C(\mathbf{X}^{n+1}; \boldsymbol{\beta}, \boldsymbol{\theta})$ satisfies the marginal coverage requirement in (1) for any fixed choice of $(\boldsymbol{\beta}, \boldsymbol{\theta})$.

To address heterogeneity in scale and variability across dimensions, a natural choice for $(\boldsymbol{\beta}, \boldsymbol{\theta})$ is the pair of coordinate-wise population mean and standard deviation vectors of $\mathbf{E}(\mathbf{X}, \mathbf{Y})$, denoted by $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$:

$$\begin{aligned} \boldsymbol{\mu}^* &= (\mu_1^*, \ldots, \mu_d^*)^\top, \qquad \boldsymbol{\sigma}^* = (\sigma_1^*, \ldots, \sigma_d^*)^\top \\ \mu_j^* &= \mathbb{E}[E_j(\mathbf{X}, Y_j)], \qquad \sigma_j^* = \sqrt{\mathrm{Var}[E_j(\mathbf{X}, Y_j)]} > 0. \end{aligned} \tag{8}$$

Then, by replacing $(\boldsymbol{\beta}, \boldsymbol{\theta})$ in (7) with the population parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$, we obtain a population-standardized oracle prediction set $C(\mathbf{X}^{n+1}; \boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$.

For this construction to be valid mathematically and extendable in practice, we do need to introduce additional regularity conditions on the residuals:

Assumption 1 *For every $j \in [d]$, the distribution of $E_j(\mathbf{X}, Y_j)$ has no point masses and satisfies: $\mathbb{E}\left[|E_j(\mathbf{X}, Y_j)|\right] < \infty, \quad 0 < \mathrm{Var}\left[E_j(\mathbf{X}, Y_j)\right] < \infty$. Moreover, for every $j \in [d]$, the residuals $E_j^1, \ldots, E_j^{n+1}$ are almost surely distinct.*

These moment conditions hold automatically for many bounded, nondegenerate residuals in practice. The distinctness and absence-of-point-masses assumptions are standard in the conformal prediction literature. If needed, any ties between residuals can be removed by adding small i.i.d. bounded continuous perturbations, independent of the data; the same rule is applied for calibration and test observations simultaneously to ensure valid construction of prediction sets (Vovk et al., 2005).

These additional assumptions are presented for mathematical completeness of the subsequent theoretical development; they are not needed for the coverage guarantee in Theorem 1. We have also performed a batch of sensitivity tests against these assumptions in Appendix D.

Under Assumption 1, the oracle prediction set $C(\mathbf{X}^{n+1}; \boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ has four appealing properties: (i) it guarantees finite-sample marginal coverage, by Theorem 1; (ii) it puts all residual components on the same mean and variance scale, which can help balance coordinate-wise coverage across output dimensions; (iii) it is fast to compute once the population parameters are supplied; (iv) it is an easy-to-interpret hyper-rectangle, at least for standard choices of residuals.

In the following, we will describe practical approximations of the oracle prediction set, starting from intuitive but not fully satisfactory solutions before presenting our method.

2.5 Simple but Unsatisfactory Practical Approaches

Naive Approach An intuitive *plug-in* approximation of the ideal oracle is obtained by replacing the population parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ with their empirical counterparts $(\hat{\boldsymbol{\mu}}^{\mathrm{pi}}, \hat{\boldsymbol{\sigma}}^{\mathrm{pi}})$, evaluated on the observed calibration data, and then plugging $(\boldsymbol{\beta}, \boldsymbol{\theta}) = (\hat{\boldsymbol{\mu}}^{\mathrm{pi}}, \hat{\boldsymbol{\sigma}}^{\mathrm{pi}})$ into the prediction set formula (7): $\hat{C}^{\mathrm{pi}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\mathrm{pi}}, \hat{\boldsymbol{\sigma}}^{\mathrm{pi}})$.

This approach, outlined by Algorithm S3 in Appendix B, tends to perform well in large samples, as shown empirically in Appendix D, but does not generally guarantee marginal coverage because its parameter estimation step can break exchangeability between the transformed calibration and test scores. For $d = 1$, the plug-in method recovers ordinary univariate split CP because the effects of standardization cancel in the one-dimensional case.

Data-Splitting Approach An alternative solution that is also intuitive yet unsatisfactory consists of combining the above plug-in approach with an additional data-splitting step: the location and scale parameters, denoted here by $(\hat{\boldsymbol{\mu}}^{\mathrm{ds}}, \hat{\boldsymbol{\sigma}}^{\mathrm{ds}})$, are estimated empirically using only a subset of n_1 calibration observations, with $2 \leq n_1 < n$, selected at random, and the remaining $n_2 = n - n_1$ observations are used to construct $\hat{C}^{\mathrm{ds}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\mathrm{ds}}, \hat{\boldsymbol{\sigma}}^{\mathrm{ds}})$, with the calibration sample in (7) restricted to these n_2 observations. Conditional on the observations used to estimate the location and scale parameters, the remaining n_2 calibration observations and the test sample remain exchangeable, hence recovering marginal coverage. This approach is outlined by Algorithm S4 in Appendix B.

While this split (inductive) approach satisfies the key assumption of Theorem 1, it uses the data inefficiently. It reduces the effective number of calibration samples from n to n_2, roughly by half for an even split, and tends to produce more unstable and often wider-than-necessary prediction sets, especially in small-sample regimes.

As we mentioned earlier in Section 2.3, one main contribution of this paper is to overcome this inefficiency. We develop a *transductive* conformal prediction method that uses a similar standardization idea but avoids this additional data split by reusing the entire calibration sample jointly with a candidate test residual, all while retaining marginal coverage at manageable computational costs.

In Section 3, we will first present our methodology with a transductive oracle construction that treats all $n+1$ residual vectors symmetrically, then derive a computable conservative approximation that does not require the unknown test outcome. Achieving this requires resolving several technical challenges, as explained step by step below.

3 Methodology

According to Theorem 1, prediction sets of the form (5) achieve finite-sample marginal coverage if the scalarization map Φ is fixed or independent of the entire collection $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$, or, more generally, if $(\Phi(\mathbf{E}^1), \ldots, \Phi(\mathbf{E}^{n+1}))$ remains *exchangeable conditional on* Φ.

This latter requirement, which is strictly weaker than independence from the calibration and test residuals, is the foundation of the method developed in this section. We begin by showing how this conditional-exchangeability condition naturally leads to a different *transductive* oracle, which is much more closely aligned with the practical method introduced below.

3.1 A Transductive Oracle That Knows the Test Residual

Consider an imaginary oracle that knows the unordered values in $\{\mathbf{E}^1, \ldots, \mathbf{E}^{n+1}\}$; unlike the previous oracle discussed in Section 2.4, no population-level information is directly accessible. Instead, we estimate the population location and scale parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ defined in (8) using these $n+1$ residual vectors:

$$\begin{aligned} &\hat{\boldsymbol{\mu}}^{\mathrm{oracle}} = (\hat{\mu}_1^{\mathrm{oracle}}, \ldots, \hat{\mu}_d^{\mathrm{oracle}})^\top, \qquad \hat{\boldsymbol{\sigma}}^{\mathrm{oracle}} = (\hat{\sigma}_1^{\mathrm{oracle}}, \ldots, \hat{\sigma}_d^{\mathrm{oracle}})^\top, \\ &\hat{\mu}_j^{\mathrm{oracle}} = \frac{1}{n+1} \sum_{i=1}^{n+1} E_j^i, \qquad \hat{\sigma}_j^{\mathrm{oracle}} = \sqrt{\frac{1}{n} \sum_{i=1}^{n+1} \big(E_j^i - \hat{\mu}_j^{\mathrm{oracle}}\big)^2}. \end{aligned} \tag{9}$$

Since Assumption 1 ensures the residual vectors are almost surely distinct coordinatewise, we know $\hat{\sigma}_j^{\text{oracle}} > 0$ almost surely for all $j \in [d]$. Then substituting $(\boldsymbol{\beta}, \boldsymbol{\theta}) = (\hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ into (6) gives the transductive oracle transformation $\Phi^{\text{oracle}}(\mathbf{t}) := \Phi(\mathbf{t}; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$.

In addition, since $(\hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ are *symmetric* in all $n+1$ data points and thus invariant to their ordering, conditioning on the random function Φ^{oracle} maintains the exchangeability of $(S_{\text{oracle}}^1, \ldots, S_{\text{oracle}}^{n+1})$, where $S_{\text{oracle}}^i := \Phi^{\text{oracle}}(\mathbf{E}^i)$.

Let $\hat{Q}_{1-\alpha}^{\text{oracle}} := \hat{Q}_{1-\alpha}(S_{\text{oracle}}^1, \ldots, S_{\text{oracle}}^n)$. Substituting $(\boldsymbol{\beta}, \boldsymbol{\theta}) = (\hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ into (7), we obtain the oracle prediction set $C^{\text{oracle}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ that satisfies the marginal coverage requirement in (1):

$$C^{\text{oracle}}(\mathbf{X}^{n+1}) = \left\{ \mathbf{y} : a_j(\mathbf{X}^{n+1}) \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}_{1-\alpha}^{\text{oracle}} \cdot \hat{\sigma}_j^{\text{oracle}} + \hat{\mu}_j^{\text{oracle}} \text{ for all } j \in [d] \right\}.$$

This construction applies to any generalized residuals satisfying the stated Assumption 1. Of course, this oracle is still impractical because both $(\hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ and $\hat{Q}_{1-\alpha}^{\text{oracle}}$ depend on the unknown test outcome $\mathbf{Y}^{n+1}$ through $\mathbf{E}^{n+1}$.

In the following, we will translate this oracle into a practical method producing a prediction set that depends only on the observed residuals $(\mathbf{E}^1, \ldots, \mathbf{E}^n)$ and the test input $\mathbf{X}^{n+1}$.

3.2 From the Transductive Oracle to a Practical Construction

We first describe the high-level idea of the practical approach in four steps. The technical details of these steps are developed separately in the following Sections 3.3–3.7.

Step 1: Finding Suitable Link Functions First, we will find suitable *monotonically nondecreasing* link functions $\omega_1, \ldots, \omega_d : \overline{\mathbb{R}} \to \overline{\mathbb{R}}$ that are computable from the observed residuals $(\mathbf{E}^1, \ldots, \mathbf{E}^n)$ and satisfy

$$S_{\text{oracle}}^{n+1} \leq c \iff a_j(\mathbf{X}^{n+1}) \leq E_j^{n+1} \leq \omega_j(c) \text{ for all } j \in [d], \qquad c \in \overline{\mathbb{R}}. \tag{10}$$

Using these link functions, we can define the *transductive oracle family* of prediction sets for difference scalar $c \in \overline{\mathbb{R}}$ by

$$\hat{C}(\mathbf{X}^{n+1}; c) = \left\{ \mathbf{y} : a_j(\mathbf{X}^{n+1}) \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \omega_j(c) \text{ for all } j \in [d] \right\}. \tag{11}$$

Substituting $c = \hat{Q}_{1-\alpha}^{\text{oracle}}$ into (10) and (11) gives the equivalence of events

$$\mathbf{Y}^{n+1} \in C^{\text{oracle}}(\mathbf{X}^{n+1}) \iff S_{\text{oracle}}^{n+1} \leq \hat{Q}_{1-\alpha}^{\text{oracle}} \iff \mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}}). \tag{12}$$

We remark that the equivalence relation in (12) matches the probabilities (at least $1-\alpha$ by Theorem 1) of the two events; it does not imply that the two oracle prediction sets are equal almost surely.

Although $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ is still impractical, it is closer to being computable as the *only unknown quantity* in this expression is $\hat{Q}_{1-\alpha}^{\text{oracle}}$. The effort remains to replace $\hat{Q}_{1-\alpha}^{\text{oracle}}$ by slightly larger but computable quantities, and this is the focus of the next step.

Step 2: Upper-bounding the Unknown Quantity First, suppose that we can replace the unknown $\hat{Q}^{\text{oracle}}_{1-\alpha}$ with a conservative data-driven upper bound $\hat{Q}^{\text{prac}}_{1-\alpha} \geq \hat{Q}^{\text{oracle}}_{1-\alpha}$.

Then, because $\omega_1(c), \ldots, \omega_d(c)$ are nondecreasing in c, we know that $\omega_j(\hat{Q}^{\text{prac}}_{1-\alpha}) \geq \omega_j(\hat{Q}^{\text{oracle}}_{1-\alpha})$ for all $j \in [d]$. Hence by substituting $c = \hat{Q}^{\text{prac}}_{1-\alpha}$ into (11), we can obtain a practical prediction set $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{prac}}_{1-\alpha})$ that satisfies

$$\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}) \implies \mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{prac}}_{1-\alpha}), \tag{13}$$

which inherits the coverage lower bound from Theorem 1.

However, $\hat{Q}^{\text{oracle}}_{1-\alpha}$ is a random variable that depends on the unknown test residual $\mathbf{E}^{n+1}$; hence, using a single upper bound $\hat{Q}^{\text{prac}}_{1-\alpha}$ to accommodate all possible values of $\mathbf{E}^{n+1}$, including those unassociated with the transductive oracles, may be overly conservative.

To reduce such conservatism, we will instead construct a collection of *pseudo* upper bounds $\hat{Q}^{\mathcal{R}}_{1-\alpha}$ for different hypothesized subregions $\mathcal{R} \subseteq \mathbb{R}^d$ of the test residual $\mathbf{E}^{n+1}$. These pseudo upper bounds satisfy $\hat{Q}^{\mathcal{R}}_{1-\alpha} \geq \hat{Q}^{\text{oracle}}_{1-\alpha}$ if $\mathbf{E}^{n+1} \in \mathcal{R}$, but not necessarily for $\mathbf{E}^{n+1}$ outside $\mathcal{R}$. Then similar to the single upper bound case, we guarantee the following event-based implication:

$$\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}), \quad \mathbf{E}^{n+1} \in \mathcal{R} \implies \mathbf{Y}^{n+1} \in \hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1}), \tag{14}$$

where $\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$ is a regional prediction set dedicated to the subregion $\mathcal{R}$ and defined as

$$\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\mathcal{R}}_{1-\alpha}) \cap \left\{\mathbf{y} : \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in \mathcal{R}\right\}. \tag{15}$$

The inner piece $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\mathcal{R}}_{1-\alpha})$ is obtained by substituting $c = \hat{Q}^{\mathcal{R}}_{1-\alpha}$ into (11).

Let $\mathcal{P}$ denote the collection of hypothesized subregions. If we can ensure that this collection is rich enough to cover all possible values of $\mathbf{E}^{n+1}$ associated with the oracle coverage event, i.e., if we can guarantee that

$$\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}) \implies \mathbf{E}^{n+1} \in \bigcup_{\mathcal{R} \in \mathcal{P}} \mathcal{R}, \tag{16}$$

then we can make sure $\mathbf{Y}^{n+1}$ belongs to $\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$ for some $\mathcal{R} \in \mathcal{P}$ whenever $\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha})$. The union of these regional prediction sets therefore preserves the oracle coverage event.

Step 3: Enforcing a Coordinatewise Structure If we simply set the union of those regional prediction sets to be our final prediction set, which we denote by $\hat{C}^{\cup}(\mathbf{X}^{n+1})$ here, then we will violate one of the most appealing properties of the oracle prediction set: its coordinatewise structure in the residual space, as outlined in Section 2.3. This is because the union of the regional prediction sets $\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$ may not be rectangular in the residual space, even if each individual $\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$ is. Therefore, we will aim to preserve the coordinatewise structure by enclosing the union of the regional prediction sets with a hyper-rectangle in the residual space. We will choose our collection of subregions in a *data-dependent* way so that such hyper-rectangular relaxation is minimal (empirically) as the sample size n grows.

As a preview of our practical construction, we will organize the hypothesized regions chosen from the previous step into separate groups and use each group to obtain an upper bound

for each different coordinate. Combining all of these upper bounds gives a coordinatewise enclosure of the union.

Figure 2 supplies an example of the nonrectangular geometry of the union in the residual space and how such a union can be enclosed using hyper-rectangles to retrieve the coordinatewise structure. We also remark that our rectangular enclosure of the union in practice does not need to be tight, although it is tight in the example shown in Figure 2. There is a sufficient condition that one can check to verify that the enclosure is tight, and even if this condition is not satisfied, our experiments (see Appendix D) show that the difference between the tightest enclosure and our practical enclosure is minimal.

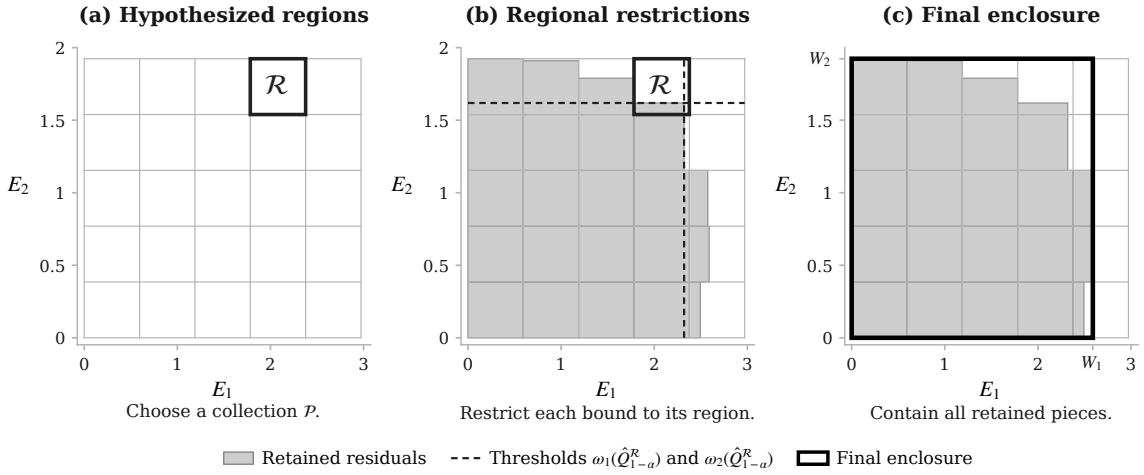


Figure 2: An illustration of a regional-union construction in the residual space using absolute residuals with $n = 10$ and $\alpha = 0.1$. (a) A collection $\mathcal{P}$ of hypothesized residual regions, with one region $\mathcal{R}$ highlighted. (b) Each shaded piece is the intersection of its region with the constraints $0 \le z_j \le \omega_j(\hat{Q}^{\mathcal{R}}_{1-\alpha})$ for $j = 1, 2$; the dashed lines show the thresholds for the highlighted region. (c) A coordinatewise enclosure of this union contains every retained piece. This figure only illustrates the idea of the regional-union argument; the practical construction described below uses a different $\mathcal{P}$.

The main technical challenges from these three steps are (1) how to define suitable link functions, (2) how to find a suitable collection of subregions $\mathcal{P}$, (3) how $\hat{Q}^{\mathcal{R}}_{1-\alpha}$ is defined, and (4) how to obtain a final prediction set with a simple and interpretable coordinatewise structure with marginal coverage. We will address these challenges in the following subsections, and then summarize the final practical construction at the end.

3.3 Explicit Form of the Link Functions

In this section, we present the explicit form of the link functions $\omega_1, \ldots, \omega_d$ defined implicitly in (10), along with their guarantee. For the constructions below, take $n \ge 2$. Define the observed statistics $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and $\hat{\boldsymbol{\sigma}}^{\text{obs}}$ as

$$\hat{\boldsymbol{\mu}}^{\text{obs}} = (\hat{\mu}_1^{\text{obs}}, \ldots, \hat{\mu}_d^{\text{obs}})^\top, \quad \hat{\boldsymbol{\sigma}}^{\text{obs}} = (\hat{\sigma}_1^{\text{obs}}, \ldots, \hat{\sigma}_d^{\text{obs}})^\top,$$
$$\hat{\mu}_j^{\text{obs}} = \frac{1}{n}\sum_{i=1}^{n} E_j^i, \quad \hat{\sigma}_j^{\text{obs}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\big(E_j^i - \hat{\mu}_j^{\text{obs}}\big)^2}. \tag{17}$$

Note that we use the factor $1/n$ instead of $1/(n-1)$ in the definition of $\hat{\boldsymbol{\sigma}}^{\text{obs}}$ for algebraic simplicity. Assumption 1 ensures $\hat{\sigma}_j^{\text{obs}} > 0$ almost surely for all $j \in [d]$.

The explicit form of $\omega_j(c)$ is given in the following lemma.

Lemma 2 *Under Assumption 1, suitable link functions $\omega_1, \ldots, \omega_d$ satisfying* (10) *are*

$$\omega_j(c) = \begin{cases} -\infty, & \text{if } c \leq -T, \\ \hat{\mu}_j^{\text{obs}} + c\hat{\sigma}_j^{\text{obs}}\sqrt{\dfrac{n+1}{T^2 - c^2}}, & \text{if } -T < c < T, \\ +\infty, & \text{if } c \geq T, \end{cases} \qquad T = \frac{n}{\sqrt{n+1}}.$$

The first and last cases also define $\omega_j(-\infty) = -\infty$ and $\omega_j(+\infty) = +\infty$.

Substituting this expression for the link functions into (11) and setting $c = \hat{Q}_{1-\alpha}^{\text{oracle}}$ gives an oracle upper bound $\omega_j(\hat{Q}_{1-\alpha}^{\text{oracle}})$ for $E_j(\mathbf{X}^{n+1}, y_j)$. These upper bounds are real-valued when $-T < \hat{Q}_{1-\alpha}^{\text{oracle}} < T$. The following lemma gives a sufficient sample-size condition on this.

Lemma 3 *Under Assumption 1, if $n \geq \max\{3, 1/\alpha - 1\}$, then $|\hat{Q}_{1-\alpha}^{\text{oracle}}| < n/\sqrt{n+1}$ almost surely.*

Thus, the oracle residual thresholds $\omega_j(\hat{Q}_{1-\alpha}^{\text{oracle}})$ are finite almost surely under the stated sample-size condition, which is mild and commonly satisfied in practice. The same sample size condition also ensures finiteness of the practical thresholds introduced in the next section, so we will revisit this lemma again but in a slightly different format.

3.4 Approximation of the Transductive Oracle with Pseudo Upper Bounds

We obtain the *pseudo upper bound* $\hat{Q}_{1-\alpha}^{\mathcal{R}}$ by assuming $\mathbf{E}^{n+1} \in \mathcal{R}$ for some subregion $\mathcal{R} \subseteq \mathbb{R}^d$ and taking worst-case upper bounds of the standardized residuals used to compute $\hat{Q}_{1-\alpha}^{\text{oracle}}$ over all possible values of $\mathbf{E}^{n+1} \in \mathcal{R}$.

To simplify the derivation, we also require $\mathcal{R}$ to be a nonempty hyper-rectangle, that is, $\mathcal{R} = \bigtimes_{j=1}^{d} I_j$, where each $I_j \subseteq \mathbb{R}$ is a nonempty interval with endpoints $l_j := \inf I_j$ and $u_j := \sup I_j$. Finite endpoints may be included or excluded, and unbounded intervals are allowed. By continuity, the inclusion or exclusion of a finite endpoint does not change the supremum below.

Let $\mathbf{z}$ denote a placeholder for the test residual $\mathbf{E}^{n+1}$. Define the pseudo location and scale parameters $\hat{\boldsymbol{\mu}}(\mathbf{z}) := (\hat{\mu}_1(z_1), \ldots, \hat{\mu}_d(z_d))^\top$ and $\hat{\boldsymbol{\sigma}}(\mathbf{z}) := (\hat{\sigma}_1(z_1), \ldots, \hat{\sigma}_d(z_d))^\top$ as if $\mathbf{E}^{n+1}$ in (9) were replaced by $\mathbf{z}$:

$$\hat{\mu}_j(z_j) = \frac{\sum_{i=1}^{n} E_j^i + z_j}{n+1}, \qquad \hat{\sigma}_j(z_j) = \sqrt{\frac{\big(z_j - \hat{\mu}_j(z_j)\big)^2 + \sum_{i=1}^{n}\big(E_j^i - \hat{\mu}_j(z_j)\big)^2}{n}}. \tag{18}$$

By plugging in $\mathbf{z} = \mathbf{E}^{n+1}$, we have $\hat{\boldsymbol{\mu}}(\mathbf{E}^{n+1}) = \hat{\boldsymbol{\mu}}^{\text{oracle}}$ and $\hat{\boldsymbol{\sigma}}(\mathbf{E}^{n+1}) = \hat{\boldsymbol{\sigma}}^{\text{oracle}}$.

Suppose now that $\mathbf{E}^{n+1} \in \mathcal{R}$ for some hyper-rectangle $\mathcal{R} \subseteq \mathbb{R}^d$, which we refer to as our working hypothesis. We can then define a pseudo transformation $\Phi^{\mathcal{R}} : \mathbb{R}^d \to \mathbb{R}$, tailored to this hypothesis, by considering the most unfavorable (largest) possible value of the standardized ratio as a function of $E_j^{n+1} = z_j \in I_j$:

$$\Phi^{\mathcal{R}}(\mathbf{t}) := \max_{1 \le j \le d} \sup_{z_j \in I_j} \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)}, \qquad \mathbf{t} \in \mathbb{R}^d. \tag{19}$$

This definition does not require the working hypothesis. Under the working hypothesis, however, the supremum includes $z_j = E_j^{n+1}$ for every $j \in [d]$, so $\Phi^{\mathcal{R}}$ is guaranteed to be at least as large as the oracle transformation:

$$\Phi^{\mathcal{R}}(\mathbf{t}) \ge \Phi^{\text{oracle}}(\mathbf{t}), \qquad \forall \mathbf{t} \in \mathbb{R}^d.$$

Conveniently, $\Phi^{\mathcal{R}}$ admits a closed-form expression.

Lemma 4 *Under Assumption 1, for each $j \in [d]$, define*

$$M_j^{\mathcal{R}}(t_j) := \begin{cases} \sqrt{\dfrac{(t_j - \hat{\mu}_j^{\text{obs}})^2}{(\hat{\sigma}_j^{\text{obs}})^2} + \dfrac{1}{n+1}}, & \text{if } t_j > \hat{\mu}_j^{\text{obs}} \text{ and} \\ & l_j \le \hat{\mu}_j^{\text{obs}} - \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}} \le u_j, \\ -\infty, & \text{otherwise.} \end{cases}$$

The pseudo transformation $\Phi^{\mathcal{R}}$ defined in (19) *can be written as*

$$\Phi^{\mathcal{R}}(\mathbf{t}) = \max_{1 \le j \le d} \max \left\{ \frac{t_j - \hat{\mu}_j(l_j)}{\hat{\sigma}_j(l_j)}, \quad M_j^{\mathcal{R}}(t_j), \quad \frac{t_j - \hat{\mu}_j(u_j)}{\hat{\sigma}_j(u_j)} \right\}.$$

In addition, if $l_j = -\infty$ or $u_j = +\infty$ for any $j \in [d]$, the corresponding endpoint term is interpreted using

$$\lim_{z_j \to -\infty} \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = \frac{1}{\sqrt{n+1}}, \qquad \lim_{z_j \to +\infty} \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = -\frac{1}{\sqrt{n+1}}.$$

Using Lemma 4, we can define the pseudo upper bound $\hat{Q}_{1-\alpha}^{\mathcal{R}} := \hat{Q}_{1-\alpha}(S_{\mathcal{R}}^1, \ldots, S_{\mathcal{R}}^n)$, where $S_{\mathcal{R}}^i = \Phi^{\mathcal{R}}(\mathbf{E}^i)$ for all $i \in [n]$. Then, under the working hypothesis $\mathbf{E}^{n+1} \in \mathcal{R}$, we have $\hat{Q}_{1-\alpha}^{\mathcal{R}} \ge \hat{Q}_{1-\alpha}^{\text{oracle}}$ because $\Phi^{\mathcal{R}} \ge \Phi^{\text{oracle}}$ uniformly. Substituting $c = \hat{Q}_{1-\alpha}^{\mathcal{R}}$ into (11) and intersecting the resulting prediction set with $\{\mathbf{y} : \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in \mathcal{R}\}$, we obtain the regional prediction set $\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$ defined in (15). Algorithm 1 summarizes the procedure in detail.

The following lemma establishes the guarantee of the regional prediction set $\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$ under the working hypothesis $\mathbf{E}^{n+1} \in \mathcal{R}$, as promised in Section 3.2:

Lemma 5 *Under Assumption 1, the regional prediction set $\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$ constructed by Algorithm 1 satisfies* (14).

Algorithm 1 Construction of regional prediction sets under the working hypothesis $\mathbf{E}^{n+1} \in \mathcal{R}$

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; test input $\mathbf{X}^{n+1}$; target miscoverage level $\alpha \in (0,1)$; nonempty hyper-rectangle $\mathcal{R} = \bigtimes_{j=1}^d I_j$.

1: Compute $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and $\hat{\boldsymbol{\sigma}}^{\text{obs}}$ using (17).
2: Compute $S_{\mathcal{R}}^i = \Phi^{\mathcal{R}}(\mathbf{E}^i)$ **for** $i \in [n]$ using $\Phi^{\mathcal{R}}$ defined in Lemma 4.
3: Compute $\hat{Q}_{1-\alpha}^{\mathcal{R}} := \hat{Q}_{1-\alpha}(S_{\mathcal{R}}^1, \ldots, S_{\mathcal{R}}^n)$.
4: Compute $\omega_1(\hat{Q}_{1-\alpha}^{\mathcal{R}}), \ldots, \omega_d(\hat{Q}_{1-\alpha}^{\mathcal{R}})$ using $\omega_1, \ldots, \omega_d$ defined in Lemma 2.
5: Construct

$$\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\mathcal{R}}) = \left\{ \mathbf{y} : a_j(\mathbf{X}^{n+1}) \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \omega_j(\hat{Q}_{1-\alpha}^{\mathcal{R}}) \text{ for all } j \in [d] \right\}.$$

6: **return**

$$\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\mathcal{R}}) \cap \left\{ \mathbf{y} : \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in \mathcal{R} \right\}.$$

Proof If $\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$, then $E_j^{n+1} \leq \omega_j(\hat{Q}_{1-\alpha}^{\text{oracle}})$ for all $j \in [d]$. Under the additional hypothesis that $\mathbf{E}^{n+1} \in \mathcal{R}$, we know that $\hat{Q}_{1-\alpha}^{\mathcal{R}} \geq \hat{Q}_{1-\alpha}^{\text{oracle}}$. Hence, $\omega_j(\hat{Q}_{1-\alpha}^{\text{oracle}}) \leq \omega_j(\hat{Q}_{1-\alpha}^{\mathcal{R}})$ for all $j \in [d]$ because $\omega_1, \ldots, \omega_d$ are nondecreasing functions. Therefore, we have $E_j^{n+1} \leq \omega_j(\hat{Q}_{1-\alpha}^{\mathcal{R}})$ for all $j \in [d]$, which is exactly the condition $\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\mathcal{R}})$, under the hypothesis that $\mathbf{E}^{n+1} \in \mathcal{R}$. Intersecting these two events yields $\mathbf{Y}^{n+1} \in \hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$. ■

In general, guaranteeing (14) does not imply marginal coverage for $\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$, because the working hypothesis $\mathbf{E}^{n+1} \in \mathcal{R}$ may occur with very low probability. As we previewed in Section 3.2, we will resolve this by finding a collection of subregions $\mathcal{R}$ whose union contains $\mathbf{E}^{n+1}$ whenever the oracle coverage event occurs, described in (16). If such a collection can be found, Lemma 5 then further implies that the union of their corresponding regional prediction sets preserves the same oracle coverage event.

The shortcoming of doing so naively is that such a union might not be a rectangle even for standard choices of residuals, which would violate our promise of interpretability (revisit Figure 2 for an example). We will use a full-cell regional union as a benchmark and construct practical coordinatewise upper bounds that enclose it. The coverage proof for the practical bounds will follow directly from preservation of the same oracle coverage event.

Before constructing these regions, we give the separate guarantee needed for finite practical residual thresholds, as promised earlier.

Lemma 6 *Under Assumption 1, if* $n \geq \max\{3, 1/\alpha - 1\}$*, then* $|\hat{Q}_{1-\alpha}^{\mathcal{R}}| < n/\sqrt{n+1}$ *almost surely for every nonempty rectangular region* $\mathcal{R}$*, including random regions. Consequently, all corresponding link values* $\omega_j(\hat{Q}_{1-\alpha}^{\mathcal{R}})$ *are finite almost surely.*

3.5 A Data-Dependent Collection and Hyper-Rectangular Enclosure

The goal in this section is to find a collection of subregions $\mathcal{P}$ that satisfies (16) and to outline the structure of the final prediction set. We first construct a coarse prediction set using a single region containing all possible test residuals. Its retained residuals then provide

a potentially smaller region within which we can refine the construction of subregions while preserving the oracle coverage event.

For simplicity of derivation, we denote the retained residuals corresponding to $\hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$ by

$$Z^{\mathcal{R}} := \left\{ \mathbf{z} \in \mathbb{R}^d : a_j(\mathbf{X}^{n+1}) \leq z_j \leq \omega_j(\hat{Q}^{\mathcal{R}}_{1-\alpha}) \text{ for all } j \in [d] \right\} \cap \mathcal{R}, \tag{20}$$

which satisfies $\mathbf{y} \in \hat{C}^{\mathcal{R}}(\mathbf{X}^{n+1})$ if and only if $\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in Z^{\mathcal{R}}$.

Before we discuss the construction of the collection of subregions, we introduce the following definition.

Definition 7 (Finite Inner Cover) *A finite inner cover of a set A is a collection of sets $\{P_1, P_2, \ldots, P_k\}$ such that $P_i \subseteq A$ for all $i \in [k]$ and $A = \bigcup_{i=1}^{k} P_i$. We allow the empty collection as a finite inner cover of the empty set.*

Therefore, to guarantee (16), we can take $\mathcal{P}$ to be a finite inner cover of D, with the restriction that all of its elements are hyper-rectangles. However, D is still an unbounded region to consider initially, so we want to find a smaller subset of D such that (16) is preserved if we choose $\mathcal{P}$ to be a finite inner cover of that smaller subset. Fortunately, our regional prediction set provides this refinement.

One-step Refinement Let D be the smallest *fixed* domain of residuals out of our knowledge. For example, we would choose $D = \mathbb{R}^d_+$ for absolute residuals and $D = \mathbb{R}^d$ for quantile residuals. Our method would still work if $D = \mathbb{R}^d$ is selected for absolute residual case, at a cost of unnecessary potential conservatism. The purpose of having a fixed domain D is to avoid recomputation for every new test input.

Applying the regional construction in Algorithm 1 with $\mathcal{R} = D$ gives the prediction set $\hat{C}^D(\mathbf{X}^{n+1})$, whose retained residuals are, by (20),

$$Z^D = \bigtimes_{j=1}^{d} Z_j^D, \qquad Z_j^D := \left[a_j(\mathbf{X}^{n+1}), \omega_j(\hat{Q}^D_{1-\alpha}) \right] \cap \mathbb{R}.$$

This is a hyperrectangle, possibly empty. Under the sample-size condition in Lemma 6, it is bounded above almost surely. Since the test residual belongs to D almost surely, Lemma 5 gives

$$\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}) \implies \mathbf{E}^{n+1} \in Z^D \subseteq D.$$

Hence, by choosing $\mathcal{P}$ to be a finite inner cover of Z^D, we preserve (16). If $Z^D = \varnothing$, we set $\mathcal{P} = \varnothing$ and $W_j = -\infty$ for every j, and return the empty prediction set. No further regional thresholds need to be computed. In the remainder of the construction, suppose that $Z^D \neq \varnothing$.

The choice of a fixed initial domain D also has a computational benefit. By substituting $\mathcal{R}$ with D, rather than the input-dependent lower-bound region, the coordinatewise upper residual thresholds $\omega_j(\hat{Q}^D_{1-\alpha})$ remain fixed for all test inputs. Having this structure allows us to compute these quantities once from the observed residuals and reuse them across test inputs, which is crucial for practical implementation. The lower bounds $a_j(\mathbf{X}^{n+1})$ may still depend on the test input, but they are easy to compute for absolute and quantile residuals. Consequently, Z^D and the subsequent regional calculations may change even though the global upper thresholds are reused.

We will construct the finite inner covers using the observed calibration residuals, with the aim of obtaining tighter regional bounds while keeping the number of additional regional calculations manageable.

For each $j \in [d]$, let N_j denote the number of calibration residual values $E_j^1, \ldots, E_j^n$ strictly inside Z_j^D, and let $\eta_j^1 < \cdots < \eta_j^{N_j}$ denote these values in increasing order (they are distinct under Assumption 1). Thus, $0 \le N_j \le n$ and $a_j(\mathbf{X}^{n+1}) =: \eta_j^0 \le \eta_j^1 < \cdots < \eta_j^{N_j} \le \eta_j^{N_j+1} := \omega_j(\hat{Q}^D_{1-\alpha})$. Define

$$I_j^h := [\eta_j^{h-1}, \eta_j^h) \cap \mathbb{R}, h \in [N_j], \qquad I_j^{N_j+1} := [\eta_j^{N_j}, \eta_j^{N_j+1}] \cap \mathbb{R}, \tag{21}$$

then $\{I_j^h\}_{h=1}^{N_j+1}$ forms a partition of Z_j^D into $N_j + 1$ disjoint intervals. In particular, when $N_j = 0$, this partition consists of the single interval $I_j^1 = Z_j^D$.

These breakpoints use the observed residual values to provide a data-dependent partition in each coordinate, with at most $n+1$ intervals. Below, we first provide a direct construction of the finite inner cover whose explicit evaluation can be computationally expensive.

The Full-Cell Benchmark Because each $\{I_j^h\}_{h=1}^{N_j+1}$ partitions Z_j^D, taking all products of the coordinate intervals gives a finite inner cover of Z^D:

$$\mathcal{P}_{\text{full}} := \left\{ \bigtimes_{j=1}^{d} I_j^{h_j} : h_j \in [N_j + 1] \text{ for all } j \in [d] \right\}. \tag{22}$$

For each cell $\mathcal{R} \in \mathcal{P}_{\text{full}}$, we can apply the regional construction in Algorithm 1 and retain its corresponding residuals $Z^{\mathcal{R}}$ defined in (20). Taking their union and its coordinatewise upper bounds gives

$$\mathcal{U}_{\text{full}} := \bigcup_{\mathcal{R} \in \mathcal{P}_{\text{full}}} Z^{\mathcal{R}} \subseteq Z^D, \qquad W_j^{\text{full}} := \sup_{\mathbf{z} \in \mathcal{U}_{\text{full}}} z_j, \quad j \in [d], \tag{23}$$

with $\sup \varnothing := -\infty$. Since $\mathcal{P}_{\text{full}}$ is a finite inner cover of Z^D, Lemma 5 ensures that this union contains the test residual whenever the oracle coverage event occurs. Therefore, $\bigtimes_{j=1}^{d} \left([a_j(\mathbf{X}^{n+1}), W_j^{\text{full}}] \cap \mathbb{R} \right)$ gives a hyper-rectangular enclosure that preserves the same event. It is also the smallest closed hyper-rectangle of this form enclosing this union.

However, explicitly evaluating every cell requires considering $|\mathcal{P}_{\text{full}}| = \prod_{j=1}^{d}(N_j + 1) \le (n+1)^d$ regions, which can be computationally infeasible even for moderate d. We therefore use this construction as a benchmark and seek an enclosure without explicitly evaluating every cell. Our practical construction restricts only one coordinate at a time, while allowing the other coordinates to range over Z^D. In general, doing so would result in looser upper bounds on the residuals because we shift from restricting all coordinates together in each cell to just one of them at a time, but we will show empirically that under our particular construction, the difference is minimal. Moreover, Proposition **??** will give a sufficient condition under which the thresholds from our construction will agree exactly with those of the full-cell benchmark.

3.6 Efficient Approximation of the Thresholds

For each coordinate $j \in [d]$, we construct a *coordinatewise cover* of Z^D by partitioning that coordinate only. Specifically, for $h \in [N_j + 1]$, define

$$\mathcal{R}_j^h := Z_1^D \times \cdots \times Z_{j-1}^D \times I_j^h \times Z_{j+1}^D \times \cdots \times Z_d^D. \tag{24}$$

The collection $\mathcal{P}_j(Z^D) := \{\mathcal{R}_j^h : h \in [N_j + 1]\}$ clearly satisfies $\bigcup_{h=1}^{N_j+1} \mathcal{R}_j^h = Z^D$ and hence forms a finite inner cover of Z^D. Together, we have constructed d finite inner covers, using at most $\sum_{j=1}^{d}(N_j + 1) \le d(n + 1)$ regions.

We use the cover $\mathcal{P}_j(Z^D)$ to obtain an upper bound for coordinate j only. For this purpose, define the union of its retained residual sets and the corresponding upper bound by

$$\mathcal{U}_j := \bigsqcup_{h=1}^{N_j+1} Z^{\mathcal{R}_j^h}, \qquad W_j := \sup_{\mathbf{z} \in \mathcal{U}_j} z_j, \quad j \in [d], \tag{25}$$

again with $\sup \varnothing := -\infty$. See Figure 3 for illustration.

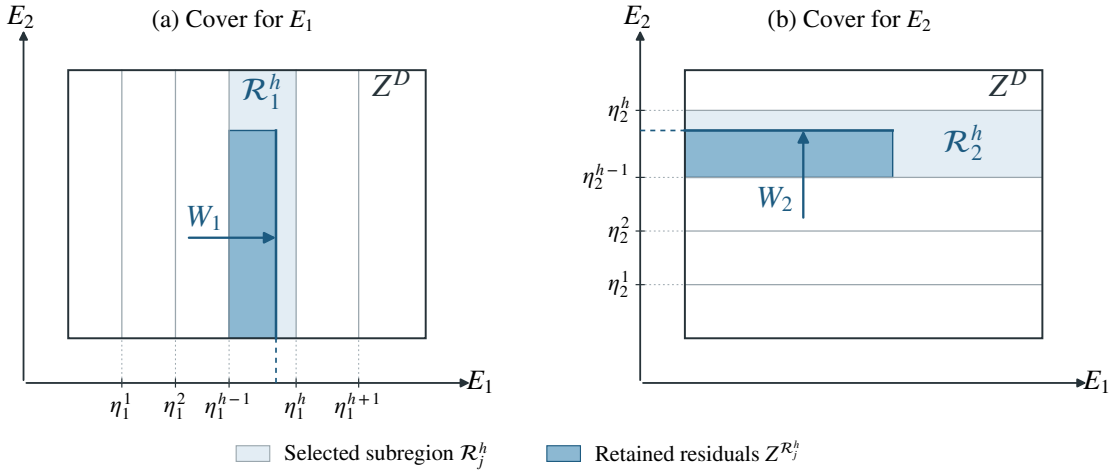


Figure 3: The data-dependent coordinatewise covers of Z^D in two dimensions. The calibration residuals inside each coordinate interval Z_j^D determine the interior breakpoints η_j^h, producing vertical slabs for the first coordinate (left) and horizontal slabs for the second coordinate (right). The arrows mark the resulting coordinatewise upper bounds W_1 and W_2.

For every $j \in [d]$, the cover $\mathcal{P}_j(Z^D)$ satisfies (16), so the test residual belongs to $\mathcal{U}_j$ for every j whenever the oracle coverage event occurs. This further implies

$$\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}}) \implies \mathbf{E}^{n+1} \in \bigcap_{j=1}^{d} \mathcal{U}_j \implies \mathbf{E}^{n+1} \in \bigtimes_{j=1}^{d} [a_j(\mathbf{X}^{n+1}), W_j]. \tag{26}$$

The last piece in (26) where the test residual belongs is clearly a hyper-rectangle in the residual space, so we can complete our prediction set construction by finding $(W_1, \ldots, W_d)$. The following lemma provides an explicit algorithm for this task.

Lemma 8 *Assume Assumption 1 holds. For every $j \in [d]$, let $h^* \in [N_j + 1]$ be the largest index such that $Z^{\mathcal{R}_j^{h^*}} \neq \emptyset$, then W_j defined in (25) can be described explicitly by*

$$W_j = \sup_{\mathbf{z} \in Z^{\mathcal{R}_j^{h^*}}} z_j = \min \left\{ \eta_j^{h^*}, \omega_j \left(\hat{Q}_{1-\alpha}^{\mathcal{R}_j^{h^*}} \right) \right\}.$$

If no such h^* exists, $W_j = -\infty$. Moreover, $W_j = W_j^{\text{full}}$ for all $j \in [d]$.

Using Lemma 8, we can construct the prediction set of our *transductively standardized conformal prediction* (TSCP) method as

$$\hat{C}^{\text{TSCP}}(\mathbf{X}^{n+1}) := \left\{ \mathbf{y} \in \mathbb{R}^d : a_j(\mathbf{X}^{n+1}) \leq E_j(\mathbf{X}^{n+1}, y_j) \leq W_j \text{ for all } j \in [d] \right\}. \tag{27}$$

Algorithm 2 summarizes this entire procedure. This algorithm can be implemented at a cost of $\mathcal{O}(dn^2)$. See Appendix B.3.3 for details.

Algorithm 2 Transductively Standardized Conformal Prediction (TSCP)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; test input $\mathbf{X}^{n+1}$; target miscoverage level $\alpha \in (0,1)$; Algorithm 1; fixed residual domain D.

Compute $\Phi^D(t) = \max_{1 \leq j \leq d} \Phi_j^D(t)$ by Lemma 4.
Construct $\hat{C}^D(\mathbf{X}^{n+1})$ using Algorithm 1 and $Z^D = \bigtimes_{j=1}^d Z_j^D$ using (20).
if $Z^D = \varnothing$ **then**
    return $\hat{C}^{\text{TSCP}}(\mathbf{X}^{n+1}) := \varnothing$.
end if
for $j = 1, \ldots, d$ **do**
    $W_j \leftarrow -\infty$.
    Compute $N_j = \sum_{i=1}^n \mathbb{I}\{a_j(\mathbf{X}^{n+1}) < E_j^i < \omega_j(\hat{Q}_{1-\alpha}^D)\}$.
    Construct $I_j^1, \ldots, I_j^{N_j+1}$ using (21).
    for $h = N_j + 1, N_j, \ldots, 1$ **do**
        Construct $\mathcal{R}_j^h$ using (24).
        Construct $\hat{C}^{\mathcal{R}_j^h}(\mathbf{X}^{n+1})$ using Algorithm 1 and $Z^{\mathcal{R}_j^h}$ using (20).
        if $Z^{\mathcal{R}_j^h} \neq \varnothing$ **then**
            $W_j \leftarrow \min \left\{ \eta_j^h, \omega_j(\hat{Q}_{1-\alpha}^{\mathcal{R}_j^h}) \right\}$.
            break.
        end if
    end for
end for
return

$$\hat{C}^{\text{TSCP}}(\mathbf{X}^{n+1}) := \left\{ \mathbf{y} \in \mathbb{R}^d : a_j(\mathbf{X}^{n+1}) \leq E_j(\mathbf{X}^{n+1}, y_j) \leq W_j \text{ for all } j \in [d] \right\}.$$

Proposition 9 establishes the fact that $\mathbf{Y}^{n+1} \in \hat{C}^{\text{TSCP}}(\mathbf{X}^{n+1})$ whenever the oracle coverage event occurs, hence inheriting its marginal coverage from Theorem 1.

Proposition 9 *Under Assumption 1, the TSCP prediction set* $\hat{C}^{\mathrm{TSCP}}(\mathbf{X}^{n+1})$ *defined in* (27) *satisfies*

$$\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\mathrm{oracle}}_{1-\alpha}) \implies \mathbf{Y}^{n+1} \in \hat{C}^{\mathrm{TSCP}}(\mathbf{X}^{n+1}).$$

3.7 Tightness of TSCP

4 Numerical Experiments

We evaluate Transductively Standardized Conformal Prediction (TSCP) on simulated and real multivariate regression problems, studying joint coverage, coordinate-wise interval length, prediction-set volume, and computational cost. We first consider absolute-residual scores under independent Gaussian noise to establish the effect of unequal outcome scales, followed by dependent and partially heteroskedastic noise to examine cross-output and input-dependent variation. We then study quantile-regression scores and six real datasets. Appendix D provides additional noise-distribution comparisons, sensitivity analyses, heavy-tail and contamination stress tests, and comparisons with the oracle and alternative constructions introduced in the paper.

4.1 Methods and Evaluation Metrics

The primary comparisons use four methods. *Unscaled Max* applies split conformal prediction to the maximum coordinate-wise residual without additional scaling. *Point CHR* estimates coordinate scales using an additional data split (Sampson and Chan, 2024). *Emp. Copula* estimates a dependence model from the calibration residuals and then uses the same observations for calibration; this implementation does not have the finite-sample guarantee enjoyed by the other methods (Messoudi et al., 2021). *TSCP* denotes the rectangular local-worst-case construction proposed in Algorithm **??**. For quantile-regression experiments, we replace Point CHR with the native quantile-hyperrectangle method *CQHR* of Sampson and Chan (2024). Appendix D.8 contains a separate low-dimensional comparison with the hyperrectangular shape-template method of Tumu et al. (2024).

For a test sample of size m, we report the empirical joint coverage

$$\widehat{\mathrm{Cov}} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}\left\{\boldsymbol{Y}_i \in \widehat{C}(\boldsymbol{X}_i)\right\}.$$

For absolute-residual experiments, all methods return a rectangle with residual half-widths $(\mathcal{L}_1, \ldots, \mathcal{L}_d)$. We therefore report the normalized residual-space volume $\prod_{j=1}^{d} \mathcal{L}_j$, which equals 2^{-d} times the full rectangular volume in the outcome space, and the full coordinate-wise interval lengths $2\mathcal{L}_j$. For quantile-regression experiments, the base interval depends on the test covariates, so volume and coordinate lengths are averaged over the test observations in outcome space. Runtime is the wall-clock time needed to construct the conformal region after residuals have been computed; fitting the underlying regression model is excluded. Unless stated otherwise, absolute-residual results are averaged over 200 repetitions (30 repetitions for quantile-regression experiments) and the target coverage is $1 - \alpha = 0.9$. All methods in each repetition use the same fitted regression model, calibration observations, and test observations. Sample sizes and model settings are specified below.

4.2 Absolute Mean-Regression Residuals

We first generate 10-dimensional features $\boldsymbol{X} \sim \mathcal{N}(\mathbf{0}, I_{10})$ and outcomes

$$Y_j = f_j(\boldsymbol{X}) + \epsilon_j, \qquad j \in [d], \tag{28}$$

where $d = 10$ and every f_j is linear, with coefficients sampled once from $\mathrm{Unif}(-10, 10)$ and held fixed across repetitions. In each repetition, we independently generate 7,200 training observations, 800 test observations, and a calibration sample of size $n \in \{30, 50, 100, 300, 500\}$. We estimate the regression coefficients using ordinary least squares, and all methods in each repetition share the same fitted predictor.

4.2.1 Independent Gaussian Noise

Figure 4 reports results under independent Gaussian noise with $\epsilon_j \sim \mathcal{N}(0, (11-j)^2)$ for $j \in [10]$. TSCP combines near-nominal joint coverage with smaller prediction-set volumes than Point CHR and Unscaled Max. The volume penalty for Point CHR is particularly pronounced when $n < 100$, where its additional split substantially reduces the number of observations available for calibration. Unscaled Max assigns one residual threshold to outputs with unequal noise scales and therefore produces substantially larger rectangles. Emp. Copula yields smaller regions but undercovers; its reuse of calibration observations for dependence estimation precludes the split-conformal guarantee enjoyed by the other benchmarks.

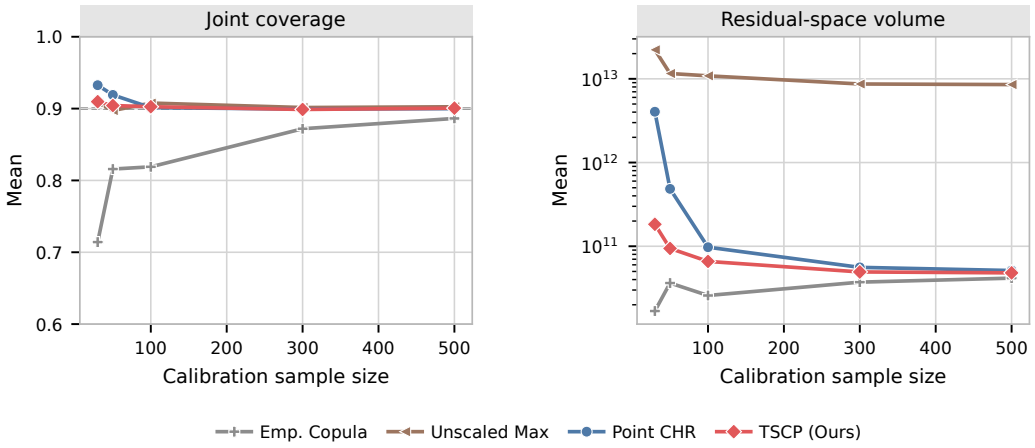


Figure 4: Absolute-residual experiment with $d = 10$, independent Gaussian noise, and heterogeneous scales $(10, 9, \ldots, 1)$. Panels report mean joint coverage and residual-space volume over 200 repetitions. The dashed line marks the target coverage 0.9. TSCP attains near-nominal coverage with smaller residual-space volumes than Point CHR and Unscaled Max.

Figure 5 reports the corresponding homogeneous setting, with $\epsilon_j \sim \mathcal{N}(0, 1)$ for all $j \in [10]$. Unscaled Max is the most efficient method with near-nominal coverage, while TSCP remains competitive. Together, these two settings show that coordinate adaptation is useful when noise scales differ, whereas unscaled calibration can be more efficient when a common scale is appropriate.

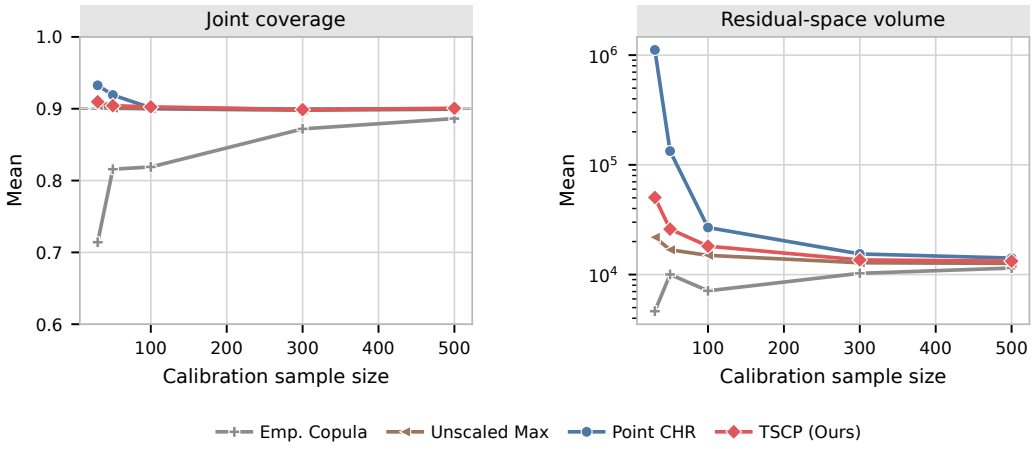


Figure 5: Absolute-residual experiment with $d = 10$, independent Gaussian noise, and unit scale in every coordinate. Panels report mean joint coverage and residual-space volume over 200 repetitions. The dashed line marks the target coverage 0.9. Among methods with near-nominal coverage, Unscaled Max has the smallest residual-space volume, while TSCP remains competitive.

4.2.2 Heterogeneous and Dependent Noise

The dependent-noise experiment uses $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, DRD)$, where $D = \mathrm{diag}(10, 9, \ldots, 1)$ and R is equicorrelated with $\rho = 0.6$. This simultaneously tests heterogeneous coordinate scales and cross-output dependence.

Figure 6 reports joint coverage, residual-space volume, and construction time. As in the independent Gaussian setting with heterogeneous scales, TSCP attains near-nominal coverage with smaller volumes than Point CHR and Unscaled Max.

Figure 7 reports the coordinate-wise interval lengths for all methods at $n = 100$. The mean TSCP full interval length decreases from 49.4 on the noisiest coordinate to 5.0 on the quietest coordinate. In contrast, Unscaled Max assigns the same length, approximately 38.0, to every coordinate. Thus, similar overall coverage does not imply similar coordinate-wise informativeness: TSCP allocates width according to coordinate scale instead of allowing the noisiest output to determine all ten intervals. The smaller Emp. Copula bars must be read together with its joint undercoverage in Figure 6.

4.2.3 Partial Heteroskedasticity

To distinguish heterogeneity across coordinates from heteroskedasticity with respect to the input, we next make only the first five coordinates heteroskedastic. For $j \leq 5$, conditional on $X_1 = x_1$, the noise standard deviation is

$$\sigma_j(x_1) = (11 - j)\sqrt{\frac{1 + 1.5x_1^2}{2.5}},$$

while $\sigma_j(x_1) = 11 - j$ remains constant for $j > 5$. The normalization preserves $\mathbb{E}[\sigma_j^2(X_1)] = (11 - j)^2$, so the comparison isolates input-dependent variance and does not change the

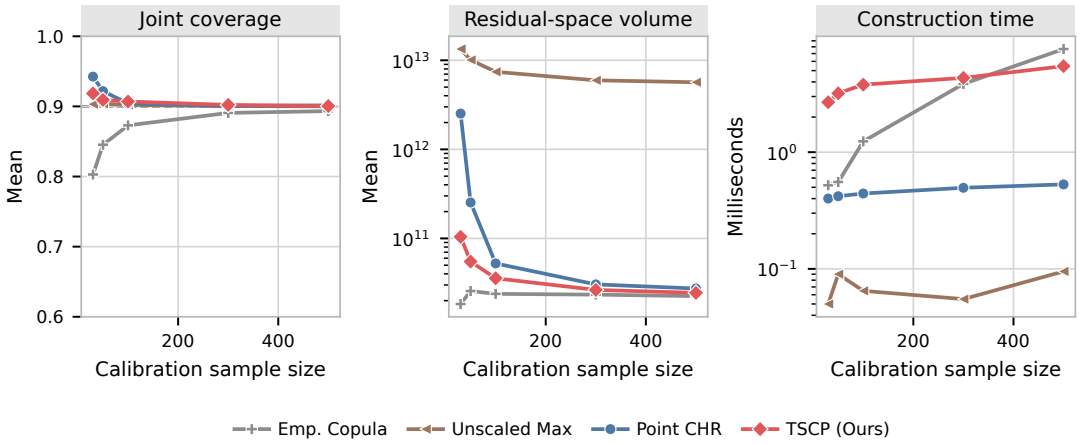


Figure 6: Absolute-residual experiment with $d = 10$, heterogeneous Gaussian scales $(10, 9, \ldots, 1)$, and equicorrelation $\rho = 0.6$. Panels report mean joint coverage, residual-space volume, and construction time over 200 repetitions. The dashed line marks the target coverage 0.9. Runtime excludes model fitting and residual computation.

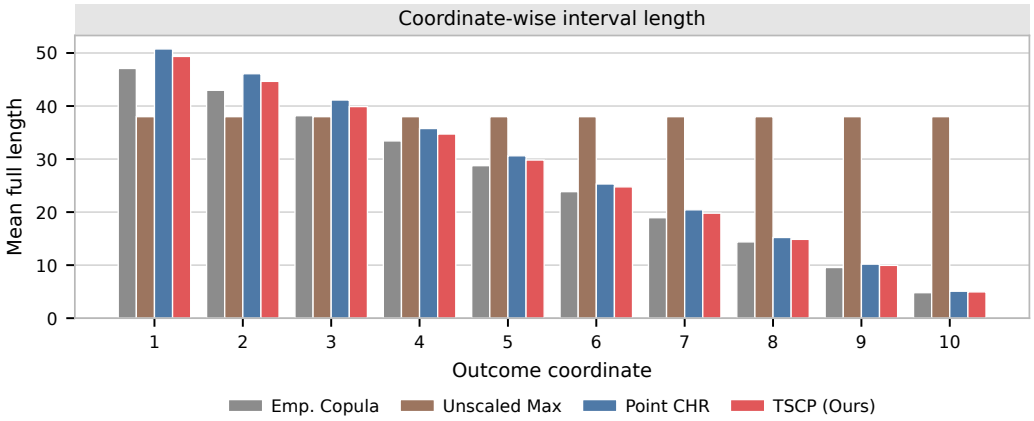


Figure 7: Coordinate-wise mean full interval lengths for the absolute-residual experiment with $d = 10$, $n = 100$, heterogeneous Gaussian scales $(10, 9, \ldots, 1)$, and equicorrelation $\rho = 0.6$. Residual-space half-widths are multiplied by two to report full outcome intervals. Results average 200 independent repetitions. Unscaled Max assigns the same length to every coordinate, while others adapt to the coordinate scales.

marginal second moments. Figure 8 shows that TSCP remains close to the target coverage 0.9 across calibration sizes and continues to adapt coordinate-wise. This experiment does not assert conditional coverage at each value of X_1.

4.3 Quantile-Regression Residuals

We next replace absolute mean-regression residuals with coordinate-wise quantile regression residuals. Let $\widehat{q}_{\ell,j}(x)$ and $\widehat{q}_{u,j}(x)$ denote fitted lower and upper conditional quantiles, ordered

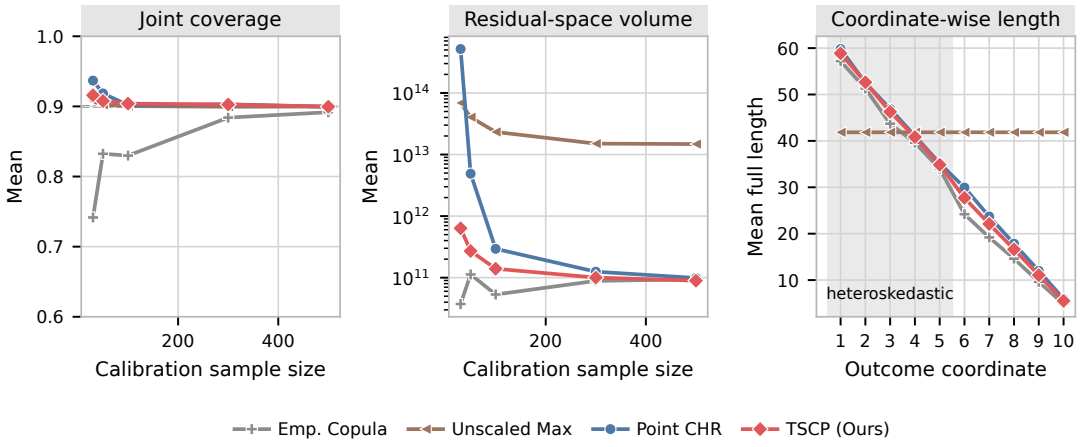


Figure 8: Performance when only the first five of ten outcomes are heteroskedastic in X_1. The shaded coordinates in the right panel are the heteroskedastic ones; full interval lengths are shown for $n = 100$. Results are averaged over 200 repetitions, and the dashed line marks the target joint coverage 0.9.

pointwise. The signed residual is

$$E_j(x, y) = \max\{\widehat{q}_{\ell,j}(x) - y_j, y_j - \widehat{q}_{u,j}(x)\}.$$

Negative values correspond to observations strictly inside the base interval $[\widehat{q}_{\ell,j}(x), \widehat{q}_{u,j}(x)]$. Because the TSCP theory is stated for nonnegative residuals, we use the capped scores $\max\{E_j, 0\}$. Capping prevents the calibrated interval from shrinking the fitted base interval. We therefore use base-interval miscoverage 0.5 to limit the prevalence of negative signed scores. This choice is not required by the method; Appendix D.7 examines other base-interval levels and a shifted-score transformation. Capping can introduce ties at zero, but we can break them by adding a small random noise as a standard procedure in conformal inference.

Unscaled Max, CQHR, and Emp. Copula use the raw signed residuals for fair comparison. CQHR uses the first coordinate as reference and a numerical floor of 10^{-12} on base-interval lengths when forming width ratios.

We use the same linear signal and ten-dimensional Gaussian features in (28) with $d = 3$ Gaussian outcomes and independent noise scales $(3, 2, 1)$, fit separate gradient-boosted quantile regressions using 2,400 training observations, and evaluate on 600 test observations. Each quantile model uses 100 boosting stages, maximum tree depth 5, learning rate 0.05, and minimum leaf size 5. The base-interval miscoverage is 0.5 and the final target miscoverage is 0.1. Calibration sizes are $n \in \{30, 50, 100, 200\}$. Each of the 30 repetitions uses independently generated training, test, and calibration samples.

Figure 9 shows that TSCP, CQHR, and Unscaled Max attain near-nominal coverage, with TSCP producing the smallest mean outcome-space volume among these three methods in the displayed settings. CQHR's volume is sensitive to the fitted base interval, as examined in Appendix D.7. Emp. Copula undercovers in this comparison.

Figure 10 reports the corresponding coordinate-wise comparison at $n = 100$. In particular, TSCP preserves a substantially shorter third-coordinate interval than Unscaled Max, while

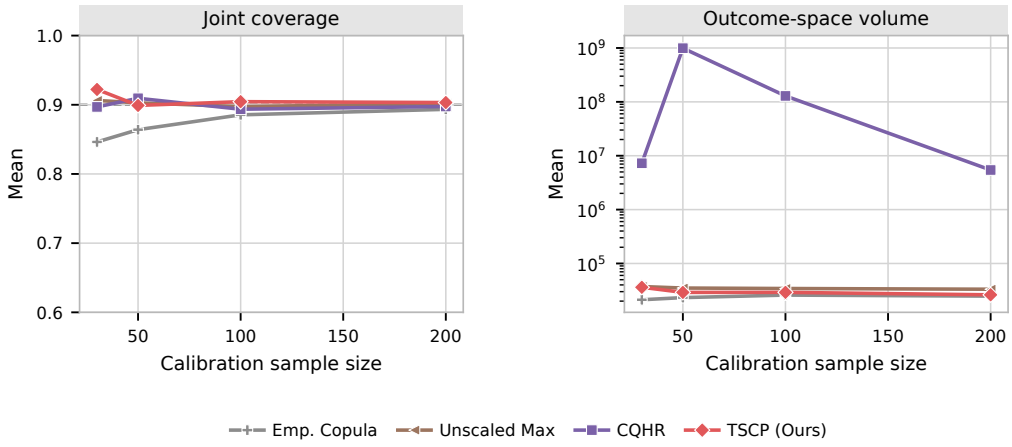


Figure 9: CQR experiment with $d = 3$ and base-interval miscoverage 0.5. TSCP uses capped nonnegative scores, Unscaled Max, CQHR, and Emp. Copula use raw signed residuals. Results are averaged over 30 independent repetitions.

CQHR is much wider on the second and third coordinates in this fitted-quantile design. As above, the coordinate-wise lengths should be interpreted together with the joint coverage panel rather than compared without regard to calibration.

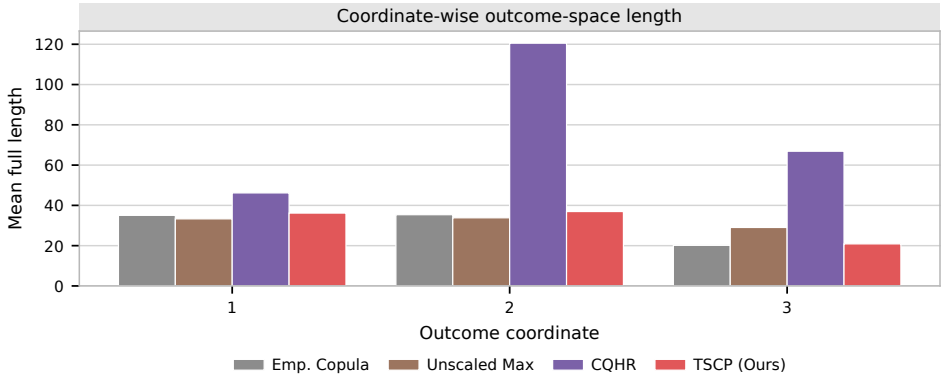


Figure 10: Coordinate-wise mean full outcome-space lengths for the CQR experiment with $d = 3$, $n = 100$, base-interval miscoverage 0.5, and final target coverage 0.9. TSCP uses capped residuals, Unscaled Max and Emp. Copula use raw signed residuals, and CQHR uses its native construction. Results average 30 independent repetitions.

4.4 Real-Data Experiments

We evaluate six real-data benchmarks: `rf2` (Tsoumakas et al., 2022), `scm1d` (Contributors, 2019a), `scm20d` (Contributors, 2019b), `stock` (Yeh, 2015), `student` (Cortez, 2008), and `energy` (Tsanas and Xifara, 2012). Each result averages 200 random training/calibration/test splits using proportions 75%/5%/20%. The calibration sizes therefore range from 15 on `stock` to 490 on `scm1d`. A multitask lasso with regularization parameter 10^{-4} is used for

`stock`; random forests are used for the other datasets. We use absolute-residual scores and target coverage 0.9 throughout. All methods share the fitted model in each split. We report joint coverage, residual-space volume, and construction time here; coordinate-wise comparisons and search diagnostics appear in Appendix D.10. Table 1 reports coverage and residual-space volume for the four primary methods.

Dataset	Method	d	Coverage	Volume
`stock`	Unscaled Max	6	0.940 (0.057)	6.03e−2 (1.57e−1)
	Emp. Copula	6	0.727 (0.113)	1.01e−5 (9.80e−6)
	Point CHR	6	1.000 (0.000)	∞
	TSCP	6	0.956 (0.059)	**1.86e−3** (**8.55e−3**)
`rf2`	Unscaled Max	8	0.900 (0.016)	4.14e3 (2.90e3)
	Emp. Copula	8	0.893 (0.016)	5.35e1 (4.69e1)
	Point CHR	8	0.901 (0.022)	**6.85e1** (**7.55e1**)
	TSCP	8	0.901 (0.017)	5.42e2 (1.50e3)
`scm1d`	Unscaled Max	16	0.900 (0.016)	2.74e39 (4.02e39)
	Emp. Copula	16	0.893 (0.016)	1.09e39 (1.34e39)
	Point CHR	16	0.905 (0.020)	2.84e39 (5.43e39)
	TSCP	16	0.901 (0.015)	**1.52e39** (**2.61e39**)
`scm20d`	Unscaled Max	16	0.901 (0.016)	2.70e40 (2.92e40)
	Emp. Copula	16	0.892 (0.017)	1.18e40 (1.19e40)
	Point CHR	16	0.902 (0.023)	3.00e40 (8.00e40)
	TSCP	16	0.901 (0.016)	**1.53e40** (**1.50e40**)
`energy`	Unscaled Max	2	0.917 (0.049)	1.58e1 (6.80e0)
	Emp. Copula	2	0.886 (0.061)	5.41e0 (4.49e0)
	Point CHR	2	0.899 (0.069)	8.81e0 (2.42e1)
	TSCP	2	0.922 (0.048)	**6.95e0** (**4.43e0**)
`student`	Unscaled Max	3	0.904 (0.055)	**1.51e2** (**1.27e2**)
	Emp. Copula	3	0.861 (0.073)	1.08e2 (7.19e1)
	Point CHR	3	0.939 (0.057)	6.89e2 (1.07e3)
	TSCP	3	0.912 (0.056)	1.98e2 (1.77e2)

Table 1: Performance on real datasets, averaged over 200 random splits. One standard deviation is reported in parentheses. Bold volume denotes the smallest average volume among TSCP, Point CHR, and Unscaled Max. Red coverage values are more than two Monte Carlo standard errors below 0.9, where the standard error is the across-split standard deviation divided by $\sqrt{200}$. This is a pointwise descriptive flag, not a simultaneous test or a proof of validity. The calibration sizes are 15, 384, 490, 448, 38, and 32 in the order shown.

Among TSCP, Point CHR, and Unscaled Max, TSCP has the smallest average volume on `stock`, `scm1d`, `scm20d`, and `energy`. Point CHR has the smallest volume on `rf2`, while Unscaled Max has the smallest volume on `student`. Thus, the coordinate-adaptive construction is beneficial in several datasets but is not uniformly preferable. Emp. Copula produces smaller regions with empirical coverage below the target.

Figure 11 compares construction times. TSCP constructs a region in 0.74–7.46 milliseconds on these datasets. It is slower than the algebraically simpler Unscaled Max and Point CHR

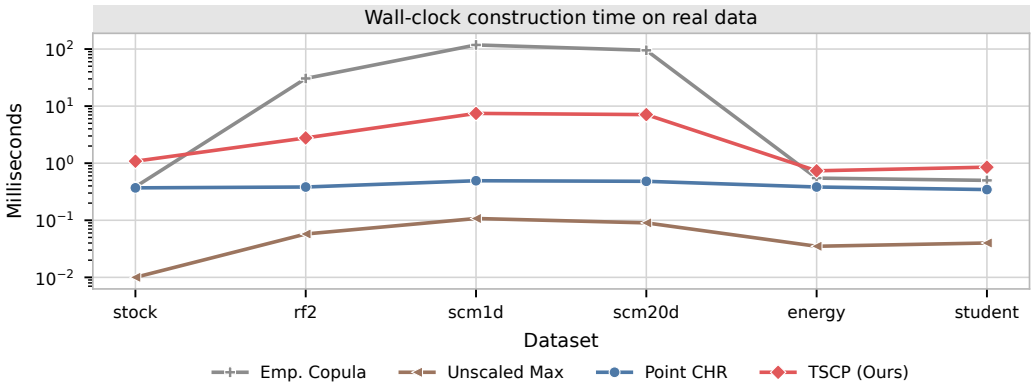


Figure 11: Average wall-clock time for conformal region construction on the six real datasets. The vertical axis is logarithmic and reports milliseconds. All methods use the same fitted regression model in each split; model fitting and residual computation are excluded.

implementations, but substantially faster than Emp. Copula on the two 16-dimensional SCM datasets.

The infinite Point CHR volume on `stock` should not be interpreted as an intrinsic failure of the method. With only $n = 15$ calibration observations, its additional split leaves too few observations for a finite 0.9 conformal quantile. A practitioner committed to Point CHR would allocate a larger fraction of the data to its final calibration stage, at the cost of fewer observations for model and scale estimation.

4.4.1 Interpretability of Rectangular Prediction Sets

To emphasize their interpretability, Table 2 displays the rectangular prediction sets produced by different methods for two representative test points from the *energy* dataset, where $d = 2$. In this setting, rectangular prediction sets consist of pairs of simultaneous prediction intervals that aim to jointly contain the true outcomes with probability 90%.

5 Discussion and Future Work

The core idea underlying the method presented in this paper is the explicit standardization of residuals, which contributes both to the transparency and to the computational efficiency of the resulting prediction sets. As with other standardization-based methods, however, this strategy introduces some sensitivity to outliers. This behavior is visible in the numerical experiments based on the *rf2* dataset, where a small number of influential observations can substantially expand the resulting prediction sets. The contamination study in Appendix D.6 likewise distinguishes stable empirical coverage from substantial volume inflation. An important direction for future work is therefore the development of robust extensions of our method, for example through rescaling based on median absolute deviation or quantile-based normalizations, while preserving the exchangeability required for conformal validity.

Another direction is a direct treatment of signed residuals without capping or an additive shift. The CQR comparisons in Section 4.3 and Appendix D.7 illustrate the limitations of

Test 1/Methods	Outcome 1	Outcome 2	Volume	Ground Truth
Unscaled Max	[31.70, 38.94]	**[34.74, 41.98]**	13.09	(35.40, 39.22)
Emp. Copula	**[34.36, 36.28]**	**[34.74, 41.98]**	3.47	
Point CHR	[34.29, 36.36]	[31.45, 45.27]	7.14	
TSCP	[34.28, 36.36]	[33.63, 43.08]	4.93	

Test 2/Methods	Outcome 1	Outcome 2	Volume	Ground Truth
Unscaled Max	[25.44, 32.67]	[29.34, 36.58]	13.09	(29.88, 28.31)
Emp. Copula	**[28.10, 30.02]**	[29.34, 36.58]	3.47	
Point CHR	[28.02, 30.09]	[26.05, 39.87]	7.14	
TSCP	[28.01, 30.10]	**[28.24, 37.69]**	4.93	

Table 2: Illustration of rectangular conformal prediction sets for two test points from the *energy* dataset. Volumes are computed in residual space. Intervals that fail to cover the corresponding true outcomes are highlighted in red; the shortest interval containing each true outcome is highlighted in **bold**. This example is illustrative: conformal coverage guarantees hold on average, not for individual test points.

these transformations. A direct extension could enable shrinkage of baseline prediction sets when starting from quantile regression models that already exhibit conservative coverage. It would also be of interest to adapt the proposed approach to settings with non-numerical outcomes, such as categorical or ordinal responses. Finally, our method could be extended to handle partly observed outcomes (Braun et al., 2025b).

Software Availability

A software implementation of the methods described in this paper, along with code necessary to reproduce the numerical experiments, is available online at `https://github.com/OTheMagic/multi-target-scaling.git`.

Acknowledgments and Disclosure of Funding

The authors thank Rundong Ding and Dr. Yizhe Zhu for helpful suggestions during an oral presentation of an earlier version of this work. M.S. was partly supported by NSF grant DMS 2210637, a USC-Capital One CREDIF award, and a Google Research Scholar award.

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Appendix: Algorithms, Proofs, and Numerical Results

Appendix A. A Full-Cell Construction and Its Relation to the Envelope

This appendix develops a full-cell construction that explains the coordinate-wise envelope in Section **??**. We first combine local prediction sets over a rectangular cover and show that their union preserves the oracle coverage event. We then describe its tightest coordinate-wise upper enclosure and prove that the practical envelope encloses this union. This establishes the mathematical relationship between the two constructions; the practical method can be computed using the coordinate-wise steps in the main text.

We retain Assumption 1 and $n \geq 2$. Recall that $\hat{Q}^{\mathcal{R}}_{1-\alpha}$ is the pseudo-upper bound obtained by taking the supremum of each standardized calibration score over a candidate residual region $\mathcal{R}$ and then taking the conformal quantile. On the event $\mathbf{E}^{n+1} \in \mathcal{R}$, it satisfies $\hat{Q}^{\mathcal{R}}_{1-\alpha} \geq \hat{Q}^{\text{oracle}}_{1-\alpha}$, and the corresponding pseudo prediction set is

$$\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\mathcal{R}}_{1-\alpha}) = \left\{ \mathbf{y} \in \mathbb{R}^d : E_j(\mathbf{X}^{n+1}, y_j) \leq \omega_j(\hat{Q}^{\mathcal{R}}_{1-\alpha}) \text{ for all } j \in [d] \right\}.$$

A.1 Combining Local Prediction Sets

A natural way to make the global prediction set less conservative is to cover residual space with smaller hyper-rectangles. Let $\mathcal{P}$ be a finite collection of nonempty hyper-rectangles such that $\mathbb{R}^d = \bigcup_{\mathcal{R} \in \mathcal{P}} \mathcal{R}$. We allow neighboring regions to share their boundaries; this retains all boundary points, as in the grid used by the practical construction. For each $\mathcal{R} \in \mathcal{P}$, we can repeat the construction of the pseudo prediction set defined in (15) and combine these sets to obtain a prediction set that preserves the oracle coverage event:

$$\hat{C}^{\mathcal{P}}(\mathbf{X}^{n+1}) = \bigcup_{\mathcal{R} \in \mathcal{P}} \left(\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\mathcal{R}}_{1-\alpha}) \cap \left\{ \mathbf{y} \in \mathbb{R}^d : \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in \mathcal{R} \right\} \right). \tag{S1}$$

The intersection inside the union restricts each pseudo prediction set to candidate outcomes whose residuals lie in the corresponding region. There is no need to consider candidate outcomes $\mathbf{y}$ for which $\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \notin \mathcal{R}$ within that region, because we already operate under the hypothesis that $\mathbf{E}^{n+1} \in \mathcal{R}$. This refinement preserves the guarantee, as demonstrated in the following lemma.

Lemma S1 *Let $\mathcal{P}$ be a finite, possibly data-dependent cover of $\mathbb{R}^d$ by nonempty hyper-rectangles. Under Assumption 1, with $n \geq 2$, the prediction set in* (S1) *satisfies*

$$\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}) \implies \mathbf{Y}^{n+1} \in \hat{C}^{\mathcal{P}}(\mathbf{X}^{n+1}).$$

The same result holds if $\mathbb{R}^d$ is replaced by any possibly data-dependent subset $K \subseteq \mathbb{R}^d$ such that

$$\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}) \implies \mathbf{E}^{n+1} \in K \quad \textit{almost surely.}$$

Proof On the oracle coverage event, the true residual vector belongs to at least one region $\mathcal{R} \in \mathcal{P}$. Lemma **??** then places $\mathbf{Y}^{n+1}$ in the corresponding term of the union in (S1). The argument does not require the regions to be disjoint or independent of the data. ■

The second part of this lemma outlines an additional way to control the volume of the resulting prediction set. If such a subset K exists and $\mathcal{P}$ covers K, then

$$\hat{C}^{\mathcal{P}}(\mathbf{X}^{n+1}) \subseteq \left\{\mathbf{y} \in \mathbb{R}^d : \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in K\right\}$$

by a simple set-inclusion argument. For the practical construction, recall from (**??**) that

$$D = \bigtimes_{j=1}^{d} D_j, \qquad D_j = [a_j, +\infty) \cap \mathbb{R},$$

$$U_j = \omega_j(\hat{Q}^D_{1-\alpha}), \qquad K = \bigtimes_{j=1}^{d} \big(D_j \cap (-\infty, U_j]\big).$$

The domains D_j contain every residual value attainable at the test input. Thus $\mathbf{E}^{n+1} \in D$ automatically and $\hat{Q}^D_{1-\alpha} \geq \hat{Q}^{\text{oracle}}_{1-\alpha}$. Monotonicity of the link functions then implies that the oracle coverage event places $\mathbf{E}^{n+1}$ in K, as required by Lemma S1. Moreover,

$$\left\{\mathbf{y} \in \mathbb{R}^d : \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in K\right\} = \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^D_{1-\alpha}).$$

Since $D \subseteq \mathbb{R}^d$, monotonicity of the supremum and the empirical quantile also gives $\hat{Q}^D_{1-\alpha} \leq \hat{Q}^{\mathbb{R}^d}_{1-\alpha}$. Restricting the global calculation to the known residual domains therefore cannot enlarge the global prediction set.

When the grid in Section **??** is constructed, each nonempty K_j is covered by the closed intervals $I_j^h = [\eta_j^{h-1}, \eta_j^h] \cap \mathbb{R}$, for $h \in [N_j]$. Their interior endpoints are the distinct calibration order statistics strictly inside K_j, and singleton K_j are retained as single intervals. The full cells and their cover are

$$\mathcal{E}^{\mathbf{h}} = \bigtimes_{j=1}^{d} I_j^{h_j}, \qquad \mathcal{P}_{n,d} = \left\{\mathcal{E}^{\mathbf{h}} : \mathbf{h} \in \bigtimes_{j=1}^{d} [N_j]\right\}.$$

There are at most $(n+1)^d$ cells. Their interiors are disjoint, but neighboring cells share their boundaries, so the union in (S1) is an ordinary union rather than a disjoint union. Lemma S1 applies to this data-driven cover of K.

A.2 The Residual Image and Its Upper-Bound Enclosure

The prediction set in (S1) is not necessarily a hyper-rectangle, even for standard choices such as absolute and quantile residuals, due to the intersections and union in its construction.

To keep the interpretability of the prediction set, we can enclose its residual image using coordinate-wise upper bounds. Specifically, let

$$\mathcal{E}^{\mathcal{P}} := \left\{\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) : \mathbf{y} \in \hat{C}^{\mathcal{P}}(\mathbf{X}^{n+1})\right\}, \qquad W_j^{\mathcal{P}} := \sup_{\mathbf{z} \in \mathcal{E}^{\mathcal{P}}} z_j,$$

with the convention $\sup \varnothing = -\infty$. These thresholds give the *tightest* enclosure among regions defined solely by coordinate-wise upper bounds. In particular,

$$\hat{C}^{\mathcal{P}}(\mathbf{X}^{n+1}) \subseteq \left\{\mathbf{y} \in \mathbb{R}^d : E_j(\mathbf{X}^{n+1}, y_j) \leq W_j^{\mathcal{P}} \text{ for all } j \in [d]\right\}.$$

Each upper bound is the supremum of the corresponding coordinate over the residual image, so any enclosure defined solely by coordinate-wise upper bounds must use thresholds at least as large. For absolute and quantile residuals, this upper-bound enclosure gives a hyper-rectangle in outcome space whenever it is nonempty. Computing these thresholds directly, however, may require checking all combinations of coordinate intervals.

The practical construction instead uses the computable upper bounds $\overline{W}_j^{\mathcal{P}_{n,d}}$ in (25). Its envelope bounds one coordinate locally while allowing the others to range over their global regions K_j. This avoids enumerating all full cells, but it can give a larger set than the tightest upper-bound enclosure of $\mathcal{E}^{\mathcal{P}_{n,d}}$. In particular, the coordinate-wise accepted pieces need not verify that a residual vector belongs to a jointly accepted full cell.

A.3 Containment in the Practical Envelope

For every full cell $\mathcal{E}^{\mathbf{h}}$ and coordinate j, the inclusions

$$\mathcal{E}^{\mathbf{h}} \subseteq \mathcal{R}_j^{h_j} \subseteq K \subseteq D$$

imply, by monotonicity of the supremum and the empirical quantile, that

$$\hat{Q}_{1-\alpha}^{\mathcal{E}^{\mathbf{h}}} \leq \hat{Q}_{1-\alpha}^{\mathcal{R}_j^{h_j}} \leq \hat{Q}_{1-\alpha}^{D}. \tag{S2}$$

This is the key connection between the coordinate-wise envelope and the full-cell construction. It gives an upper bound for each relevant cell quantile without enumerating all combinations of the other coordinates.

Lemma S2 *Under Assumption 1, with $n \geq 2$, the construction above satisfies*

$$\hat{C}^{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1}) \subseteq \hat{C}_{\mathrm{TSCP}}(\mathbf{X}^{n+1}) \subseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{D})$$

whenever the grid is constructed. In particular,

$$W_j^{\mathcal{P}_{n,d}} \leq \overline{W}_j^{\mathcal{P}_{n,d}} \leq U_j, \qquad j \in [d].$$

Proof Take $\mathbf{y} \in \hat{C}^{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$ and write $\mathbf{e} = \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y})$. For at least one full cell $\mathcal{E}^{\mathbf{h}}$, we have $\mathbf{e} \in \mathcal{E}^{\mathbf{h}}$ and $e_j \leq \omega_j(\hat{Q}_{1-\alpha}^{\mathcal{E}^{\mathbf{h}}})$ for every j. By (S2) and monotonicity of the link functions,

$$e_j \in I_j^{h_j}, \qquad e_j \leq \omega_j(\hat{Q}_{1-\alpha}^{\mathcal{R}_j^{h_j}}).$$

Thus $e_j \in A_j^{h_j}$ and $e_j \leq B_j^{h_j} \leq \overline{W}_j^{\mathcal{P}_{n,d}}$ for every j. This proves the first inclusion and, on taking suprema over the residual image, the bound $W_j^{\mathcal{P}_{n,d}} \leq \overline{W}_j^{\mathcal{P}_{n,d}}$. Since every accepted piece lies in K_j, its upper endpoint is at most U_j, proving the remaining bounds and the second inclusion.

Lemma S1 now shows that the final set preserves the oracle coverage event. If K is empty, that event cannot occur, while the case $r = n + 1$ returns the entire outcome space. The oracle coverage guarantee from Section 3.1 therefore gives the marginal coverage in Theorem **??** in all cases. ■

Figure S1 illustrates the full-cell construction and its tight upper-bound enclosure in a two-dimensional toy example. The envelope in Section **??** gives a computable enclosure that need not coincide with this tight enclosure.

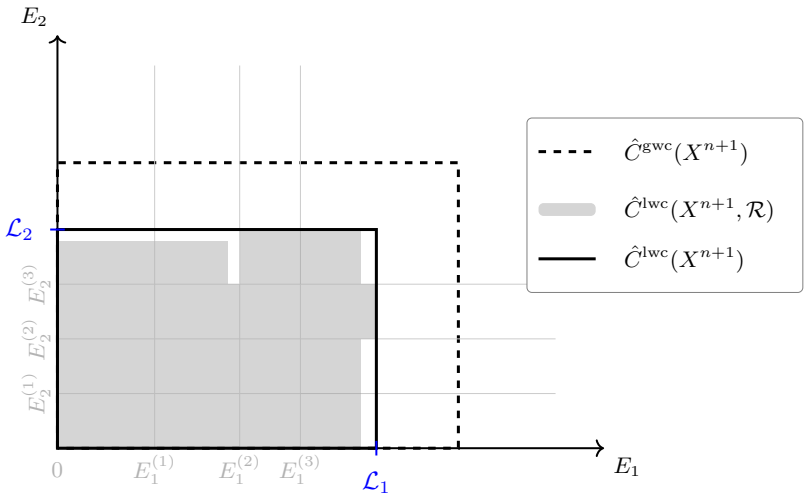


Figure S1: Illustration of the local prediction set in (S1), using the residual-space cover $\mathcal{P}_{n,d}$ in (**??**), in a toy example with a two-dimensional outcome variable. The gray grid lines are determined by the calibration order statistics. The shaded regions represent the intersections of each cell $\mathcal{R} \in \mathcal{P}_{n,d}$ with its local residual-space region $\{\mathbf{z} \in \mathbb{R}^d : z_j \leq \omega_j(\hat{Q}^{\mathcal{R}}_{1-\alpha})$ for all $j \in [d]\}$. For standard residuals with exact domains D_j, the solid boundary indicates the tight upper bounds $W_j^{\mathcal{P}_{n,d}}$ of this union. The larger dashed boundary shows the global region K. The practical envelope thresholds $\overline{W}_j^{\mathcal{P}_{n,d}}$ need not coincide with the tight bounds shown here.

Appendix B. Additional Algorithmic Details

B.1 Implementation of Alternative Methods

Algorithm S1 Separate Prediction Intervals with Bonferroni Correction

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

1: Compute $\alpha^* \leftarrow \alpha/d$.
2: **for** $j = 1$ to d **do**
3: Compute $\hat{Q}_{1-\alpha^*,j} \leftarrow \hat{Q}_{1-\alpha^*}(E_j^1, \ldots, E_j^n)$.
4: **end for**
5: **return** $\hat{C}^{\text{Bonf.}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}_{1-\alpha^*,j}, \forall j \in [d]\right\}$.

Algorithm S2 Prediction Intervals via Dimension Reduction with ℓ_∞-Norm (Unscaled)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

Compute $S^i_{\text{unscaled}} \leftarrow \|\mathbf{E}^i\|_\infty$ for $i = 1, \ldots, n$.
Compute $\hat{Q}^{\text{unscaled}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{unscaled}}, \ldots, S^n_{\text{unscaled}})$
return $\hat{C}^{\text{unscaled}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\text{unscaled}}_{1-\alpha}, \forall j \in [d]\right\}$.

Algorithm S3 Heuristic Plug-In Standardization

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

for $j = 1$ to d **do**
    Compute the location and scale parameters

$$\hat{\mu}^{\text{pi}}_j \leftarrow \frac{1}{n}\sum_{i=1}^n E^i_j, \qquad \hat{\sigma}^{\text{pi}}_j \leftarrow \sqrt{\frac{1}{n-1}\sum_{i=1}^n (E^i_j - \hat{\mu}^{\text{pi}}_j)^2}.$$

end for
Compute $S^i_{\text{pi}} \leftarrow \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$ for $i \in [n]$, using Φ defined in (6).
Compute $\hat{Q}^{\text{pi}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{pi}}, \ldots, S^n_{\text{pi}})$.
Compute

$$\hat{C}^{\text{pi}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\text{pi}}_{1-\alpha} \cdot \hat{\sigma}^{\text{pi}}_j + \hat{\mu}^{\text{pi}}_j, \forall j \in [d]\right\},$$

which is hyper-rectangular for the standard choices of $\mathbf{E}$ discussed in Section 2.3.
return Prediction set $\hat{C}^{\text{pi}}(\mathbf{X}^{n+1})$.

Algorithm S4 Standardization with Data Splitting

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.
Output: $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1})$

Randomly split $\{1, \ldots, n\}$ into two disjoint subsets, $\mathcal{I}_1$ and $\mathcal{I}_2$.
for $j = 1$ to d **do**
    Compute the location and scale parameters

$$\hat{\mu}^{\text{ds}}_j \leftarrow \frac{1}{|\mathcal{I}_1|}\sum_{i \in \mathcal{I}_1} E^i_j, \qquad \hat{\sigma}^{\text{ds}}_j \leftarrow \sqrt{\frac{1}{|\mathcal{I}_1|-1}\sum_{i \in \mathcal{I}_1} (E^i_j - \hat{\mu}^{\text{ds}}_j)^2}.$$

end for
Compute $S^i_{\text{ds}} \leftarrow \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{ds}}, \hat{\boldsymbol{\sigma}}^{\text{ds}})$ for $i \in \mathcal{I}_2$, using Φ defined in (6).
Compute $\hat{Q}^{\text{ds}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(\{S^i_{\text{ds}}\}_{i \in \mathcal{I}_2})$.
return $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\text{ds}}_{1-\alpha} \cdot \hat{\sigma}^{\text{ds}}_j + \hat{\mu}^{\text{ds}}_j, \forall j \in [d]\right\}$.

B.2 Auxiliary Algorithms

Algorithm S5 Global-Worst-Case TSCP (TSCP-GWC)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

Compute $\hat{\boldsymbol{\mu}}^{\text{obs}}, \hat{\boldsymbol{\sigma}}^{\text{obs}}$ using (17).
Compute $S^i_{\text{gwc}} \leftarrow \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ using $\hat{\Phi}^{\text{gwc}}$ defined in Lemma **??**.
Compute $\hat{Q}^{\text{gwc}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{gwc}}, \ldots, S^n_{\text{gwc}})$.
Compute $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \ldots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$ using ω defined in Lemma 2.
return $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha}), \forall j \in [d]\right\}$.

Algorithm S6 Backward Search in Algorithm **??**

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; global bounds $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \ldots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$; target miscoverage level $\alpha \in (0,1)$; calibration mean $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and standard deviation $\hat{\boldsymbol{\sigma}}^{\text{obs}}$; mean index $\mathbf{h}^*$; current coordinate j.

for $\mathbf{h} \in \text{Row}_j(\mathbf{h}^*)$ with $h_j = h^*_j - 1$ to 1 **do**
    Compute $\omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}}))$ using Algorithm **??** and $B^{\mathbf{h}}_j$ using (**??**).
    if $B^{\mathbf{h}}_j > 0$ **then**
        return $\mathbf{h}^{[j]} = \mathbf{h}$.
    end if
end for

Algorithm S7 Binary Search in Algorithm **??**

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; global bounds $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \ldots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$; target miscoverage level $\alpha \in (0,1)$; calibration mean $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and standard deviation $\hat{\boldsymbol{\sigma}}^{\text{obs}}$; mean index $\mathbf{h}^*$; current coordinate j.

Let $I_{\min} \leftarrow h^*_j$ and $I_{\max} \leftarrow n+1$.
while $I_{\min} < I_{\max}$ **do**
    Compute $m_j \leftarrow \lceil (I_{\min} + I_{\max})/2 \rceil$ and $m_k = h^*_k$ for all $k \neq j$.
    Compute $\omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{m}}))$ and B^m_j using (**??**). ▷ Algorithm **??**
    if $B^m_j > 0$ **then**
        $I_{\min} \leftarrow m_j$.
    else
        $I_{\max} \leftarrow m_j - 1$.
    end if
    if $I_{\max} = I_{\min}$ **then**
        return $h^{[j]}_j = I_{\min}$ and $h^{[j]}_k = h^*_k$ for all $k \neq j$.
    end if
end while

B.3 Computational Cost Analysis

B.3.1 Algorithm S5

Since the calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$ are stored as an (n,d)-array, we know

1. Computing $(\hat{\mu}_j^{\text{obs}}, \hat{\sigma}_j^{\text{obs}})_{j=1}^d$ using (17) requires $\mathcal{O}(nd)$.
2. Evaluating $S_{\text{gwc}}^i \leftarrow \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ for all $i \in [n]$ is also $\mathcal{O}(nd)$.
3. Computing the quantile $\hat{Q}_{1-\alpha}^{\text{gwc}}$ costs $\mathcal{O}(n)$ using a selection algorithm such as quickselect.
4. Computing $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}), \ldots, \omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$ costs $\mathcal{O}(d)$ once $\hat{Q}_{1-\alpha}^{\text{gwc}}$ and $(\hat{\mu}_j^{\text{obs}}, \hat{\sigma}_j^{\text{obs}})_{j=1}^d$ are known.

Hence, the overall computational complexity of Algorithm S5 is $\mathcal{O}(nd)$.

B.3.2 Algorithm ??

Algorithm **??** shares the same structure and computational cost as Algorithm S5: $\mathcal{O}(nd)$.

B.3.3 Algorithm ??

Since the calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$ are stored as an (n, d)-array, we know

1. Computing $(\hat{\mu}_j^{\text{obs}}, \hat{\sigma}_j^{\text{obs}})_{j=1}^d$ using (17) and $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}), \ldots, \omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$ using Algorithm S5 requires $\mathcal{O}(nd)$.
2. Sorting $\{E_j^i\}_{i=1}^n$ for all $j \in [d]$ costs $\mathcal{O}(dn \log n)$;
3. Computing $\mathcal{Y}^{\mathbf{h}^*}$ relies on computing $\mathbf{h}^*$, which requires masking all values below $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and reporting the index of the lowest value from the sorted residuals list $\{E_j^i\}_{i=1}^n$, each of which costs $\mathcal{O}(n)$, so together this operation contributes $\mathcal{O}(dn)$.
4. If $\mathcal{Y}^{\mathbf{h}^*} = \emptyset$, then the cost is only $\mathcal{O}(d)$ since all global bounds have been computed.
5. If $\mathcal{Y}^{\mathbf{h}^*} \neq \emptyset$, then for each coordinate, we compute the local bounds using Algorithm **??** and $B_j^{\mathbf{h}^*}$ using (**??**), which costs $\mathcal{O}(dn)$. If $B_j^{\mathbf{h}^*} = 0$, then we use the backward search detailed in Algorithm S6, which costs $\mathcal{O}(dn^2)$; see Appendix B.3.4. If $B_j^{\mathbf{h}^*} > 0$, then we use the binary search detailed in Algorithm S7, which costs $\mathcal{O}(dn \log n)$; see Appendix B.3.5.

Let $T \leq d$ be the number of coordinates where binary searches are enabled. Then the total cost of Algorithm **??** is $\mathcal{O}(Tdn \log n + (d - T)dn^2)$.

B.3.4 Algorithm S6

Running Algorithm **??** and forming $B_j^{\mathbf{h}}$ costs $\mathcal{O}(dn)$ for each $\mathbf{h}$; there are $n + 1$ indices in $\text{Row}_j(\mathbf{h}^*)$, so the computational complexity of Algorithm S6 never exceeds $\mathcal{O}(dn^2)$.

B.3.5 Algorithm S7

Running Algorithm **??** and forming $B_j^{\mathbf{h}}$ costs $\mathcal{O}(dn)$ for each $\mathbf{h}$. The number of indices $\mathbf{h}$ encountered in this algorithm is $\mathcal{O}(\log_2 n)$ because each step retains only half of the current index range at each step, as in a standard binary search. Therefore, the total computational complexity of Algorithm S7 never exceeds $\mathcal{O}(dn \log n)$.

Appendix C. Mathematical Proofs

C.1 Auxiliary Lemmas

Lemma S3 *Let $x^1, \dots, x^n \in \mathbb{R}$ and $\bar{x}^n = \frac{1}{n}\sum_{i=1}^n x^i$. For any $x^{n+1} \in \mathbb{R}$, let $\hat{\sigma}(x^{n+1})$ be the sample standard deviation of $\{x^1, \dots, x^n, x^{n+1}\}$. Then*

$$\hat{\sigma}^2(x^{n+1}) = \frac{(x^{n+1} - \bar{x}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.$$

Proof Denote $\bar{x}^{n+1} = \frac{1}{n+1}\sum_{i=1}^{n+1} x^i$. We notice that

$$\begin{aligned}
\hat{\sigma}^2(x^{n+1}) =& \frac{1}{n}\sum_{i=1}^{n+1}(x^i - \bar{x}^{n+1})^2 \\
=& \frac{1}{n}\left[(x^{n+1} - \bar{x}^{n+1})^2 + \sum_{i=1}^n (x^i - \bar{x}^n + \bar{x}^n - \bar{x}^{n+1})^2\right] \\
=& \frac{(x^{n+1} - \bar{x}^{n+1})^2}{n} + \frac{1}{n}\sum_{i=1}^n \left[(x^i - \bar{x}^n)^2 + 2(x^i - \bar{x}^n)(\bar{x}^n - \bar{x}^{n+1}) + (\bar{x}^n - \bar{x}^{n+1})^2\right] \\
=& \frac{(x^{n+1} - \bar{x}^{n+1})^2}{n} + (\bar{x}^n - \bar{x}^{n+1})^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.
\end{aligned}$$

Substituting $\bar{x}^{n+1} = \frac{x^{n+1} + n\bar{x}^n}{n+1}$, we get

$$\begin{aligned}
\hat{\sigma}^2(x^{n+1}) =& \frac{1}{n}\left(x^{n+1} - \frac{x^{n+1} + n\bar{x}^n}{n+1}\right)^2 + \left(\bar{x}^n - \frac{n\bar{x}^n + x^{n+1}}{n+1}\right)^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2 \\
=& \frac{n}{(n+1)^2}(x^{n+1} - \bar{x}^n)^2 + \frac{1}{(n+1)^2}(\bar{x}^n - x^{n+1})^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2 \\
=& \frac{(x^{n+1} - \bar{x}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.
\end{aligned}$$

■

Lemma S4 *Let $X^1, \dots, X^n \in \mathbb{R}$ be real-valued random variables with sample mean $\bar{X}^n$. Let X^{n+1} and $\tilde{X}^{n+1}$ be two additional random variables such that*

$$|X^{n+1} - \bar{X}^n| \leq |\tilde{X}^{n+1} - \bar{X}^n| \quad a.s.$$

Then,

$$\hat{\sigma}(X^{n+1}) \leq \hat{\sigma}(\tilde{X}^{n+1}) \quad a.s.,$$

where $\hat{\sigma}(X^{n+1})$ is the sample standard deviation of $\{X^1, \dots, X^n, X^{n+1}\}$, as in Lemma S3.

Proof This is a direct application of Lemma S3:

$$\begin{aligned}\hat{\sigma}^2(X^{n+1}) &= \frac{(X^{n+1}-\bar{X}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^{n}(X^i-\bar{X}^n)^2 \\ &\leq \frac{(\tilde{X}^{n+1}-\bar{X}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^{n}(X^i-\bar{X}^n)^2 \quad \text{[Assumption]} \\ &= \hat{\sigma}^2(\tilde{X}^{n+1}),\end{aligned}$$

and the claim follows by taking square roots. ■

Lemma S5 *Let $X^1,\ldots,X^{n+1}\in\mathbb{R}$ be real-valued random variables with sample mean $\bar{X}^{n+1}$ and sample standard deviation $\bar{S}^{n+1}>0$. Define $Z^i=(X^i-\bar{X}^{n+1})/\bar{S}^{n+1}$ for $i\in[n+1]$. Then,*

$$|Z^i|\leq\frac{n}{\sqrt{n+1}} \quad \textit{almost surely,} \quad \forall i\in[n+1].$$

This inequality is strict if $X^1,\ldots,X^{n+1}$ are almost surely distinct.

Proof Fix any $i\in[n+1]$. Let $\bar{X}^{-i}=\frac{1}{n}\sum_{k\neq i}X^k$ be the sample mean of the random variables excluding the i-th one, and

$$\bar{S}^{-i}=\sqrt{\frac{1}{n}\sum_{k\neq i}(X^k-\bar{X}^{-i})^2}.$$

Then, it follows from Lemma S3 that

$$|Z^i|=\frac{|X^i-\bar{X}^{n+1}|}{\bar{S}^{n+1}}=\frac{\left|X^i-\frac{1}{n+1}\sum_{k=1}^{n+1}X^k\right|}{\sqrt{\frac{1}{n}\sum_{k=1}^{n+1}(X^k-\bar{X}^{n+1})^2}}=\frac{\frac{n}{n+1}|X^i-\bar{X}^{-i}|}{\sqrt{\frac{1}{n+1}(X^i-\bar{X}^{-i})^2+(\bar{S}^{-i})^2}}.$$

But $\bar{S}^{-i}\geq 0$, so we must have

$$|Z^i|\leq\frac{\frac{n}{n+1}|X^i-\bar{X}^{-i}|}{\sqrt{\frac{1}{n+1}(X^i-\bar{X}^{-i})^2}}=\frac{n}{\sqrt{n+1}}.$$

If $X^1,\ldots,X^{n+1}$ are almost surely distinct, we know $\bar{S}^{-i}>0$ almost surely for any $i\in[n+1]$, so this inequality becomes strict. ■

C.2 Proofs of Main Results

Proof [of Theorem 1] This is a well-known result, whose proof is included for completeness. By definition of $\hat{C}(\mathbf{X}^{n+1})$ in (5),

$$\mathbf{Y}^{n+1}\notin\hat{C}(\mathbf{X}^{n+1})\iff S^{n+1}>\hat{Q}_{1-\alpha}(S^1,\ldots,S^n).$$

Moreover, $S^1, \ldots, S^{n+1}$ are exchangeable because $\mathbf{E}^1, \ldots, \mathbf{E}^{n+1}$ are exchangeable and Φ is fixed. Therefore, the proof is complete by applying the fundamental "quantile inflation lemma" of conformal prediction (e.g., see Angelopoulos et al. (2024)), which states

$$\mathbb{P}\left[S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n)\right] \leq \alpha.$$

Moreover, it is also a standard result that, if $S^1, \ldots, S^{n+1}$ are almost surely distinct,

$$\mathbb{P}\left[S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n)\right] \geq \alpha - \frac{1}{n+1}.$$

■

Proof [of Lemma 2] From Assumption 1, we know that for all $j \in [d]$, $E_j^1, \ldots, E_j^{n+1}$ are almost surely distinct hence $0 < \hat{\sigma}_j^{\text{oracle}} < \infty$ almost surely. This ensures S_{oracle}^{n+1} is well-defined. To prove this lemma, fix any $c \in \mathbb{R}$ and note that

$$\begin{aligned}
&S_{\text{oracle}}^{n+1} \leq c \\
&\iff E_j^{n+1} \leq c\,\hat{\sigma}_j^{\text{oracle}} + \hat{\mu}_j^{\text{oracle}}, \quad \forall j \in [d] \\
&\iff E_j^{n+1} - c\sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i\right)^2} - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i \leq 0, \quad \forall j \in [d] \\
&\iff \frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) - c\sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i\right)^2} \leq 0, \quad \forall j \in [d] \\
&\iff \frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) - c\sqrt{\frac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2} \leq 0, \quad \forall j \in [d] \\
&\iff \frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) \leq c\sqrt{\frac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2}, \quad \forall j \in [d]
\end{aligned}$$

where the second to last implication follows from Lemma S3. Everything except $\mathbf{E}^{n+1}$ here is observed, so we can solve this inequality for $\mathbf{E}^{n+1}$ coordinate-wise. Fix any $j \in [d]$. The solution of this inequality in E_j^{n+1} depends on the range of c. There are two cases:

1. if $c \geq 0$, then this inequality holds if and only if *either* $E_j^{n+1} < \hat{\mu}_j^{\text{obs}}$ *or* $E_j^{n+1} \geq \hat{\mu}_j^{\text{obs}}$ and
$$\left(\frac{n}{n+1}\right)^2\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 \leq c^2\left(\frac{1}{n+1}(E_j^{n+1} - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2\right).$$

2. if $c < 0$, then this inequality holds if and only if $E_j^{n+1} < \hat{\mu}_j^{\text{obs}}$ *and*
$$\left(\frac{n}{n+1}\right)^2\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 \geq c^2\left(\tfrac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2\right).$$

The solution in both cases relies on solving a quadratic equality in the form $h(x) = 0$, where:

$$h(x) := \left(\frac{n}{n+1}\right)^2 \left(x - \hat{\mu}_j^{\text{obs}}\right)^2 - \tfrac{c^2}{n+1} \left(x - \hat{\mu}_j^{\text{obs}}\right)^2 - c^2(\hat{\sigma}_j^{\text{obs}})^2 = 0.$$

This equation can be written in the standard quadratic form $h(x) = Ax^2 + Bx + K = 0$, where

$$\begin{aligned}
A &= \left(\frac{n}{n+1}\right)^2 - \frac{c^2}{n+1}, \\
B &= -2\hat{\mu}_j^{\text{obs}} \left(\frac{n}{n+1}\right)^2 + 2\hat{\mu}_j^{\text{obs}} \frac{c^2}{n+1}, \\
K &= \left(\frac{n}{n+1}\right)^2 (\hat{\mu}_j^{\text{obs}})^2 - \frac{c^2}{n+1}(\hat{\mu}_j^{\text{obs}})^2 - c^2(\hat{\sigma}_j^{\text{obs}})^2.
\end{aligned}$$

The solution set is:

$$h(x) = 0 \iff x = \begin{cases} \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}} * c\sqrt{\frac{(n+1)^2}{n^2-(n+1)c^2}}, & c^2 < \frac{n^2}{n+1}, \\ \hat{\mu}_j^{\text{obs}}, & c^2 = \frac{n^2}{n+1} \text{ and } \hat{\sigma}_j^{\text{obs}} = 0, \\ \text{no real solution}, & \text{otherwise.} \end{cases}$$

We first note that the event $\hat{\sigma}_j^{\text{obs}} = 0$ occurs if and only if $E_j^1 = \ldots = E_j^n$, which has probability zero by Assumption 1. For the rest, we note that $c^2 < \frac{n^2}{n+1}$ implies $A > 0$, so $h(x) = Ax^2 + Bx + K$ is convex for all $x \geq 0$. Analyzing the sign of $h(x)$ in the two cases shows that $h(E_j^{n+1}) \leq 0$ holds almost surely if and only if

$$\begin{cases} \hat{\mu}_j^{\text{obs}} \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}}|c|\sqrt{\frac{(n+1)^2}{n^2-(n+1)c^2}}, & c \geq 0,\ c^2 < \frac{n^2}{n+1}, \\ E_j^{n+1} < \infty, & c \geq 0,\ c^2 \geq \frac{n^2}{n+1}, \\ E_j^{n+1} < \hat{\mu}_j^{\text{obs}}, & c \geq 0, \\ E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}}|c|\sqrt{\frac{(n+1)^2}{n^2-(n+1)c^2}}, & c < 0,\ c^2 < \frac{n^2}{n+1}, \\ \text{no solution}, & c < 0,\ c^2 \geq \frac{n^2}{n+1}. \end{cases}$$

$$\iff \begin{cases} E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}} * c\sqrt{\frac{(n+1)^2}{n^2-(n+1)c^2}}, & -\frac{n}{\sqrt{n+1}} \leq c < \frac{n}{\sqrt{n+1}}, \\ E_j^{n+1} < \infty, & c \geq \frac{n}{\sqrt{n+1}}, \\ \text{no solution}, & c \leq -\frac{n}{\sqrt{n+1}}. \end{cases}$$

Interpreting the "no-solution" case as $E_j^{n+1} < -\infty$, we obtain precisely the expression for $\omega_j(c)$ given in the statement of the lemma. This notation adjustment is harmless because both $E_j^{n+1} < -\infty$ is a probability-zero event. The monotonicity of $\omega_j(c)$ follows immediately from the construction. ■

Proof [of Lemma 3] From Assumption 1, $E_j^1, \ldots, E_j^{n+1}$ are almost surely distinct for every $j \in [d]$ hence $0 < \hat{\sigma}_j^{\text{oracle}} < \infty$ almost surely for all $j \in [d]$. This ensures S_{oracle}^i is well-defined for all $i \in [n+1]$. Also because of the distinctness assumption, we know from Lemma S5 that

$$\left| \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| < \frac{n}{\sqrt{n+1}},$$

almost surely for all $i \in [n+1]$. Aggregating this inequality for all $j \in [d]$, we obtain

$$|S_{\text{oracle}}^i| = \left| \max_{1 \le j \le d} \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| \le \max_{1 \le j \le d} \left| \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| < \max_{1 \le j \le d} \frac{n}{\sqrt{n+1}} = \frac{n}{\sqrt{n+1}},$$

almost surely for all $i \in [n+1]$. Now, recall that the oracle quantile $\hat{Q}_{1-\alpha}^{\text{oracle}}$ is defined as

$$\hat{Q}_{1-\alpha}^{\text{oracle}} := \lceil (1-\alpha)(n+1) \rceil \text{th smallest value in } \{S_{\text{oracle}}^1, \ldots, S_{\text{oracle}}^n, +\infty\},$$

so $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost surely if $n \ge 1/\alpha - 1$ and $\hat{Q}_{1-\alpha}^{\text{oracle}} = +\infty$ otherwise. ■

Proof [of Lemma 4] Fix any $j \in [d]$ and let

$$g(t_j, z_j) := \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)}.$$

Finding the supremum of this g over $z_j \in (l_j, u_j)$ is a simple univariate optimization problem. Expanding g in full, we get

$$g(t_j, z_j) = \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = \frac{t_j - \frac{1}{n+1}(z_j + n\hat{\mu}_j^{\text{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}}, \tag{S3}$$

where the last equality follows from Lemma S3. Differentiating g with respect to z_j yields

$$g_{z_j}(t_j, z_j) = \frac{-\frac{1}{n+1}\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2} - \frac{\left(t_j - \frac{1}{n+1}(z_j + n\hat{\mu}_j^{\text{obs}})\right)\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}}}{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}.$$

Solving $g_{z_j}(t_j, z_j^*) = 0$ with respect to z_j^* gives:

$$(\hat{\sigma}_j^{\text{obs}})^2 + (t_j - \hat{\mu}_j^{\text{obs}})(z_j^* - \hat{\mu}_j^{\text{obs}}) = 0 \quad \Longrightarrow \quad z_j^* = \hat{\mu}_j^{\text{obs}} - \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}.$$

There are three cases here for any fixed $t_j \in \mathbb{R}$:

- Case 1: if $t_j > \hat{\mu}_j^{\text{obs}}$, then $g(t_j, z_j^*)$ is indeed the global maximum with

$$\sup_{z_j \in (l_j, u_j)} g(t_j, z_j) = g(t_j, z_j^*) = \sqrt{\frac{(t_j - \hat{\mu}_j^{\text{obs}})^2}{(\hat{\sigma}_j^{\text{obs}})^2} + \frac{1}{n+1}}.$$

- Case 2: if $t_j > \hat{\mu}_j^{\text{obs}}$, then $g(t_j, z_j^*)$ is the global minimum, then the supremum must occur at one of the endpoints due to the fact that g_{z_j} has only one critical point. Hence,

$$\sup_{z_j \in (l_j, u_j)} g(t_j, z_j) = \max \left\{ \lim_{z_j \to l_j} g(t_j, z_j), \quad \lim_{z_j \to u_j} g(t_j, z_j) \right\}.$$

- Case 3: if $t_j = \hat{\mu}_j^{\text{obs}}$, then the numerator of g_{z_j} reduces to $-\frac{1}{n+1}(\hat{\sigma}_j^{\text{obs}})^2 < 0$ while its denominator stays strictly positive. This implies that g is monotonically decreasing thus we take

$$\sup_{z_j \in (l_j, u_j)} g(\hat{\mu}_j^{\text{obs}}, z_j) = \lim_{z_j \to l_j} g(\hat{\mu}_j^{\text{obs}}, z_j).$$

Collecting all candidates, we obtain the closed form:

$$\sup_{z_j \in (l_j, u_j)} g(t_j, z_j) = \max \left\{ \lim_{z_j \to l_j} g(t_j, z_j), \sqrt{\frac{(t_j - \hat{\mu}_j^{\text{obs}})^2}{(\hat{\sigma}_j^{\text{obs}})^2} + \frac{1}{n+1}}, \lim_{z_j \to u_j} g(t_j, z_j) \right\}.$$

The fact that for any t_j fixed,

$$\lim_{z_j \to -\infty} g(t_j, z_j) = \frac{1}{\sqrt{n+1}}, \quad \lim_{z_j \to +\infty} g(t_j, z_j) = -\frac{1}{\sqrt{n+1}}$$

is just standard calculus by knowing (S3). ■

Proof [of Lemma 6] For every $i \in [n]$ and $j \in [d]$, coordinatewise distinctness and $n \geq 3$ give

$$\frac{(E_j^i - \hat{\mu}_j^{\text{obs}})^2}{(\hat{\sigma}_j^{\text{obs}})^2} < n - 1.$$

Equality in the usual non-strict bound would require all other $n-1$ calibration residuals in that coordinate to be equal. By (**??**) and the Cauchy–Schwarz inequality, for every $z \in \mathbb{R}$,

$$\left| \frac{E_j^i - \hat{\mu}_j(z)}{\hat{\sigma}_j(z)} \right| \leq \sqrt{\frac{(E_j^i - \hat{\mu}_j^{\text{obs}})^2}{(\hat{\sigma}_j^{\text{obs}})^2} + \frac{1}{n+1}} < T.$$

The bound is uniform in z, so all calibration scores $S_{\mathcal{R}}^i$ lie strictly between $-T$ and T, simultaneously for every nonempty $\mathcal{R}$. The sample-size condition also ensures that $\lceil (1-\alpha)(n+1) \rceil \leq n$, so the quantile selects one of these scores. ■

Proof [of Proposition **??**] Throughout this proof we write $\tilde{C}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ for the estimated oracle prediction set defined in (11). Let $\mathcal{P}$ be any (random) partition of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$, so that $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) = \bigsqcup_{\mathcal{R} \in \mathcal{P}} \mathcal{R}$ almost surely. Because $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \supseteq \tilde{C}(\mathbf{X}^{n+1})$ almost surely, we know

$$\begin{aligned}
\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) &\iff \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}), \mathbf{Y}^{n+1} \in \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \\
&\iff \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}), \mathbf{Y}^{n+1} \in \mathcal{R} \\
&\iff \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) \cap \mathcal{R}.
\end{aligned}$$

If $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ satisfies the containment on the right-hand side of (**??**),

$$\mathcal{R} \supseteq \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \supseteq \tilde{C}(\mathbf{X}^{n+1}) \cap \mathcal{R},$$

then

$$\begin{aligned}
\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) &\implies \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \\
&\iff \mathbf{Y}^{n+1} \in \bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \\
&\implies \mathbf{Y}^{n+1} \in \text{Rect}_{\mathcal{P}}\left(\bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \right) = \hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}),
\end{aligned}$$

where the last implication holds because $\text{Rect}_{\mathcal{P}}(A) \supseteq A$ by definition. Since we already know that $\mathbb{P}[\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1})] \geq 1 - \alpha$, by Theorem 1 and the dual relation (12), this proves that

$$\mathbb{P}\left[\mathbf{Y}^{n+1} \in \hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})\right] \geq 1 - \alpha.$$

Finally, the containment $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ almost surely is immediate from the definition of $\text{Rect}_{\mathcal{P}}$ in (**??**), because

$$\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \subseteq \mathcal{R} \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}),$$

for every $\mathcal{R} \in \mathcal{P}$. ■

Proof [of Lemma **??**] Fix any $\mathbf{h} \in [n+1]^d$. Suppose $\mathcal{Y}^{\mathbf{h}} \neq \emptyset$. Then $\mathcal{E}^{\mathbf{h}} \neq \emptyset$ and, by (**??**),

$$\mathcal{E}^{\mathbf{h}} = \bigtimes_{j=1}^{d} \mathcal{E}_j^{\mathbf{h}}, \quad \text{where} \quad \mathcal{E}_j^{\mathbf{h}} := \left[E_j^{(h_j-1)}, U_j^{h_j}\right) \neq \emptyset, \quad \forall j \in [d].$$

Fix any $\mathbf{t} \in \mathbb{R}^d_+$. Similar to the proof of Lemma **??**, we have to solve a univariate optimization problem. We first notice that for any $j \in [d]$,

$$\sup_{\mathbf{z} \in \mathcal{E}^{\mathbf{h}}} \frac{t_j}{\hat{\sigma}_j(z_j)} = \sup_{z_j \in \mathcal{E}_j^{\mathbf{h}}} \frac{t_j}{\hat{\sigma}_j(z_j)} = \frac{t_j}{\inf_{z_j \in \mathcal{E}_j^{\mathbf{h}}} \hat{\sigma}_j(z_j)}.$$

By Lemma S3, we know that

$$\hat{\sigma}_j(z_j) = \sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2},$$

which is continuous in z_j, so the infimum above is attained on the closure $\overline{\mathcal{E}_j^{\mathbf{h}}} = \left[E_j^{(h_j-1)}, U_j^{h_j}\right]$; clearly $\hat{\sigma}_j(z_j)$ is minimized there at

$$z_j^* = \operatorname*{argmin}_{z_j \in \overline{\mathcal{E}_j^{\mathbf{h}}}} |z_j - \hat{\mu}_j^{\text{obs}}| = \begin{cases} \hat{\mu}_j^{\text{obs}}, & \text{if } \hat{\mu}_j^{\text{obs}} \in \overline{\mathcal{E}_j^{\mathbf{h}}}, \\ E_j^{(h_j-1)}, & \text{if } \hat{\mu}_j^{\text{obs}} < E_j^{(h_j-1)}, \\ U_j^{h_j}, & \text{if } \hat{\mu}_j^{\text{obs}} > U_j^{h_j}. \end{cases}$$

Hence, $r_j(h_j) = \hat{\sigma}_j(z_j^*)$ is exactly the infimum proposed. For the second part, denote

$$g(z_j) := \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = \frac{\frac{1}{n+1}(z_j + n\hat{\mu}_j^{\text{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}} = \frac{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}}) + \hat{\mu}_j^{\text{obs}}}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}},$$

and consider

$$\inf_{\mathbf{z} \in \mathcal{E}^{\text{gwc}}} g(z_j) = \inf_{z_j \in [0, \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})]} g(z_j).$$

Differentiating $g(z_j)$ in z_j, we get

$$g'(z_j) = \frac{\frac{1}{n+1}(\hat{\sigma}_j^{\text{obs}})^2 - \frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})\hat{\mu}_j^{\text{obs}}}{(\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2)\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}}.$$

Setting this to zero gives the critical point

$$z_j^* = \hat{\mu}_j^{\text{obs}} + \frac{(\hat{\sigma}_j^{\text{obs}})^2}{\hat{\mu}_j^{\text{obs}}} > 0.$$

For $\epsilon > 0$, we have

$$\begin{aligned}
\text{sign}(g'(z_j^* + \epsilon)) &= \text{sign}\left(-\tfrac{1}{n+1}\epsilon\hat{\mu}_j^{\text{obs}}\right) = -1, \\
\text{sign}(g'(z_j^* - \epsilon)) &= \text{sign}\left(\tfrac{1}{n+1}\epsilon\hat{\mu}_j^{\text{obs}}\right) = +1.
\end{aligned}$$

This implies that $g(z_j^*)$ is always a local maximum. Hence, if $z_j^* \in [0, \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})]$, then

$$\inf_{z_j \in [0, \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})]} g(z_j) = \min\left\{ g(0), \lim_{z_j \uparrow \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})} g(z_j) \right\};$$

if $z_j^* > \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})$, then g is nondecreasing on $[0, \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})]$ and

$$\inf_{z_j \in [0, \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})]} g(z_j) = g(0).$$

But in either case, we only need to compare the boundaries in order to find the infimum. Therefore, it follows that

$$\inf_{z_j \in [0, \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})]} g(z_j) = \min\left\{ g(0), \lim_{z_j \uparrow \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})} g(z_j) \right\}.$$

This is true for every $j \in [d]$, so we must have

$$\begin{aligned}
\hat{\Phi}^{\text{lwc}}_{\mathbf{h}}(\mathbf{t}) &= \max_{1 \le j \le d} \left\{ \sup_{\mathbf{z} \in \mathcal{E}^{\mathbf{h}}} \frac{t_j}{\hat{\sigma}_j(z_j)} - \inf_{\mathbf{z} \in \mathcal{E}^{\text{gwc}}} g(z_j) \right\} \\
&= \max_{1 \le j \le d} \left\{ \frac{t_j}{r_j(h_j)} - \min\left\{ \frac{\hat{\mu}_j(0)}{\hat{\sigma}_j(0)}, \lim_{z_j \uparrow \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})} \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \right\} \right\}.
\end{aligned}$$

■

Proof [of Lemma **??**] Fix any $j \in [d]$ and suppose $\mathcal{Y}^{\mathbf{h}^*} \neq \emptyset$. There are two parts to this lemma, which we prove separately. Throughout, for any $k \in [n+1]$ we let $\mathbf{h}^*(j,k)$ denote the unique multi-index in $\mathrm{Row}_j(\mathbf{h}^*)$ whose j-th coordinate equals k; that is, $h^*_j(j,k) = k$ and $h^*_l(j,k) = h^*_l$ for all $l \neq j$.

Part I. We aim to prove

$$\sup_{\mathbf{h}\in[n+1]^d} B_j^{\mathbf{h}} = \sup_{\mathbf{h}\in\mathrm{Row}_j(\mathbf{h}^*)} B_j^{\mathbf{h}} = B_j^{\mathbf{h}^{[j]}}. \tag{S4}$$

We first notice that the first supremum can be decomposed into suprema over surfaces:

$$\sup_{\mathbf{h}\in[n+1]^d} B_j^{\mathbf{h}} = \sup_{k\in[n+1]} \sup_{\mathbf{h}\in\mathrm{Surface}_j(k)} B_j^{\mathbf{h}},$$

where

$$\mathrm{Surface}_j(k) := \left\{\mathbf{h} \in [n+1]^d : h_j = k\right\}.$$

Fix any $k \in [n+1]$ and note that $\mathbf{h}^*(j,k) \in \mathrm{Surface}_j(k) \cap \mathrm{Row}_j(\mathbf{h}^*)$. Since this holds for every $k \in [n+1]$, to prove (S4) the main challenge is to show that:

$$\sup_{\mathbf{h}\in\mathrm{Surface}_j(k)} B_j^{\mathbf{h}} = B_j^{\mathbf{h}^*(j,k)}. \tag{S5}$$

To prove (S5), recall from the definition in (**??**) and Lemma **??** that

$$\hat{\Phi}^{\mathrm{lwc}}_{\mathbf{h}}(\mathbf{t}) = \max_{1\leq l\leq d}\left\{\frac{t_l}{r_l(h_l)} - c_l\right\},$$

where

$$r_l(h_l) = \min\left\{\hat{\sigma}_l(z_l) : z_l \in \overline{\mathcal{E}_l^{\mathbf{h}}}\right\}, \quad \overline{\mathcal{E}_l^{\mathbf{h}}} := \left[E_l^{(h_l-1)}, U_l^{h_l}\right], \qquad c_l = \inf_{\mathbf{z}\in\mathcal{E}^{\mathrm{gwc}}} \frac{\hat{\mu}_l(z_l)}{\hat{\sigma}_l(z_l)}.$$

Without loss of generality, assume $\mathcal{Y}^{\mathbf{h}} \neq \emptyset$ for all $\mathbf{h} \in \mathrm{Surface}_j(k)$. This does not affect the final result because $\mathcal{Y}^{\mathbf{h}} = \emptyset$ leads to $B_j^{\mathbf{h}} = 0$, which can be safely excluded from the supremum calculation. Then notice that, for all $\mathbf{h} \in \mathrm{Surface}_j(k)$, we have $r_j(h_j) = r_j(k)$ and $r_l(h_l) \geq \hat{\sigma}_l^{\mathrm{obs}} = r_l(h^*_l)$ for all $l \neq j$, the latter because $\hat{\mu}_l^{\mathrm{obs}} \in \overline{\mathcal{E}_l^{\mathbf{h}^*}}$ by the definition of the mean index $\mathbf{h}^*$. Therefore, for any fixed $\mathbf{t}$,

$$\hat{\Phi}^{\mathrm{lwc}}_{\mathbf{h}}(\mathbf{t}) = \max_{1\leq l\leq d}\left\{\frac{t_l}{r_l(h_l)} - c_l\right\} \leq \max\left(\max_{l\neq j}\left\{\frac{t_l}{\hat{\sigma}_l^{\mathrm{obs}}} - c_l\right\}, \frac{t_j}{r_j(k)} - c_j\right) = \hat{\Phi}^{\mathrm{lwc}}_{\mathbf{h}^*(j,k)}(\mathbf{t}).$$

Because this holds pointwise for all $\mathbf{t}$, we can extend the inequality to the corresponding conformal quantile and conclude that, almost surely for all $\mathbf{h} \in \mathrm{Surface}_j(k)$,

$$\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}}) \leq \hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,k)})$$

and therefore, by the monotonicity of the function ω_j from Lemma 2,

$$\omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}})) \leq \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,k)})).$$

Therefore, it follows that, for all $\mathbf{h} \in \mathrm{Surface}_j(k)$,

$$\begin{aligned} B_j^{\mathbf{h}} &= \begin{cases} \min\left\{U_j^k, \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}}))\right\}, & \text{if } U_j^k,\ \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}})) > E_j^{(k-1)}, \\ 0, & \text{otherwise,} \end{cases} \\ &\leq \begin{cases} \min\left\{U_j^k, \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,k)}))\right\}, & \text{if } U_j^k,\ \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,k)})) > E_j^{(k-1)}, \\ 0, & \text{otherwise,} \end{cases} \\ &= B_j^{\mathbf{h}^*(j,k)}. \end{aligned}$$

This completes the proof of (S5).

The second equality in (S4) is trivial. We can easily see that for any $k \in [n]$, if $B_j^{\mathbf{h}^*(j,k+1)} \neq 0$, then

$$U_j^{k+1},\ \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,k+1)})) \geq E_j^{(k)} \geq \min\left\{E_j^{(k)}, \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})\right\} = U_j^k.$$

Hence, it follows that $B_j^{\mathbf{h}^*(j,k+1)} \geq B_j^{\mathbf{h}^*(j,k)}$ regardless of whether $B_j^{\mathbf{h}^*(j,k)} = 0$ or not. Then finding the supremum among all $k \in [n+1]$ simplifies to finding the rightmost nonzero $B_j^{\mathbf{h}^*(j,k)}$, which is exactly how $B_j^{\mathbf{h}^{[j]}}$ is defined.

Part II. In this part, we aim to show that if $B_j^{\mathbf{h}^*(j,m_j)} = 0$ for some $m_j \geq h_j^*$, then

$$B_j^{\mathbf{h}^*(j,N)} = 0, \quad \forall N \geq m_j.$$

There are two cases that could lead to $B_j^{\mathbf{h}^*(j,m_j)} = 0$:

- If $U_j^{m_j} = \min\left\{E_j^{(m_j)}, \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})\right\} \leq E_j^{(m_j-1)}$, then $\omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}) \leq E_j^{(m_j-1)} \leq E_j^{(N-1)}$ for any $N \geq m_j$ by the nature of order statistics. Hence, $U_j^N \leq E_j^{(N-1)}$ for all $N \geq m_j$, which further implies $B_j^{\mathbf{h}^*(j,N)} = 0$ for all $N \geq m_j$.

- If $\omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,m_j)})) \leq E_j^{(m_j-1)}$, then we claim that

  $$\omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,N)})) \leq \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,m_j)})) \leq E_j^{(m_j-1)} \leq E_j^{(N-1)},$$

  which then implies $B_j^{\mathbf{h}^*(j,N)} = 0$ for all $N \geq m_j$. This is actually trivial because $m_j \geq h_j^*$ implies that both $\overline{\mathcal{E}_j^{\mathbf{h}^*(j,m_j)}}$ and $\overline{\mathcal{E}_j^{\mathbf{h}^*(j,N)}}$ lie to the right of $\hat{\mu}_j^{\mathrm{obs}}$, with the latter no closer to it than the former, so that

  $$r_j(m_j) = \min_{z_j \in \overline{\mathcal{E}_j^{\mathbf{h}^*(j,m_j)}}} \hat{\sigma}_j(z_j) \leq \min_{z_j \in \overline{\mathcal{E}_j^{\mathbf{h}^*(j,N)}}} \hat{\sigma}_j(z_j) = r_j(N),$$

by Lemma S4 and the definition of $\mathcal{E}_j^{\mathbf{h}}$. This then leads to

$$\begin{aligned}
&\hat{\Phi}^{\text{lwc}}_{\mathbf{h}^*(j,m_j)}(\mathbf{t}) \geq \hat{\Phi}^{\text{lwc}}_{\mathbf{h}^*(j,N)}(\mathbf{t}), \quad \forall \mathbf{t} \\
&\implies \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,N)}) \leq \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,m_j)}) \\
&\implies \omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,N)})) \leq \omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,m_j)})).
\end{aligned}$$

■

Appendix D. Additional Numerical Results

We examine how coordinate heterogeneity, dependence, calibration size, noise tails, and residual transformations affect the coverage and efficiency of TSCP. Comparisons with the population oracle and the unenclosed local union separate the statistical and computational effects of its approximations. Real-data marginal coverage and search-regime measurements complement the results in Section 4.

D.1 Additional Noise-Distribution Experiments

Tables S3–S7 report coverage and normalized residual-space volume for five noise laws. Gaussian errors contrast unequal and equal coordinate scales; Laplace, mixture, and Gamma errors vary the tail shape and symmetry under heterogeneous scaling. The Laplace distributions are centered at zero, with the stated scale parameter; the Gamma parameterization specifies shape followed by scale.

Gaussian Noise (Heterogeneous)

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Gaussian)	Emp. Copula	0.714 (0.079)	1.68e+10(1.16e+10)
	Point CHR	0.933 (0.066)	4.04e+12(1.38e+13)
	TSCP (Ours)	0.910 (0.053)	**1.83e+11(2.74e+11)**
	Unscaled Max	0.908 (0.047)	2.22e+13(4.77e+13)
(50, 10, Gaussian)	Emp. Copula	0.816 (0.057)	3.64e+10(2.22e+10)
	Point CHR	0.919 (0.057)	4.85e+11(1.72e+12)
	TSCP (Ours)	0.904 (0.047)	**9.42e+10(8.82e+10)**
	Unscaled Max	0.897 (0.043)	1.16e+13(1.10e+13)
(100, 10, Gaussian)	Emp. Copula	0.819 (0.039)	2.58e+10(9.90e+09)
	Point CHR	0.901 (0.041)	9.74e+10(9.22e+10)
	TSCP (Ours)	0.903 (0.034)	**6.59e+10(3.43e+10)**
	Unscaled Max	0.908 (0.027)	1.09e+13(6.93e+12)
(300, 10, Gaussian)	Emp. Copula	0.872 (0.019)	3.73e+10(8.11e+09)
	Point CHR	0.899 (0.025)	5.60e+10(2.05e+10)
	TSCP (Ours)	0.899 (0.018)	**4.93e+10(1.17e+10)**
	Unscaled Max	0.901 (0.020)	8.68e+12(3.29e+12)
(500, 10, Gaussian)	Emp. Copula	0.886 (0.018)	4.16e+10(8.27e+09)
	Point CHR	0.900 (0.022)	5.13e+10(1.49e+10)
	TSCP (Ours)	0.901 (0.016)	**4.81e+10(9.67e+09)**
	Unscaled Max	0.903 (0.016)	8.54e+12(2.46e+12)

Table S3: Performance with $d = 10$ outcome variables, as in Figure 4. Results are averaged over 200 repetitions, with one standard deviation in parentheses. The target coverage is 0.9; estimates below this target are highlighted in red. Bold values indicate the smallest mean residual-space volume among TSCP, Point CHR, and Unscaled Max. TSCP has the smallest volume among these methods in the displayed settings.

Gaussian Noise (Homogeneous)

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Unit Gaussian)	Emp. Copula	0.714 (0.079)	4.64e+03(3.20e+03)
	Point CHR	0.933 (0.066)	1.11e+06(3.80e+06)
	TSCP (Ours)	0.910 (0.053)	5.04e+04(7.56e+04)
	Unscaled Max	0.904 (0.054)	**2.19e+04(1.94e+04)**
(50, 10, Unit Gaussian)	Emp. Copula	0.816 (0.057)	1.00e+04(6.12e+03)
	Point CHR	0.919 (0.057)	1.34e+05(4.75e+05)
	TSCP (Ours)	0.904 (0.047)	2.60e+04(2.43e+04)
	Unscaled Max	0.901 (0.045)	**1.68e+04(1.15e+04)**
(100, 10, Unit Gaussian)	Emp. Copula	0.819 (0.039)	7.11e+03(2.73e+03)
	Point CHR	0.901 (0.041)	2.68e+04(2.54e+04)
	TSCP (Ours)	0.903 (0.034)	1.82e+04(9.46e+03)
	Unscaled Max	0.900 (0.035)	**1.50e+04(7.38e+03)**
(300, 10, Unit Gaussian)	Emp. Copula	0.872 (0.019)	1.03e+04(2.23e+03)
	Point CHR	0.899 (0.025)	1.54e+04(5.66e+03)
	TSCP (Ours)	0.899 (0.018)	1.36e+04(3.23e+03)
	Unscaled Max	0.898 (0.018)	**1.28e+04(3.07e+03)**
(500, 10, Unit Gaussian)	Emp. Copula	0.886 (0.018)	1.15e+04(2.28e+03)
	Point CHR	0.900 (0.022)	1.41e+04(4.11e+03)
	TSCP (Ours)	0.901 (0.016)	1.33e+04(2.67e+03)
	Unscaled Max	0.899 (0.017)	**1.26e+04(2.50e+03)**

Table S4: Performance with $d = 10$ outcome variables and homogeneous Gaussian noise, as in Figure 5. Other details are as in Table S3. Unscaled Max has the smallest volume among methods with near-nominal coverage, while TSCP remains competitive.

Laplace Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Laplace)	Emp. Copula	0.712 (0.079)	3.82e+12(4.91e+12)
	Point CHR	0.938 (0.060)	5.57e+16(2.71e+17)
	TSCP (Ours)	0.908 (0.060)	**3.14e+14(9.93e+14)**
	Unscaled Max	0.897 (0.053)	6.36e+15(1.99e+16)
(50, 10, Laplace)	Emp. Copula	0.819 (0.056)	1.36e+13(1.75e+13)
	Point CHR	0.917 (0.050)	6.64e+14(1.81e+15)
	TSCP (Ours)	0.906 (0.045)	**7.32e+13(1.60e+14)**
	Unscaled Max	0.901 (0.041)	3.34e+15(5.12e+15)
(100, 10, Laplace)	Emp. Copula	0.811 (0.039)	5.96e+12(3.94e+12)
	Point CHR	0.896 (0.039)	4.74e+13(6.07e+13)
	TSCP (Ours)	0.898 (0.030)	**2.64e+13(1.94e+13)**
	Unscaled Max	0.895 (0.029)	1.77e+15(1.78e+15)
(300, 10, Laplace)	Emp. Copula	0.871 (0.020)	1.11e+13(4.03e+12)
	Point CHR	0.895 (0.023)	2.01e+13(1.12e+13)
	TSCP (Ours)	0.896 (0.017)	**1.74e+13(6.40e+12)**
	Unscaled Max	0.896 (0.017)	1.38e+15(7.08e+14)
(500, 10, Laplace)	Emp. Copula	0.882 (0.016)	1.29e+13(4.10e+12)
	Point CHR	0.896 (0.021)	1.83e+13(8.95e+12)
	TSCP (Ours)	0.896 (0.016)	**1.64e+13(5.68e+12)**
	Unscaled Max	0.897 (0.017)	1.36e+15(5.51e+14)

Table S5: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Laplace* distribution; i.e., $\epsilon_j \sim \text{Laplace}(10 - j + 1)$ for all $j \in [10]$. Other details are as in Table S3.

Mixture Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Mixed)	Emp. Copula	0.715 (0.075)	2.69e+11(3.04e+11)
	Point CHR	0.935 (0.062)	2.56e+15(1.98e+16)
	TSCP (Ours)	0.913 (0.051)	**9.53e+12(3.12e+13)**
	Unscaled Max	0.903 (0.052)	3.46e+13(6.74e+13)
(50, 10, Mixed)	Emp. Copula	0.821 (0.058)	7.07e+11(5.96e+11)
	Point CHR	0.925 (0.048)	6.10e+13(3.06e+14)
	TSCP (Ours)	0.906 (0.044)	**2.57e+12(3.91e+12)**
	Unscaled Max	0.901 (0.044)	1.98e+13(1.68e+13)
(100, 10, Mixed)	Emp. Copula	0.819 (0.035)	3.82e+11(1.98e+11)
	Point CHR	0.898 (0.042)	2.76e+12(4.47e+12)
	TSCP (Ours)	0.900 (0.029)	**1.23e+12(7.54e+11)**
	Unscaled Max	0.905 (0.028)	1.65e+13(9.89e+12)
(300, 10, Mixed)	Emp. Copula	0.874 (0.022)	6.47e+11(2.25e+11)
	Point CHR	0.902 (0.025)	1.27e+12(7.23e+11)
	TSCP (Ours)	0.900 (0.019)	**9.05e+11(3.48e+11)**
	Unscaled Max	0.900 (0.020)	1.36e+13(5.22e+12)
(500, 10, Mixed)	Emp. Copula	0.886 (0.016)	7.26e+11(1.82e+11)
	Point CHR	0.897 (0.020)	1.01e+12(5.42e+11)
	TSCP (Ours)	0.898 (0.014)	**8.17e+11(2.05e+11)**
	Unscaled Max	0.902 (0.016)	1.35e+13(3.73e+12)

Table S6: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Mixture* distribution; i.e., $\epsilon_j \sim \frac{1}{2}\text{Laplace}(10 - j + 1) + \frac{1}{2}\mathcal{N}(0, (10 - j + 1)^2)$ for all $j \in [10]$. Other details are as in Table S3.

Gamma Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Gamma)	Emp. Copula	0.714 (0.067)	5.33e-02(6.85e-02)
	Point CHR	0.929 (0.058)	7.55e+02(4.32e+03)
	TSCP (Ours)	0.904 (0.056)	**3.77e+00(1.59e+01)**
	Unscaled Max	0.898 (0.051)	2.69e+15(6.18e+15)
(50, 10, Gamma)	Emp. Copula	0.826 (0.048)	2.20e-01(2.78e-01)
	Point CHR	0.927 (0.049)	5.01e+01(3.36e+02)
	TSCP (Ours)	0.910 (0.041)	**1.15e+00(2.56e+00)**
	Unscaled Max	0.902 (0.039)	1.94e+15(3.54e+15)
(100, 10, Gamma)	Emp. Copula	0.818 (0.038)	1.04e-01(1.01e-01)
	Point CHR	0.906 (0.038)	1.28e+00(2.38e+00)
	TSCP (Ours)	0.900 (0.030)	**4.65e-01(5.82e-01)**
	Unscaled Max	0.897 (0.030)	9.67e+14(7.86e+14)
(300, 10, Gamma)	Emp. Copula	0.873 (0.019)	1.75e-01(1.15e-01)
	Point CHR	0.895 (0.024)	3.74e-01(3.70e-01)
	TSCP (Ours)	0.898 (0.018)	**2.67e-01(1.69e-01)**
	Unscaled Max	0.895 (0.018)	7.04e+14(3.45e+14)
(500, 10, Gamma)	Emp. Copula	0.885 (0.015)	1.89e-01(1.06e-01)
	Point CHR	0.896 (0.020)	3.20e-01(2.33e-01)
	TSCP (Ours)	0.897 (0.016)	**2.36e-01(1.35e-01)**
	Unscaled Max	0.893 (0.015)	6.37e+14(2.53e+14)

Table S7: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Gamma* distribution; i.e., $\epsilon_j \sim \text{Gamma}(1, 10 - j + 1)$ for all $j \in [10]$. Other details are as in Table S3.

D.2 Dependence Strength

We vary the equicorrelation of the Gaussian noise over $\rho \in \{0, 0.3, 0.6, 0.9\}$ while retaining marginal standard deviations $(10, 9, \ldots, 1)$ and a calibration size of $n = 100$. Figure S2 shows that TSCP's empirical joint coverage is stable across different values of ρ. Its average residual-space volume decreases from 6.37×10^{10} at $\rho = 0$ to 9.47×10^{9} at $\rho = 0.9$. Point CHR and Unscaled Max also remain close to the target. Emp. Copula improves as the correlation strengthens but remains below the nominal level throughout this test. The experiment therefore does not indicate a coverage failure of TSCP under strong equicorrelation, while also showing that our method adapts well to varying dependence structures.

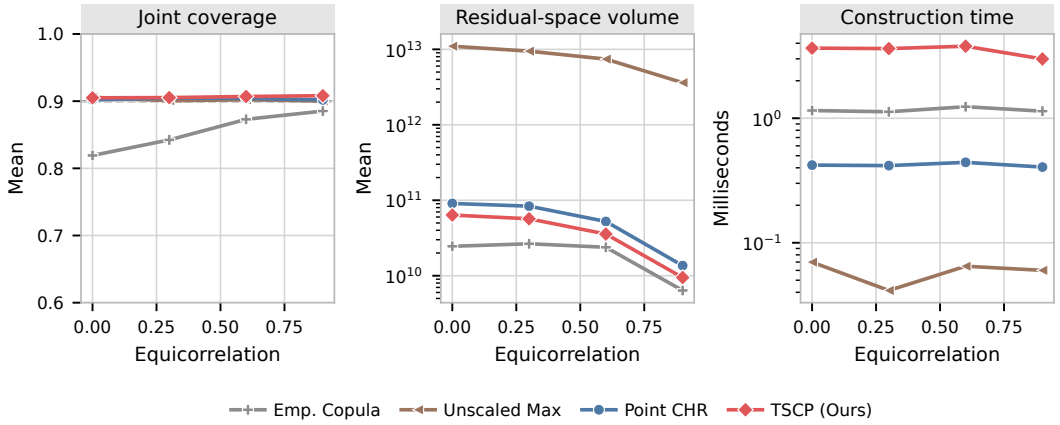


Figure S2: Sensitivity to equicorrelated Gaussian noise at $d = 10$ and $n = 100$. The three panels report joint coverage, mean residual-space volume, and mean construction time. Results average 200 repetitions; the dashed line marks the target coverage.

D.3 Severity of Coordinate Heterogeneity

To isolate heterogeneity across outcomes, we use independent Gaussian noise and choose ten coordinate scales geometrically spaced between r and 1, where $r \in \{1, 2, 5, 10, 20\}$. The calibration size is fixed at $n = 100$. Figure S3 plots joint coverage and mean residual-space volume against the scale ratio r and shows that the coverage of TSCP, Point CHR, and Unscaled Max is essentially unchanged as r grows. At $r = 1$, Unscaled Max produces slightly smaller volumes than TSCP, consistent with the unit-scale comparison in Figure 5. As r grows, Unscaled Max's volume increases more steeply than the volumes of the coordinate-adaptive methods. This widening gap is the expected benefit of coordinate adaptation: a single maximum-residual threshold becomes increasingly inefficient when one scale must be used for all coordinates.

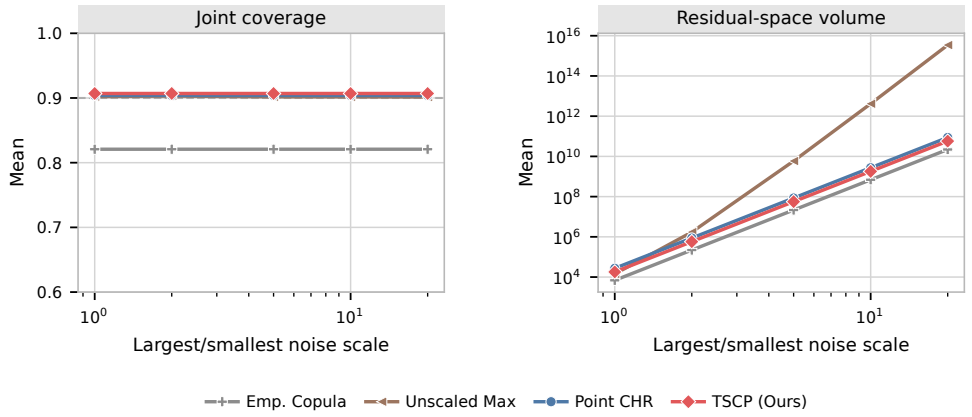


Figure S3: Sensitivity to the ratio between the largest and smallest outcome noise scales, with independent Gaussian errors and $n = 100$. The scales are geometrically spaced, and both axes of the volume panel are logarithmic. Results average 200 repetitions.

D.4 Sensitivity to the Target Coverage Level

Figure S4 repeats the dependent Gaussian experiment at nominal coverage levels 0.8, 0.9, and 0.95, with $n = 100$ and $\rho = 0.6$. TSCP achieves empirical coverages 0.811, 0.907, and 0.954. As expected, increasing the target level increases all region volumes. The relative ordering is stable: TSCP is smaller than Point CHR and dramatically smaller than Unscaled Max, while Emp. Copula is smaller but below the nominal coverage at every displayed target.

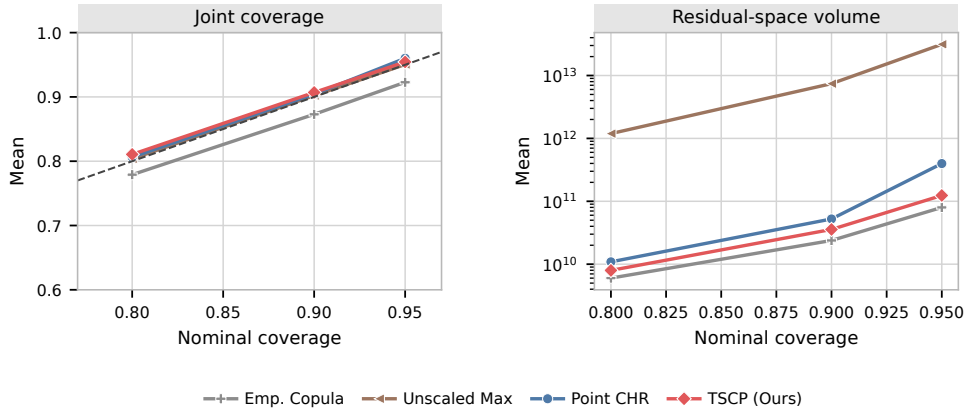


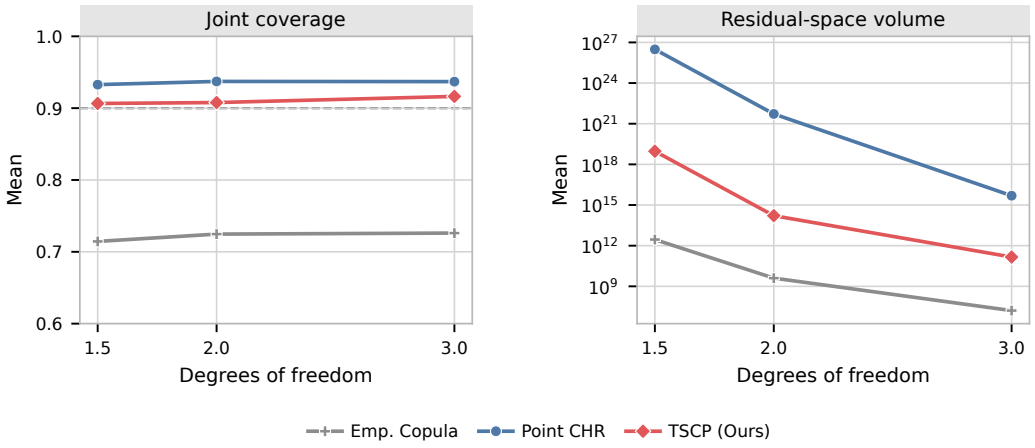
Figure S4: Sensitivity to the nominal coverage level in the $d = 10$ equicorrelated Gaussian experiment with $n = 100$ and $\rho = 0.6$. The dashed diagonal in the left panel is exact nominal calibration. Results average 200 repetitions.

D.5 Heavy-Tailed Noise

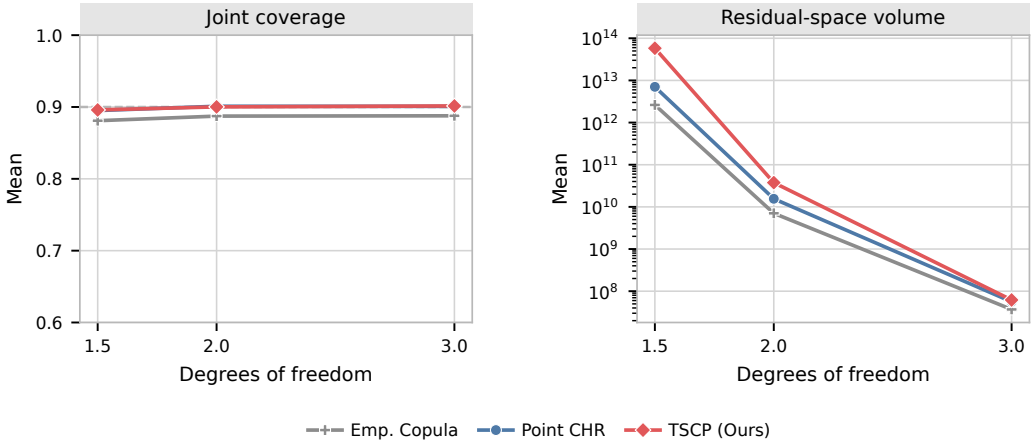
We examine independent Student t noise with $d = 10$, a common unit scale across coordinates, and degrees of freedom $\nu \in \{1.5, 2, 3\}$. The cases $\nu = 1.5$ and $\nu = 2$ have infinite variance

and therefore lie outside Assumption 1, which is invoked by the validity theorem as well as the moment-based motivation; $\nu = 3$ has finite variance but remains heavy-tailed.

Figure S5 presents the results averaged over 200 repetitions. Coverage remains near the target while interval volumes grow substantially as the degrees of freedom decrease. Point CHR has larger volume at the smaller calibration size but can outperform TSCP at the larger size.



(a) Calibration size $n = 30$.



(b) Calibration size $n = 500$.

Figure S5: Performance under Student t noise with varying degrees of freedom, using calibration sizes (a) $n = 30$ and (b) $n = 500$. The target coverage is 0.9. TSCP has near-nominal empirical coverage in these settings; the infinite-variance cases are outside the stated assumptions. Point CHR produces larger volumes at the smaller calibration size but can outperform TSCP at the larger size. Downward triangles at the coverage-axis boundary indicate values below 0.6.

D.6 Exchangeable Contamination Stress Test

To test sensitivity to atypically large residuals, we contaminate the Gaussian experiment at rates $\varepsilon \in \{0, 0.01, 0.05, 0.10\}$. Independently for each observation, all ten noise coordinates are multiplied by 10 with probability ε; the same exchangeable mechanism is used for training, calibration, and test data. The calibration size is $n = 100$. We report median rather than mean volume because the contaminated distribution produces a small number of extreme regions.

Figure S6 shows that TSCP coverage stays near nominal regardless of contamination, but its median volume grows from 5.38×10^{10} to 1.10×10^{17}. Point CHR and Unscaled Max exhibit the same qualitative inflation, and Emp. Copula undercovers at every contamination level. Thus, the conformal calibration is comparatively stable in coverage, whereas the sample mean and standard deviation used for coordinate scaling are not robust efficiency summaries. Replacing them by robust estimators would change the construction and its analysis and requires a separate validity argument.

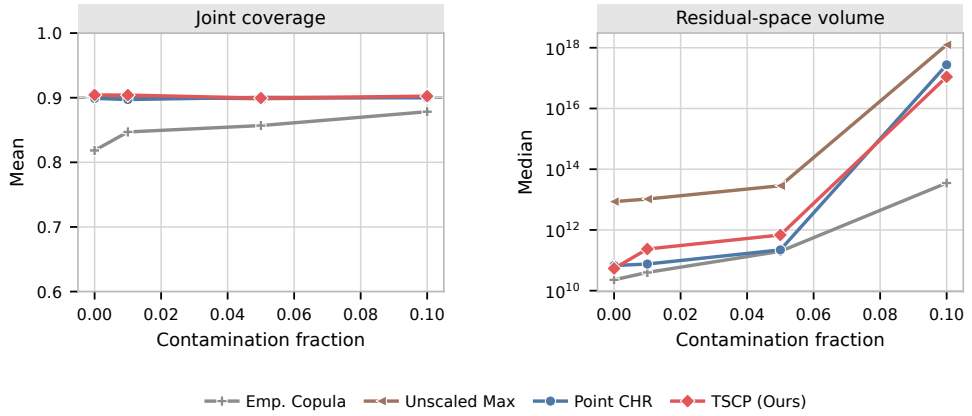


Figure S6: Row-wise contamination stress test with $d = 10$ and $n = 100$. A contaminated observation has every noise coordinate multiplied by 10. The right panel reports median residual-space volume over 200 repetitions on a logarithmic scale.

D.7 Additional CQR Sensitivity Analyses

Base-interval width. The main CQR comparison uses base-interval miscoverage 0.5. We repeat it at 0.3, 0.5, 0.7, and 0.9, fixing $n = 100$ and final miscoverage 0.1. Figure S7 shows that TSCP with capped residuals and Unscaled Max behave similarly, while Emp. Copula undercovers. CQHR remains close to nominal coverage but its volume is highly sensitive to the fitted base interval in this design.

Shifted scores. An alternative transformation replaces the signed CQR residuals E_j by $E_j + C$, apply TSCP, and subtract C from the final coordinate adjustment. We test $C \in \{100, 300, 1000\}$. As shown in Figure S8, the recorded TSCP rectangle is numerically invariant to C and, more importantly, agrees exactly with the TSCP-GWC rectangle: the maximum coordinate-length difference is zero in all 30 repetitions for every C. The proposed

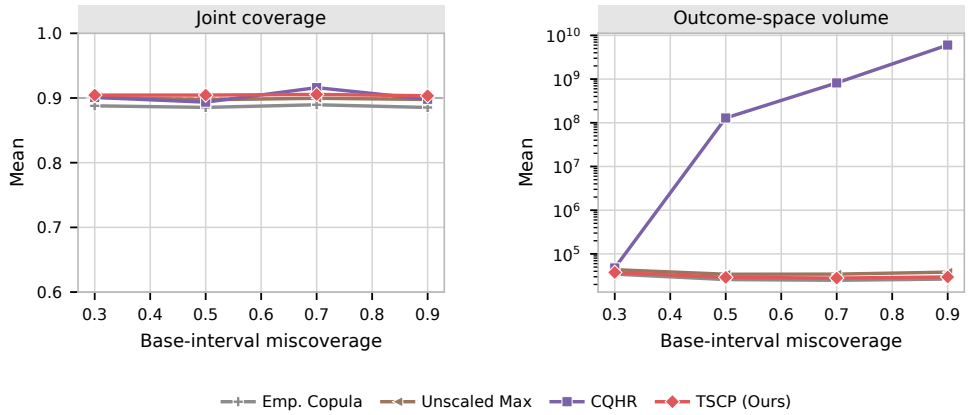


Figure S7: CQR sensitivity to the base-interval miscoverage with $d = 3$, $n = 100$, and final target coverage 0.9. TSCP uses capped scores, CQHR uses its native construction, and results average 30 independent repetitions.

explanation is that a large additive constant inflates the LWC score bound in Section **??**, so the global-worst-case bound determines the final rectangle. Equality of the recorded rectangles is empirical evidence, not a general shift-invariance result. Capped residuals $\max\{E_j, 0\}$ avoid choosing C and are used for the main comparison, with the limitations described in Section 4.3.

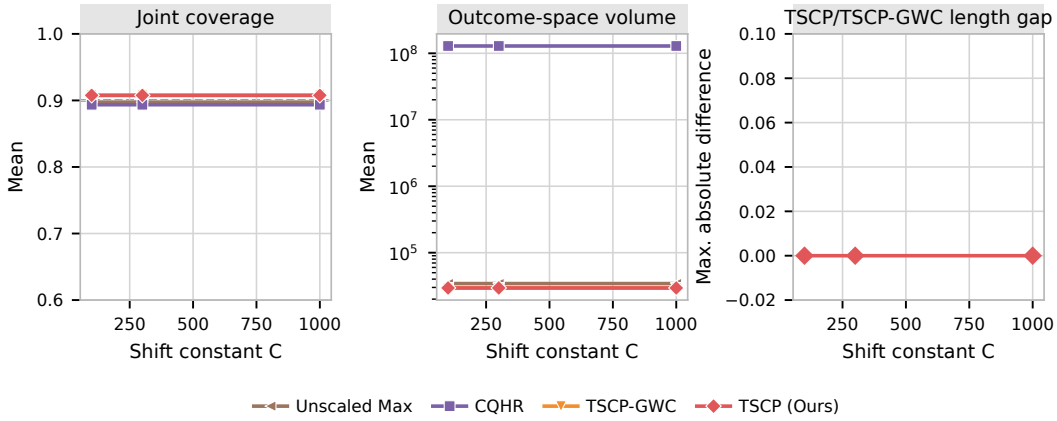


Figure S8: Sensitivity of shifted CQR scores to the additive constant C. The first two panels report coverage and mean outcome-space volume; the third reports the maximum coordinate-length difference between shifted TSCP and TSCP-GWC. Results use $d = 3$, $n = 100$, and 30 independent repetitions.

D.8 Low-Dimensional Shape-Template Baseline

We compare with the convex shape-template approach of Tumu et al. (2024). We use the public `conformal-region-designer` components: a kernel density estimate on a grid of size

20 per dimension, mean-shift clustering, and an axis-aligned hyperrectangular template. The signed calibration residuals are divided equally between fitting the density and template and conformalizing its inflation. We use a score-consistent hyperrectangle implementation and impose an aspect-ratio floor of 0.05 when a very small fitted cluster is degenerate along one coordinate. This detail keeps the conformal inflation finite.

We use $d = 2$ with independent Gaussian scales $(2, 1)$ and calibration sizes $n \in \{30, 50, 100, 200\}$. Each of the 100 repetitions uses 2,400 training observations, 600 test observations, and a separately generated calibration sample. Shape-template volume is divided by 2^d to match the residual-space volume convention used for the symmetric rectangular methods. Figure S9 shows that TSCP has smaller volume than the shape-template method at every calibration size, and its construction time is more than two orders of magnitude shorter for this implementation.

This is deliberately a low-dimensional comparison. A Cartesian KDE grid has 20^d locations and is therefore impractical at the $d = 10$ dimension used in the main absolute-residual comparisons. This limitation concerns the chosen grid implementation, not every shape-template construction.

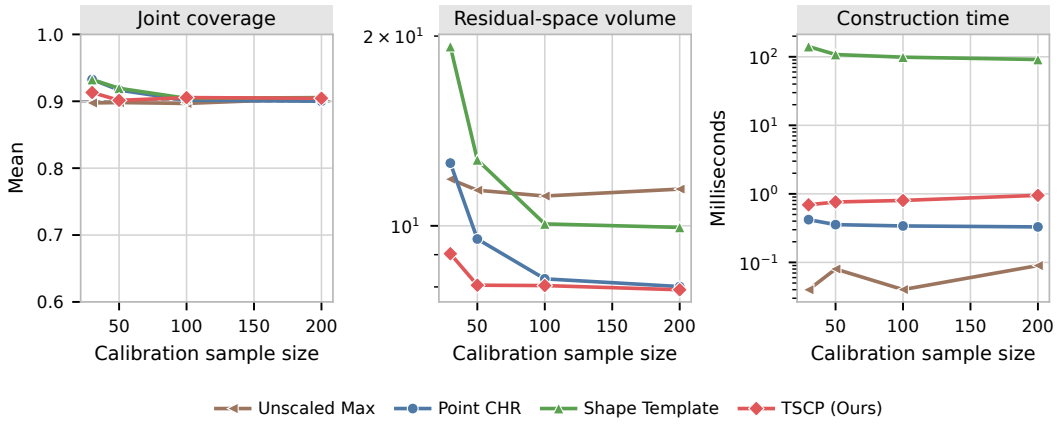


Figure S9: Comparison with a KDE/mean-shift hyperrectangular shape template for $d = 2$ Gaussian residuals with scales $(2, 1)$. Volume follows the same 2^{-d} normalization as the other residual-space experiments. Results average 100 independent repetitions. The shape-template method is slower and has larger volume than TSCP at every calibration size.

D.9 Oracle Approximation and Computational Scaling

We examine how the practical constructions approximate the population oracle under independent Laplace noise with coordinate scales $(d, d-1, \ldots, 1)$ and varying calibration size.

The additional methods considered in this section include (i) *Naive*, the heuristic plug-in rescaling method that uses location and scale parameters estimated directly from the calibration data without further adjustments, summarized in Algorithm S3; (ii) *TSCP-S*, the inefficient data-splitting variant summarized in Algorithm S4; (iii) *TSCP-GWC*, our

conservative global-worst-case method summarized in Algorithm S5; (iv) *TSCP-LWC*, the aggregated union described in Section **??** but not enclosed by a rectangle.

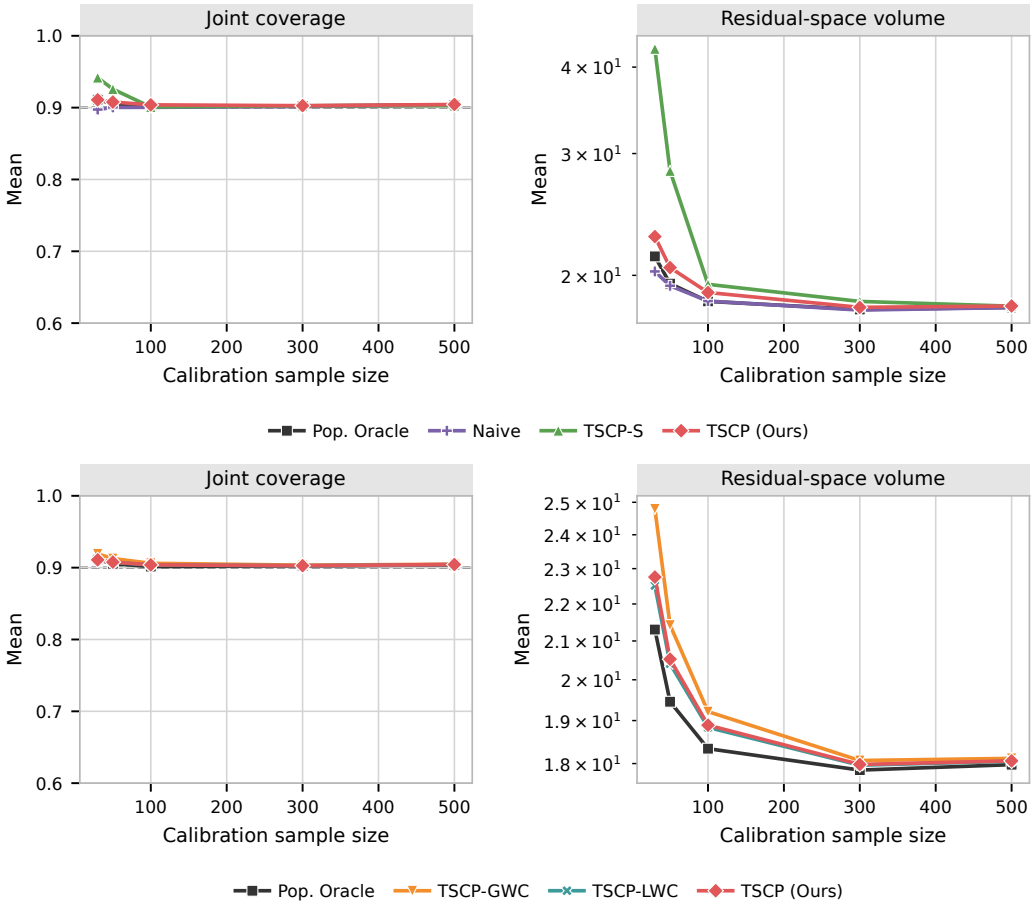


Figure S10: Performance of TSCP and alternative approaches on simulated data with $d = 2$ targets and heterogeneous Laplace noise. The target coverage level is 90%. Results are averaged over 200 trials. All methods achieve near-nominal coverage and approximate *Pop. Oracle* as the calibration sample size increases. TSCP tracks TSCP-LWC closely, with an almost indistinguishable average volume, highlighting the tightness of its rectangular enclosure.

The methods in Figure S10 approach the volume of *Pop. Oracle* as n increases. TSCP and TSCP-LWC have nearly indistinguishable mean volumes, suggesting that the rectangular enclosure adds little volume in this setting. TSCP-GWC is more conservative but produces smaller volumes than the data-splitting variant TSCP-S. Despite lacking a finite-sample coverage guarantee, the *Naive* approach performs well at larger calibration sizes, as also shown in Figure S11. Computational cost is examined separately below.

Figure S12 better highlights the prohibitive computational cost of TSCP-LWC, by reporting on the results of similar experiments where a small sample size is fixed, $|\mathcal{D}_{\text{cal}}| = 30$, while the outcome dimensions are gradually increased: $d \in \{2, 3, 4, 5, 10, 20, 30\}$.

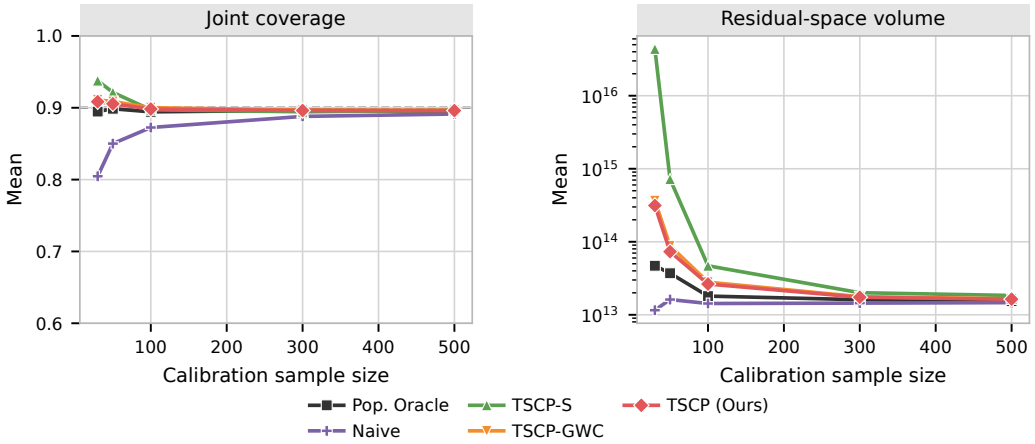


Figure S11: Performance of TSCP and alternative approaches on simulated data with $d = 10$ targets. Other details are as in Figure S10. TSCP-LWC is omitted from this figure due to its prohibitive cost with even moderately high-dimensional outcomes.

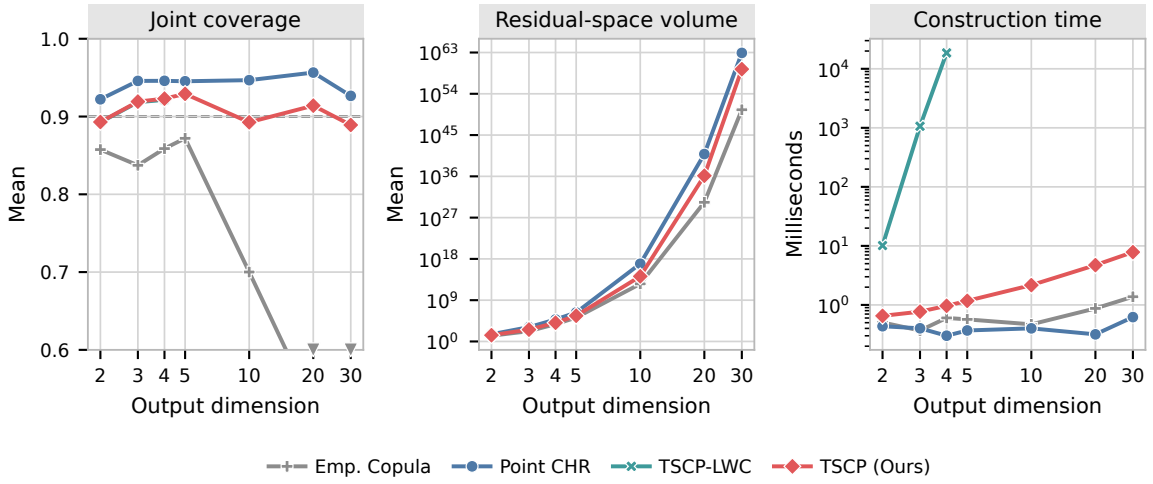


Figure S12: Absolute-residual experiment with Laplace noise, calibration size $n = 30$, and varying output dimension d. Results are averaged over 30 repetitions. The dashed line marks target coverage 0.9. TSCP-LWC is evaluated only at $d \in \{2, 3, 4\}$ because of its rapidly increasing computational cost.

D.10 Additional Real-Data Experiments

These diagnostics use 200 randomized splits per dataset with the same allocation and model settings as Section 4.4. All methods share the fitted model and observations within each split. The reported variation is over splits of each fixed dataset, not over independently sampled datasets from a population.

D.10.1 Coordinate-Wise Marginal Coverage and Interval Lengths

Figure S13 reports coordinate-wise marginal coverage for each of the six real datasets. The dashed line indicates the target joint coverage of 0.9, and each panel also reports TSCP's mean joint coverage. The coordinates follow the target-column order of each dataset, allowing for a clear comparison of how well each method performs across different dimensions of the outcome space.

To compare coordinate-wise lengths, Figure S14 reports the paired length ratio

$$A_{j,m} = \frac{1}{200} \sum_{r=1}^{200} \frac{\ell_{j,m}^{(r)}}{\ell_{j,\text{Unscaled}}^{(r)}},$$

where $\ell_{j,m}^{(r)} = 2\mathcal{L}_{j,m}^{(r)}$ is the full outcome-space interval length for coordinate j, method m, and split r. Thus, a bar below one indicates a shorter interval on average relative to Unscaled Max on the same splits. Ratios are formed before averaging; they are not ratios of mean interval lengths.

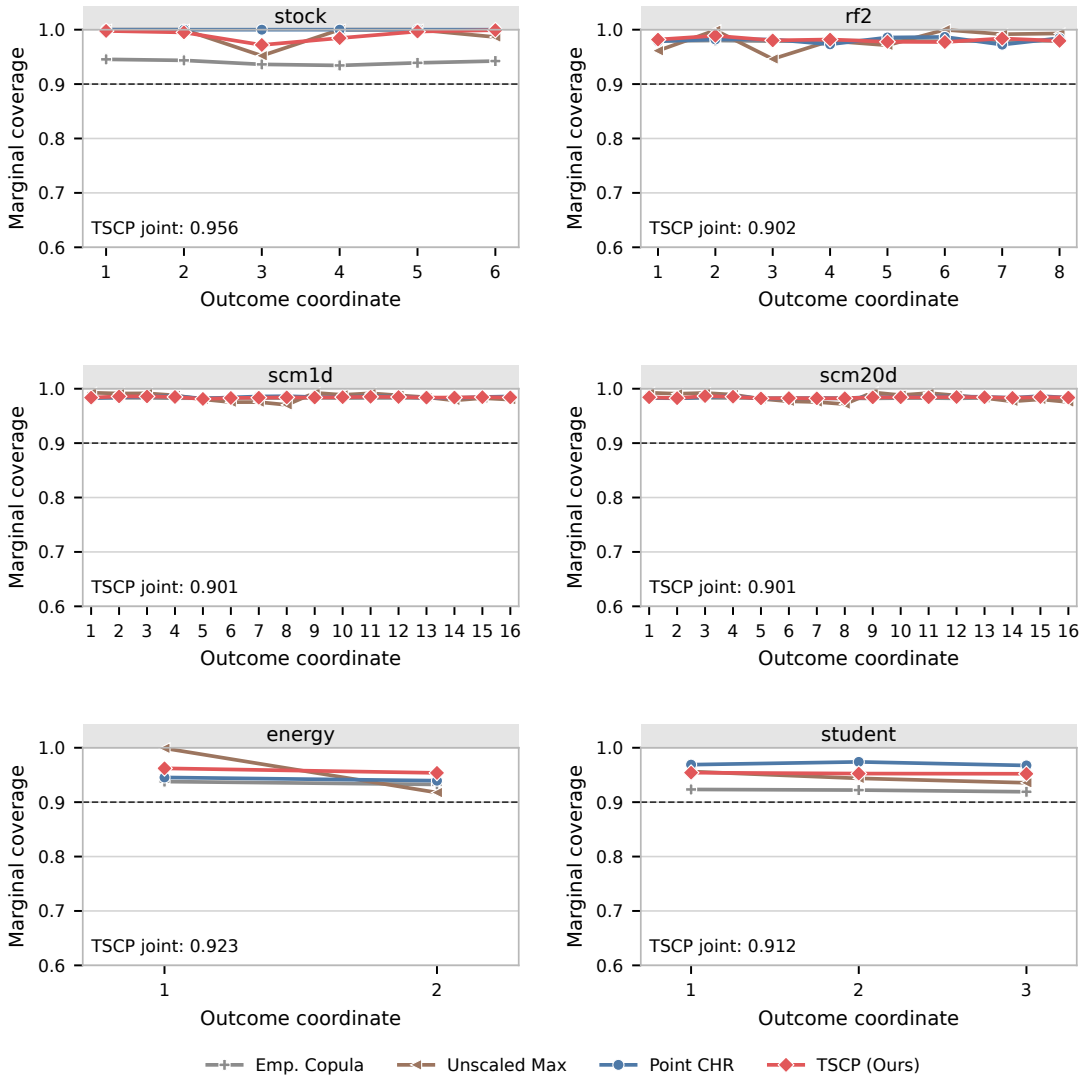


Figure S13: Coordinate-wise marginal coverage over 200 splits of each real dataset. The dashed line is the 0.9 joint target, and each panel also reports TSCP's mean joint coverage. Coordinates follow the target-column order of each dataset.

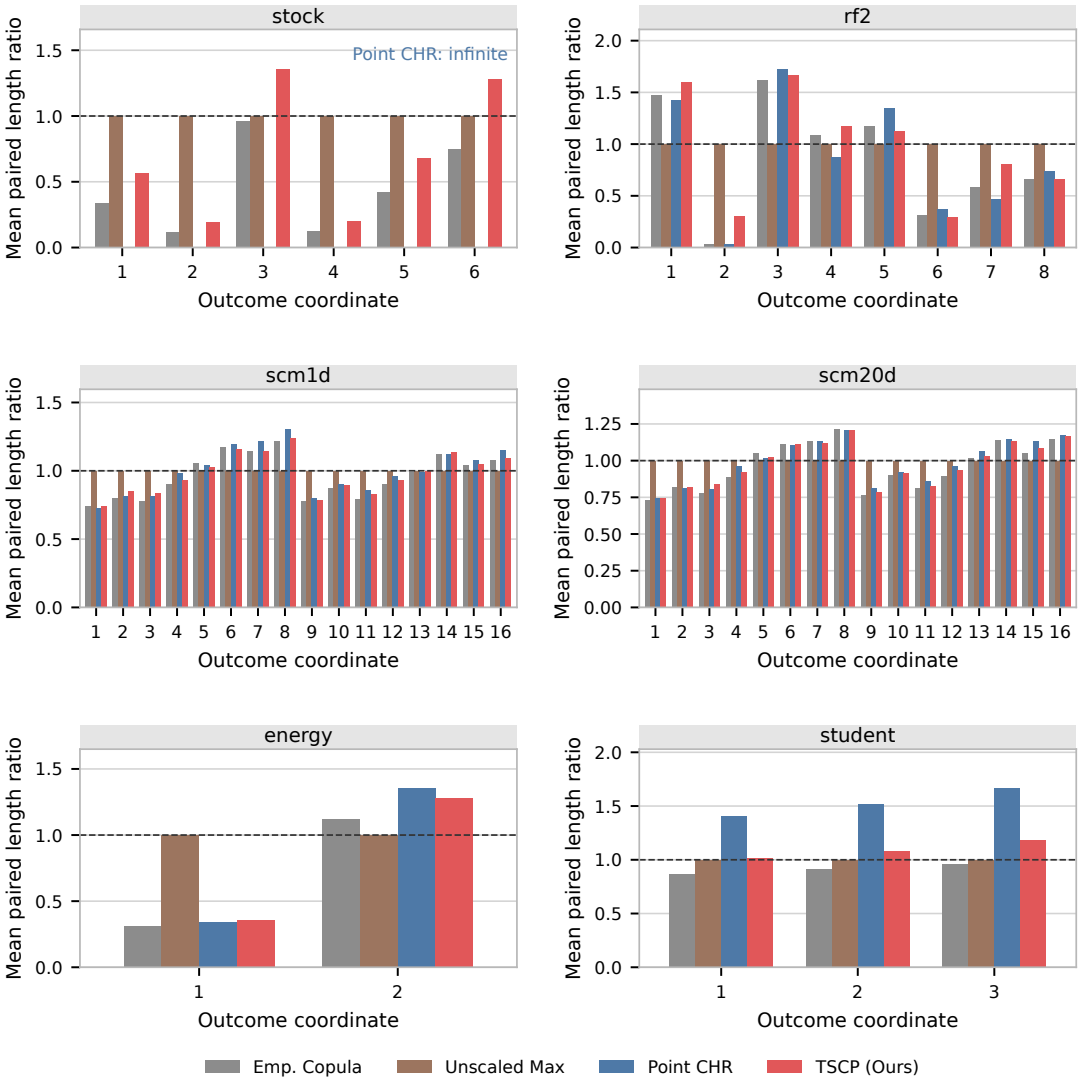


Figure S14: Coordinate-wise full interval lengths on all six real datasets, normalized within each split by the corresponding Unscaled Max length and then averaged over 200 splits. The dashed line marks a ratio of one. Point CHR is infinite on `stock`. These width comparisons should be read with the matching joint coverage and the marginal coverage in Figure S13.

D.10.2 Backward-Search and Fallback Diagnostics

We investigate the fallback and backward-search behavior of the TSCP implementation summarized in Algorithm **??**. A fallback occurs when the mean cell has empty intersection with the global rectangle; a backward-search run occurs when that intersection is nonempty but the binary-search condition fails for at least one coordinate. We also record the number of coordinates using that branch and the number of candidate cells inspected before finding the boundary.

Table S8 reports the split-level and coordinate-level counts. Figure S15 shows the number of candidate cells inspected per coordinate and the construction time of the observed search regime. For each split, we time repeated TSCP calls on the same calibration residuals after a warm-up. The mean of the per-split median runtimes is

$$\frac{1}{200}\sum_{s=1}^{200}\text{median}(T_{s1},\ldots,T_{sr}),$$

where T_{sj} is the runtime of the jth call on split s. We use $r = 7$ for $\alpha = 0.1$ and $r = 3$ for $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$. These calls are repeated timing measurements, not additional statistical repetitions. Regime-specific times average the medians only over splits exhibiting that regime. Timings exclude model fitting, residual computation, and diagnostic logging; calls run serially with native thread pools limited to one thread.

Dataset	n	Backward splits	Backward coordinates	Fallback splits	Binary only (ms)	Any backward (ms)	Fallback (ms)
`stock`	15	0/200	0/1200	0/200	1.007	–	–
`rf2`	384	0/200	0/1600	0/200	2.780	–	–
`scm1d`	490	0/200	0/3200	0/200	7.239	–	–
`scm20d`	448	0/200	0/3200	0/200	7.120	–	–
`energy`	38	0/200	0/400	0/200	0.466	–	–
`student`	32	0/200	0/600	0/200	0.620	–	–

Table S8: Executed TSCP search regimes at target coverage 0.9 on the six real datasets. Each split-count denominator is 200. The coordinate denominator is d times the number of non-fallback splits. Conditional times summarize the full TSCP call, not just the backward loop; a dash denotes an unobserved regime. n is the calibration size.

At the main target $\alpha = 0.1$, neither backward search nor fallback occurs in any of the 200 splits for any dataset. All 10,200 coordinate searches use the binary branch, with about 4.00–8.49 candidate-cell inspections per coordinate on average across datasets. Observing no events in these splits does not establish that backward search or fallback is impossible.

To explore further, we reuse the same 200 calibration residual matrices per dataset and vary the miscoverage level over $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. In this experiment, we observe that backward search and fallback occur more frequently at higher miscoverage levels. On `energy` at $\alpha = 0.5$, backward search occurs in 3/200 splits (1.5%), while fallback occurs in 140/200 (70%). The remaining 57 splits use only binary search. At $\alpha = 0.7$ and 0.9, 10 energy splits use backward search and 190 fall back. In every observed backward-search case, each of the two coordinates inspects only *one* backward candidate. No backward search is observed in the other five datasets over the displayed levels. Fallback is often faster because

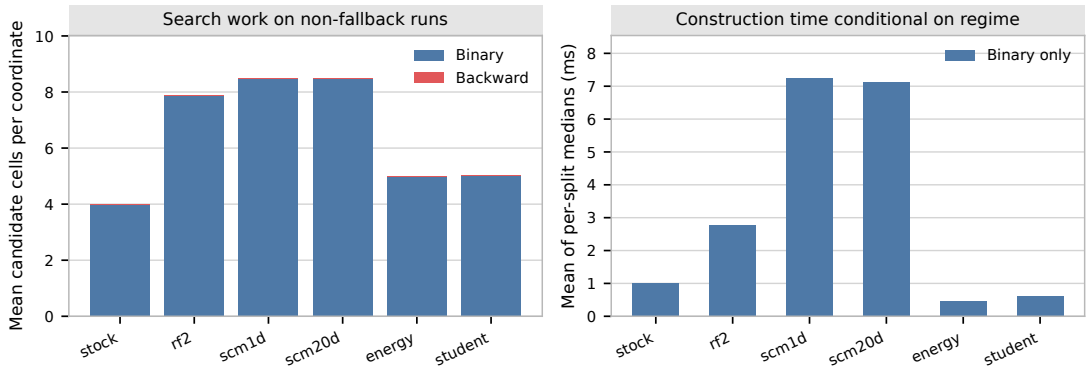


Figure S15: TSCP search work and construction time for 200 splits per real dataset at target coverage 0.9. Left: candidate cells inspected per coordinate, averaged over non-fallback runs. Counts include cells rejected by intersection checks. Right: construction times of the observed regime. Unobserved scenarios have no bars.

it returns the already computed global rectangle, so a lower average runtime at an extreme target need not indicate a more efficient local search.

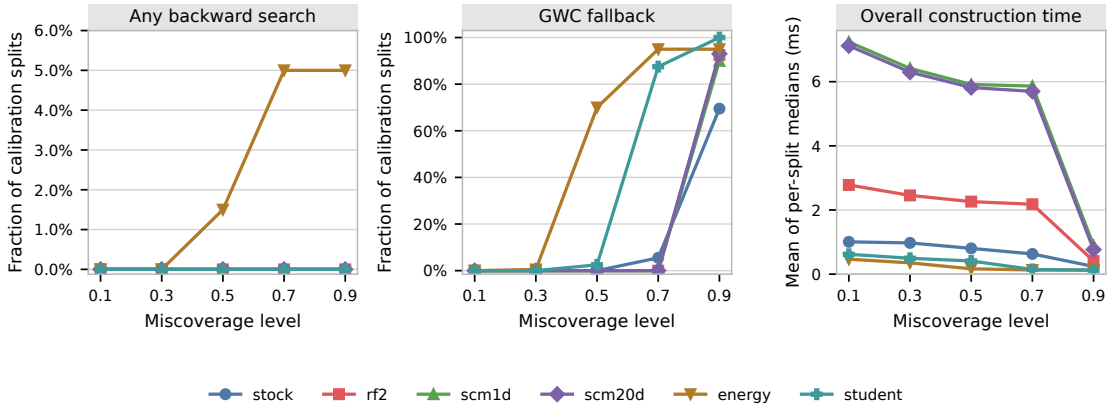


Figure S16: Computational stress test on the cached real-data calibration residuals. Each point summarizes 200 splits. The first two panels show the fractions of splits using any backward search or the global fallback. The final panel reports construction time.